

1em 0pt

Vers l'interopérabilité cognitive : jumeau numérique cognitif et architecture cognitive pour la collaboration Humain-CPS

Towards cognitive interoperability: cognitive digital twin
and cognitive architecture for Human-CPS collaboration

THÈSE

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par

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*"Peace upon a land created for peace,
yet never saw a peaceful day."
— Mahmoud Darwish*

Abstract

Industry 5.0 promotes human-centric production systems where humans and Cyber-Physical Systems (CPS) collaborate as partners rather than operating in isolated roles. In such contexts, semantic interoperability alone is insufficient, since successful collaboration requires not only a common interpretation of data, but also the alignment of situation awareness, mental models, intentions, and decision-making processes. In this context, this thesis investigates the transition towards cognitive interoperability in Cyber-Physical Enterprises (CPEs) by proposing a framework based on a Cognitive Digital Twin (CDT) and a Cognitive Architecture to support collaboration between humans and CPS.

The thesis first establishes a formal conceptualisation of cognitive systems and introduces a definition of cognitive interoperability that extends beyond semantic alignment. Based on cognitive psychology and team cognition theories, a Maturity Model of Cognitive Interoperability (MMCI) is proposed to characterise progressive levels of cognitive alignment, from shared situation awareness to joint decision-making with metacognitive regulation. The work then defines the notion of a CDT as a DT enriched with cognitive capabilities, and analyses the architectural requirements needed to operationalise cognition in such systems. To support structured reasoning and explainability, an ontology-based semantic foundation is designed to represent tasks, agents, capabilities, abilities, and contextual constraints within a unified knowledge model. In addition, the CLARION cognitive architecture is integrated as the decision-making core of the CDT, enabling hybrid neuro-symbolic operation by combining symbolic reasoning with learning mechanisms.

The proposed framework is validated through an industrial human-robot collaborative assembly use case involving a UR5e cobot. Experimental results show that embedding cognition into the CDT improves adaptability, decision-making quality, and robustness in dynamic collaboration scenarios compared to a semantic-only baseline. Overall, this thesis provides conceptual foundations, modelling tools, and an implementable architecture to support the transition from semantic to cognitive interoperability in future human-centric manufacturing systems.

Keywords: Cognitive Interoperability, Cognitive Digital Twin, Cyber-Physical Systems, Human-Robot Collaboration, Ontologies, Cognitive Architectures, Neuro-symbolic AI, Industry 5.0.

Résumé

L'Industrie 5.0 promeut des systèmes de production centrés sur l'humain, dans lesquels les humains et les systèmes cyber-physiques (Cyber-Physical Systems - CPS) collaborent en tant que partenaires plutôt que d'évoluer dans des rôles isolés. Dans de tels contextes, l'interopérabilité sémantique seule est insuffisante, car une collaboration efficace requiert non seulement une interprétation commune des données, mais aussi l'alignement de la conscience de la situation, des modèles mentaux, des intentions et des processus de prise de décision. Dans ce contexte, cette thèse étudie la transition vers l'interopérabilité cognitive au sein des Entreprises Cyber-Physiques

(Cyber-Physical Enterprises - CPE) en proposant un cadre basé sur un Jumeau Numérique Cognitif (Cognitive Digital Twin - CDT) et une Architecture Cognitive afin de soutenir la collaboration humain-CCPS.

La thèse établit d’abord une conceptualisation formelle des systèmes cognitifs et introduit une définition de l’interopérabilité cognitive allant au-delà de l’alignement sémantique. En s’appuyant sur la psychologie cognitive et les théories de la cognition en équipe, un Modèle de Maturité de l’Interopérabilité Cognitive (Maturity Model of Cognitive Interoperability - MMCI) est proposé afin de caractériser des niveaux progressifs d’alignement cognitif, allant du partage de la conscience de la situation jusqu’à la prise de décision conjointe avec régulation métacognitive. Le travail définit ensuite la notion de Jumeau Numérique Cognitif (Cognitive Digital Twin - CDT) comme un jumeau numérique enrichi de capacités cognitives, et analyse les exigences architecturales nécessaires pour opérationnaliser la cognition dans de tels systèmes. Afin de soutenir un raisonnement structuré et l’explicabilité, une base sémantique fondée sur une ontologie est conçue pour représenter les tâches, les agents, les capacités, les compétences et les contraintes contextuelles au sein d’un modèle unifié de connaissances. Par ailleurs, l’architecture cognitive CLARION est intégrée comme noyau décisionnel du CDT, permettant un fonctionnement hybride neuro-symbolique en combinant le raisonnement symbolique avec des mécanismes d’apprentissage.

Le cadre proposé est validé à travers un cas d’usage industriel d’assemblage collaboratif humain-robot impliquant un cobot UR5e. Les résultats expérimentaux montrent que l’intégration de la cognition dans le CDT améliore l’adaptabilité, la qualité de la prise de décision et la robustesse dans des scénarios de collaboration dynamique, comparativement à une approche reposant uniquement sur l’interopérabilité sémantique. Globalement, cette thèse apporte des fondements conceptuels, des outils de modélisation et une architecture implémentable permettant de soutenir la transition de l’interopérabilité sémantique vers l’interopérabilité cognitive dans les futurs systèmes de production centrés sur l’humain.

Mots-clés: Interopérabilité cognitive, Jumeau numérique cognitif, Systèmes cyber-physiques, Collaboration humain-robot, Ontologies, Architectures cognitives, IA neuro-symbolique, Industrie 5.0.

List of Publications

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- Gaffinet, B., **Al Haj Ali, J.**, Panetto, H., Naudet, Y. (2025). ([Gaffinet et al., 2025a](#)) Human Digital Twins: A systematic literature review and concept disambiguation for industry 5.0, Computers in Industry, Volume 166, 2025, 104230, ISSN 0166-3615, <https://doi.org/10.1016/j.compind.2024.104230>.
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Acronyms

ACS Action-Centred Subsystem

AI Artificial Intelligence

AI4C2PS Artificial Intelligence for Cognitive Cyber-Physical Systems Interoperability

AMNs Associative Memory Networks

ARS Action Rule Store

CCPS Cognitive Cyber-Physical Systems

CDT Cognitive Digital Twin

CPE Cyber-Physical Enterprise

CPHS Cyber-Physical-Human Systems

CPSs Cyber-Physical Systems

CPSSs Cyber-Physical-Social Systems

DT Digital Twin

EIF European Interoperability Framework

GKS General Knowledge Store

HCC Human-CCPS Collaboration

HDT Human Digital Twin

HRC Human-Robot Collaboration

IDNs Implicit Decision Networks

KGs knowledge graphs

KRR knowledge representation and reasoning

LTM Long-Term Memory

MCS Meta-Cognitive Subsystem

MS Motivational Subsystem

NACS Non-Action-Centred Subsystem

OWL Web Ontology Language

RDF Resource Description Framework

RDFS RDF Schema

SLR Systematic Literature Review

SWRL Semantic Web Rule Language

URN Uniform Resource Name

WM Working Memory

Résumé en français (French summary)

Contexte général et problématique

L'Industrie 5.0 met en avant une vision centrée sur l'humain où les systèmes de production doivent permettre une collaboration fluide, sûre et efficace entre opérateurs humains et systèmes cyber-physiques (*Cyber-Physical Systems - CPS*). Dans ce cadre, l'interopérabilité sémantique qui est déjà essentielle pour assurer un échange cohérent d'informations via des modèles partagés, des standards et des ontologies, ne suffit plus. En effet, collaborer en temps réel dans un environnement industriel dynamique requiert davantage qu'un accord sur le sens des données : cela implique une coordination basée sur la compréhension mutuelle, l'alignement des intentions, la construction de modèles mentaux partagés, ainsi que la régulation conjointe de la prise de décision.

Cette thèse s'inscrit dans le cadre du projet **AI4C2PS** dont l'objectif est d'étendre l'interopérabilité des systèmes de production du niveau sémantique vers le niveau cognitif, en s'appuyant sur des approches d'IA neuro-symbolique et sur les Jumeaux Numériques (*Digital Twins - DT*). Les DT y sont considérés comme des entités capables d'intégrer et d'héberger des mécanismes cognitifs facilitant la collaboration humain-CPS au sein d'une *Cyber-Physical Enterprise (CPE)*. Dans cette perspective, la thèse répond à un défi central : l'absence de fondements théoriques et opérationnels clairs permettant de définir, structurer et mettre en œuvre l'interopérabilité cognitive entre humains et CPS.

L'objectif général de cette thèse est donc d'établir les principes conceptuels, méthodologiques et computationnels permettant de rendre l'interopérabilité cognitive opérationnelle en environnement industriel. Trois questions de recherche structurent l'ensemble du travail :

1. **RQ1** : Comment définir et caractériser formellement l'interopérabilité cognitive au sein d'une CPE ?
2. **RQ2** : Comment définir une couche cognitive pour les CPS et formaliser les notions de systèmes cognitifs, CCPS et CDTs ?
3. **RQ3** : Quels principes de représentation et d'architecture sont nécessaires pour rendre la cognition fonctionnelle dans les CDTs, et comment les ontologies et architectures cognitives peuvent-elles y contribuer ?

Ces trois questions guident une démarche progressive allant de l'état de l'art à la conceptualisation, puis à l'implémentation et à l'évaluation sur un cas d'usage industriel.

Organisation globale du manuscrit

Le manuscrit est organisé en cinq chapitres principaux (hors introduction et conclusion), chacun apportant une contribution spécifique, depuis les fondations théoriques jusqu'à la validation expérimentale :

- **Chapitre 1** : État de l'art structuré selon les trois questions de recherche.
- **Chapitre 2** : Synthèse et conceptualisation de l'interopérabilité cognitive et proposition d'un modèle de maturité (MMCI).
- **Chapitre 3** : Conception d'une architecture cognitive de CDT fondée sur CLARION.
- **Chapitre 4** : Conception de l'ontologie CDTO pour représenter et raisonner sur les connaissances de collaboration.
- **Chapitre 5** : Implémentation et évaluation sur un cas d'usage de collaboration humain-robot en assemblage industriel.

L'introduction présente le contexte AI4C2PS et les questions de recherche, tandis que le dernier chapitre synthétise les contributions et ouvre des perspectives.

Chapitre 1 - État de l'art : fondations scientifiques

Le Chapitre 1 vise à construire le cadre conceptuel nécessaire pour répondre à l'objectif central de la thèse : définir, caractériser et opérationnaliser l'interopérabilité cognitive dans les systèmes H-CPS. Ce chapitre est structuré autour de trois revues complémentaires correspondant directement aux trois questions de recherche (RQ1–RQ3).

1.1 Analyse du concept d'interopérabilité cognitive (RQ1)

La première revue examine les définitions existantes de l'interopérabilité cognitive dans différents domaines (systèmes socio-techniques, collaboration humaine, santé, défense, etc.) et en souligne les limites pour des scénarios industriels H-CPS. L'analyse met en évidence que la plupart des travaux utilisent le terme *cognitive* de manière intuitive, sans cadre formel ni critères mesurables permettant de distinguer clairement le cognitif du sémantique.

Cette section aboutit à l'identification des notions clés associées à l'interopérabilité cognitive : compréhension mutuelle, perception partagée, modèles mentaux, intention, et prise de décision. Elle montre également que la coopération efficace implique une synchronisation des représentations internes des agents, et pas seulement un alignement des symboles ou des données.

1.2 Revue des systèmes cognitifs et des entités industrielles cognitives (RQ2)

La deuxième revue est une revue systématique portant sur les systèmes cognitifs, en croisant des perspectives issues des sciences cognitives, de l'IA et des CPS. L'objectif est de clarifier ce qui constitue une couche cognitive et comment celle-ci peut être définie, structurée et appliquée à des entités industrielles. La revue distingue notamment cognition humaine et cognition artificielle, et analyse les fonctions typiques associées à des systèmes cognitifs (perception, mémoire, apprentissage, raisonnement, décision, adaptation, explicabilité).

Cette revue permet de positionner les concepts de **Cognitive CPS (CCPS)** et de **Cognitive Digital Twin (CDT)** comme des entités capables de soutenir des mécanismes de collaboration plus avancés, notamment via l'intégration de modèles cognitifs.

1.3 Technologies habilitantes pour la cognition dans les CDTs (RQ3)

La troisième revue porte sur les technologies permettant de rendre la cognition opérationnelle dans des systèmes artificiels : représentation ontologique, raisonnement symbolique, apprentissage automatique, approches hybrides neuro-symboliques, et architectures cognitives. Cette partie justifie l'intérêt des ontologies pour structurer un espace de connaissance explicite, cohérent et réutilisable, tout en montrant la nécessité de mécanismes d'apprentissage et de décision adaptatifs pour gérer les imprévus et la variabilité des tâches industrielles.

Ce chapitre se conclut par une observation clé : le passage du sémantique au cognitif nécessite un couplage explicite entre représentation des connaissances et architecture décisionnelle. Cette conclusion prépare directement les contributions développées dans les chapitres suivants.

Chapitre 2 - Conceptualisation : interopérabilité cognitive et modèle de maturité

Le Chapitre 2 constitue le cœur conceptuel de la thèse. Il répond principalement à **RQ1** et partiellement à **RQ2** en proposant : (i) une définition formelle de l'interopérabilité cognitive et (ii) un modèle de maturité structurant ses niveaux.

2.1 Définition de l'interopérabilité cognitive

À partir des analyses du Chapitre 1, l'interopérabilité cognitive est définie comme la capacité d'agents humains et/ou artificiels à aligner leurs pensées et leur perception de l'information afin d'atteindre une compréhension mutuelle et des intentions partagées, via la construction de modèles mentaux communs et une manière commune d'utiliser, interpréter et raisonner sur la connaissance.

Cette définition vise explicitement à dépasser l'interopérabilité sémantique : deux agents peuvent partager le sens d'un concept sans pour autant partager la perception de l'état réel du système, ni la stratégie de décision à adopter dans un contexte donné.

2.2 Proposition du modèle MMCI

La contribution majeure de ce chapitre est la proposition d'un **Maturity Model of Cognitive Interoperability (MMCI)**. Ce modèle structure l'interopérabilité cognitive en niveaux progressifs, en s'appuyant sur des fondations issues :

- des hiérarchies de compétences cognitives (ex. taxonomie de Bloom),
- de domaines neuropsychologiques (ex. fonctions exécutives, métacognition),
- de la cognition en équipe et de l'action conjointe.

Le MMCI démarre au **niveau 0**, considéré comme le maximum atteignable via l'interopérabilité sémantique, puis définit quatre niveaux cognitifs :

1. **Niveau 1** : conscience de situation partagée,
2. **Niveau 2** : modèles mentaux partagés,
3. **Niveau 3** : alignement mutuel des intentions et du raisonnement,
4. **Niveau 4** : prise de décision conjointe avec régulation métacognitive.

Ce modèle fournit un cadre théorique structuré pour analyser le degré de maturité des interactions humain–CPS et permet d'évaluer la transition progressive du sémantique vers le cognitif.

2.3 Apports du chapitre

En synthèse, le Chapitre 2 :

- établit une définition formalisée et opérationnalisable de l'interopérabilité cognitive,
- propose un modèle de maturité permettant de caractériser des niveaux de collaboration,
- pose les bases d'une future évaluation basée sur des indicateurs mesurables.

Chapitre 3 - Architecture cognitive : intégration de CLARION dans un CDT

Le Chapitre 3 répond principalement à **RQ3** en s'intéressant à la manière d'implémenter des fonctions cognitives dans un CDT. Il introduit les principes architecturaux permettant de transformer un DT classique en **Cognitive Digital Twin** apte à soutenir l'interopérabilité cognitive.

3.1 Définition fonctionnelle du CDT et stratégies d'implémentation

Le CDT est défini comme une entité computationnelle reposant sur trois fonctions principales : *émulation*, *simulation* et *cognition*. La thèse se concentre sur la fonction cognitive, tout en considérant que celle-ci doit s'intégrer dans un cadre plus large où simulation et émulation soutiennent l'anticipation et la reproductibilité des comportements du système physique.

Deux stratégies d'implémentation sont comparées : une cognition externalisée et fortement couplée au système physique, et une cognition internalisée au CDT, plus modulaire, servant principalement à la prise de décision et au raisonnement.

3.2 Choix d'une architecture cognitive : CLARION

Le chapitre analyse plusieurs architectures cognitives (symboliques, sub-symboliques, hybrides) et justifie le choix de **CLARION** comme architecture pertinente pour répondre aux exigences de l'interopérabilité cognitive, notamment grâce à :

- une organisation hybride explicite/implicite (neuro-symbolique),
- la présence de sous-systèmes dédiés à l'action, la connaissance, la motivation et la métacognition,
- des mécanismes permettant à la fois décision explicable et adaptation par apprentissage.

CLARION est détaillée à travers ses sous-systèmes : **ACS** (procédural), **NACS** (déclaratif), **MS** (motivation/objectifs), et **MCS** (métacognition). Le chapitre montre comment ces composants permettent de soutenir des processus clés : sélection d’actions, gestion des objectifs, surveillance des erreurs, et arbitrage entre règles et apprentissage.

3.3 Contribution du chapitre

Ce chapitre fournit ainsi :

- une architecture cognitive de CDT cohérente avec l’objectif d’interopérabilité cognitive,
- une justification méthodologique du choix de CLARION,
- une base de couplage entre représentation symbolique et décision adaptative.

Chapitre 4 - Ontologie CDTO : représentation et raisonnement pour CDTs

Le Chapitre 4 traite le pilier symbolique du cadre proposé : une ontologie permettant de représenter les concepts nécessaires à l’interopérabilité cognitive et au fonctionnement d’un CDT.

4.1 Motivation

Afin de rendre la collaboration humain-CCPS cohérente et explicable, il est nécessaire de disposer d’un modèle partagé décrivant :

- les tâches et leur structure (workflow),
- les agents (humain, robot, systèmes),
- les capacités et compétences (capabilities/abilities),
- les objets, ressources, contraintes et conditions de contexte.

L’ontologie joue ici un rôle double : (i) structurer la connaissance de manière formelle et (ii) permettre le **raisonnement** pour déduire des informations utiles à la prise de décision.

4.2 Conception de l’ontologie CDTO

Le chapitre décrit les bases technologiques (RDF/RDFS/OWL), les mécanismes d’inférence, ainsi que l’utilisation de requêtes SPARQL et de règles SWRL. L’ontologie CDTO est organisée en modules et réutilise des ontologies externes afin de favoriser l’alignement avec des standards existants. Elle est validée dans Protégé, et la cohérence logique est vérifiée par raisonneur (ex. HermiT).

4.3 Apports principaux

L’ontologie constitue la **mémoire déclarative** et la base de raisonnement du CDT :

- elle permet de formaliser des règles d’exécution, de faisabilité et de dépendances temporelles,

- elle soutient des mécanismes de décision explicables (via déduction symbolique),
- elle prépare l'intégration du raisonnement ontologique dans une architecture cognitive hybride.

Chapitre 5 - Cas d'usage industriel : implémentation et évaluation

Le Chapitre 5 valide expérimentalement le cadre proposé à travers un cas d'usage de **collaboration humain-robot** pour l'assemblage d'un moteur, avec un cobot **UR5e**. Le but est de démontrer comment le CDT, enrichi par ontologie et architecture cognitive, permet une collaboration plus adaptative qu'une approche strictement sémantique.

5.1 Description du scénario HRC

Le chapitre décrit l'environnement de travail, la séquence d'assemblage, les rôles attribués aux agents, et la motivation : gérer les imprévus, redistribuer les tâches si nécessaire, et permettre au robot de compenser certaines limitations de compétences.

5.2 Instanciation de l'ontologie et raisonnement

L'ontologie CDTO est instanciée pour représenter la situation courante : agents, tâches, objets, contraintes. Les mécanismes d'inférence (SWRL/OWL) permettent de dériver des informations de haut niveau comme la progression du workflow, la faisabilité d'actions, les prérequis de capacités, ou encore les tâches manquantes.

5.3 Intégration de CLARION avec le raisonnement ontologique

Une contribution importante du chapitre est l'intégration concrète de l'ontologie dans CLARION : les résultats du raisonnement sont transformés en variables de perception et injectés dans les sous-systèmes de l'architecture. Les rôles sont répartis comme suit :

- **ACS** : filtrage et exécution des actions via règles et routines,
- **NACS** : stockage des connaissances déclaratives (workflow, relations, contexte),
- **MS** : gestion des objectifs et priorités de collaboration,
- **MCS** : supervision, déclenchement de l'apprentissage, régulation adaptative.

Cette structuration permet de relier décision explicable et comportement adaptatif en contexte industriel.

5.4 Apprentissage : adaptation procédurale pour la tâche de vissage

Le chapitre introduit un module d'apprentissage basé sur *Reinforcement Learning* (PPO) afin d'acquérir une compétence procédurale manquante : la réalisation d'un **vissage** robuste. L'environnement de simulation modélise la dynamique outil-vis (position, rotation, force, couple), et la politique apprise contrôle l'action via un ensemble discret (visser, dévisser, monter, descendre).

Le lien avec CLARION est clairement posé : l'apprentissage ne remplace pas la décision cognitive globale, mais intervient comme un module d'exécution de compétence lorsqu'une adaptation

motrice est nécessaire. Les décisions de haut niveau restent gérées par CLARION, garantissant cohérence du workflow et explicabilité.

5.5 Évaluation et comparaison

L'approche proposée est comparée à une solution de référence sans architecture cognitive. Les résultats montrent que le CDT cognitif améliore :

- l'adaptabilité (gestion des déviations et des tâches manquantes),
- la qualité de décision (prise en compte des contraintes, objectifs et contexte),
- la robustesse globale de l'interaction en collaboration dynamique.

Enfin, le système est évalué au regard des niveaux MMCI afin de montrer la progression au-delà de l'interopérabilité sémantique.

Conclusion générale

Cette thèse propose un cadre complet pour soutenir la transition du sémantique au cognitif dans les CPE, en combinant :

1. une **définition formelle** de l'interopérabilité cognitive,
2. un **modèle de maturité** (MMCI) structurant ses niveaux,
3. une **architecture cognitive de CDT** fondée sur CLARION,
4. une **ontologie** (CDTO) pour la représentation et le raisonnement,
5. une **validation expérimentale** en collaboration humain-robot avec intégration de simulation et apprentissage procédural.

En apportant à la fois des contributions conceptuelles et une implémentation démonstrative, ce travail constitue un socle méthodologique et technique pour le développement de systèmes de production réellement collaboratifs, explicables et adaptatifs, en cohérence avec les ambitions de l'Industrie 5.0.

Introduction

0.1 Context and motivation

Industrial production systems are currently going through a profound transformation commonly associated with Industry 4.0, driven by the spread of sensors, the omnipresence of connectivity, and the close integration of computing into physical processes ([Kagermann et al., 2013](#)). Within this vision, [Cyber-Physical Systems \(CPSs\)](#) are recognised as a central technological building block, in which physical entities are coupled with cyber components for monitoring, control, and decision support ([Lezoche and Panetto, 2020](#)). In industrial contexts, CPS provide the technological basis for real-time coordination and adaptive behaviour in production environments, and constitute a key technological backbone of smart industrialisation ([Liao et al., 2017](#)).

However, as industrial environments become increasingly interactive and adaptive, it is no longer sufficient to consider only the cyber-physical coupling. In practice, operators and decision-makers continuously influence the system: they interpret context, manage trade-offs, handle uncertainty, and adapt actions when conditions deviate from nominal expectations, which increases cognitive demands and may create overload when automation is not designed to support human work ([Kiel et al., 2017](#)). This motivates a broader paradigm that explicitly integrates the human component within system modelling and operation: [Cyber-Physical-Human Systems \(CPHS\)](#) ([Annaswamy et al., 2023](#)). CPHS consider humans as integral elements of the global system, rather than as external users, and have developed into a structured field of research, supported by reference works and specialised scientific communities.

These findings have led to a reconsideration of the role of technology and the emergence of a new paradigm: Industry 5.0. This paradigm is grounded in human-centricity. It does not succeed Industry 4.0 in a completely new way, but rather extends it by explicitly reintroducing human beings, resilience and sustainability at the heart of industrial systems ([Nahavandi, 2019](#)). The aim is no longer to fully automate tasks, but to facilitate cooperation between people and machines, leveraging human skills such as creativity, problem-solving, and adaptation to unexpected situations. Emerging technologies such as cobots (collaborative robots), exoskeletons, augmented reality and adaptive interfaces are becoming tools for cooperation rather than substitution.

In parallel, the literature has progressed towards socio-technical perspectives such as [Cyber-Physical-Social Systems \(CPSSs\)](#), which extend CPS by integrating social components and behavioural dynamics ([Yilma et al., 2021](#)). CPSS were introduced to better understand the mutual impact between CPS and humans, and to account for the additional complexity brought by human dynamics and socio-technical coupling ([Yilma et al., 2021](#)).

This development is directly aligned with the strategic focus of Industry 5.0, which prioritises human capabilities and recommends industrial technologies that support them (Nahavandi, 2019). However, despite this objective, there is still a gap: many CPSs are primarily designed as autonomous technical entities, with a limited human role in supervision or execution. At the enterprise level, this gap is particularly problematic in a **Cyber-Physical Enterprise (CPE)**, i.e., a set of interconnected technical, human, and organisational elements across all levels of the enterprise, from production processes to supply chain networks (Panetto et al., 2019). In such CPEs, the *human-in-the-loop* often remains confined to interfaces and control actions; humans are not truly integrated as collaborative partners in the system’s reasoning and adaptation mechanisms (Nahavandi, 2019; Kiel et al., 2017).

This gap is becoming increasingly apparent in literature and practice: a human cannot truly collaborate with a CPS in both directions if the CPS is unable to interpret human intentions, roles and context. Therefore, the CPS cannot adapt its behaviour as a cooperative agent. Collaboration requires shared decision-making and responsibility, as well as mutual adaptation in real time (Castañer and Oliveira, 2020). In this context, recent research has proposed **Cognitive Cyber-Physical Systems (CCPS)**, CPS enriched with cognitive capabilities (e.g., interpretation, reasoning, adaptation, and explainability) during interaction (Naudet et al., 2023a; Chervontsev et al., 2023). With this in mind, collaboration becomes bidirectional: humans adapt to the system, and the system must also be able to understand and adapt to humans through cognitive coupling mechanisms (Naudet et al., 2023a).

This is precisely the motivation behind the **Artificial Intelligence for Cognitive Cyber-Physical Systems Interoperability (AI4C2PS)** project¹, in which this thesis is situated. AI4C2PS starts from the observation that current integration and interoperability mechanisms are effective for connecting systems and aligning data, but remain insufficient to support mutual understanding and robust **Human-CCPS Collaboration (HCC)**, since humans and machines process knowledge differently (Naudet et al., 2023a). The project therefore aims to provide conceptual and computational foundations for cognitive capabilities in CCPS, and to investigate how these capabilities can support richer forms of HCC in industrial environments by combining symbolic and learning-based approaches.

0.2 Trends and industrial transformation

The transition towards human-centric industrial systems is enabled by a convergence of digital and cognitive technologies. Cyber-physical infrastructures provide the real-time data and actuation capabilities required for adaptive production, while emerging technologies such as collaborative robotics, augmented interfaces and intelligent decision support aim to improve cooperation between humans and machines (Liao et al., 2017; Nahavandi, 2019). Historically, this evolution can be understood as a gradual shift from mechanisation and electrification to digitalisation and intelligent connected systems (Mathur et al., 2022; Liao et al., 2017).

¹<https://www.ai4c2ps.eu/>

0.2.1 From challenges in Human-CPS collaboration

The effectiveness of a CPE depends on interoperability, i.e., the ability of entities to exchange information and make use of it in a consistent manner (European Commission, 2017). In highly connected environments, the key difficulty is often not the transmission of data, but the preservation of meaning across heterogeneous actors, models and contexts. This is precisely the role of *semantic interoperability*: ensuring that exchanged information is interpreted with the same meaning by all parties (Panetto, 2007; European Commission, 2017). Semantic interoperability is commonly achieved through shared vocabularies and formal knowledge models, typically implemented via ontologies. Ontologies provide a machine-processable representation of domain concepts and relations, allowing distributed cyber-physical components to align their interpretations and to perform rule-based reasoning on exchanged knowledge (Panetto, 2007). In machine-to-machine exchanges, this approach is effective: systems can query, infer, and coordinate based on formally defined semantics (Panetto, 2007; Panetto et al., 2019).

When collaboration involves humans (e.g., HCC, HRC), semantic interoperability remains a necessary foundation, but it does not, by itself, characterise the quality of collaboration. Indeed, collaboration is not only a matter of accessing the same information; it also relies on how agents interpret situations, form expectations, coordinate roles, and adapt to evolving context during joint activity (Castañer and Oliveira, 2020). In other words, two partners may access semantically aligned information, yet still diverge in their understanding of what is relevant, what is intended, or what should be done next, because interpretation in collaborative work depends on goals, perspectives, experience, and situational constraints (Tomasello, 2020; Hayes and Sanford, 2014).

This thesis therefore adopts the perspective that, in CPE, interoperability must be analysed not only at the level of shared meaning, but also at the level of the cognitive processes that support mutual coordination. Rather than stating that semantic technologies are “insufficient”, we consider that semantic alignment is a prerequisite which must be complemented by mechanisms enabling systems to represent and reason about elements that are central to collaboration, such as intentions, roles, commitments, and context (Naudet et al., 2023a). This motivates the transition from semantic interoperability to a stronger requirement: *cognitive interoperability*, understood as the ability of interacting entities to align not only the meaning of exchanged information, but also the cognitive structures that support interpretation and joint action (e.g., intentions, perspectives, roles, and context) (Naudet et al., 2023a).

0.2.2 Towards cognitive interoperability between humans and CCPs

Human cooperation is supported by cognitive capabilities that go beyond decoding information: humans coordinate by forming shared goals, anticipating each other’s intentions, managing commitments and adapting their behaviour as situations evolve (Hayes and Sanford, 2014; Tomasello, 2020). In collaborative work, cognition can be understood as the set of processes by which sensory information is transformed, reduced, processed, stored, retrieved and used to guide decision and action (Neisser, 1967). These processes include perception, attention, memory, learning, reasoning, problem solving, decision making, and situational awareness (Kiely, 2014; Vernon et al., 2007). In CPE, such cognitive functions allow partners to interpret the *why* of actions (intentions and goals), the *how* (roles, constraints, strategies), and the *when* (timing and adaptation), thereby maintaining coherence in joint activity (Naudet et al., 2023a).

Consequently, the thesis investigates how cognitive interoperability can be defined and operationalised for HCC, and how this perspective can complement semantic foundations to support more robust, interpretable and adaptive collaboration.

Operationalizing such a concept requires computational supports that can (i) represent the state of the system and the collaboration context, (ii) reason about tasks and goals, and (iii) adapt behavior and explanations to human needs. Within this perspective, Digital Twins (DTs) provide a natural infrastructure: they already represent system states and integrate heterogeneous data, and they can be extended to host reasoning and decision mechanisms (Grieves and Vickers, 2016; Semeraro et al., 2021).

This leads to the notion of a **Cognitive Digital Twin (CDT)**, a DT enriched with cognitive capabilities (Adl, 2016). When the twinned entity is technical, the CDT supports interpretation and adaptive control beyond physical monitoring, relying on cognitive functions and learning to reason about its environment and objectives (Chervontsev et al., 2023; Alohali et al., 2022). When the twinned entity is human, the CDT takes the form of an **Human Digital Twin (HDT)**, intended to represent human state and dynamics and to provide feedback to support interaction (Gaffinet et al., 2025a). Together, CDTs and HDTs create an architecture in which both technical and human entities can be represented and reasoned about in a unified framework, enabling cognitive alignment in collaborative scenarios (Naudet et al., 2023a).

From an implementation viewpoint, most CDT proposals rely on a semantic knowledge layer (ontologies, knowledge graphs) to structure heterogeneous data and support shared meaning and reasoning (Ali et al., 2021; Liu et al., 2025; Ding et al., 2025). For decision-making purposes, CDTs typically incorporate AI-based mechanisms (machine learning, reasoning engines, optimisation) to process heterogeneous data and generate adaptive behaviours (Yitmen et al., 2021). However, when the objective is cognitive interoperability with humans, semantics and AI alone remain insufficient: systems also require mechanisms to model perception, memory, goals and decision-making strategies in ways compatible with human cognitive functioning (Sandini et al., 2024; Naudet et al., 2023a). Cognitive architectures provide a promising pathway here, as they offer explicit computational models of cognitive processes (perception, memory, planning, learning) that can support interpretable and adaptive interaction (Kotseruba and Tsotsos, 2020). Architectures such as SOAR² Laird et al. (2017a), ACT-R³ Ritter et al. (2019) and CLARION⁴ Sun (2006) have been studied to model human-like cognitive processes involved in problem solving and adaptive behaviour.

0.3 Project context: AI4C2PS

This thesis is conducted in the context of the **AI4C2PS** project⁵, a bilateral initiative funded by the French National Research Agency (ANR) and the Luxembourg National Research Fund (FNR). The consortium includes the University of Lorraine, the Research Centre for Automatic COntrol (CRAN CNRS UMR 7039), the Luxembourg Institute of Science and Technology (LIST)

²[https://en.wikipedia.org/wiki/Soar_\(cognitive_architecture\)](https://en.wikipedia.org/wiki/Soar_(cognitive_architecture))

³<https://en.wikipedia.org/wiki/ACT-R>

⁴[https://en.wikipedia.org/wiki/CLARION_\(cognitive_architecture\)](https://en.wikipedia.org/wiki/CLARION_(cognitive_architecture))

⁵<https://www.ai4c2ps.eu/>

and the enterprise Orisun (ORSN).

AI4C2PS addresses the limitation identified above: while ontology-based approaches can ensure semantic interoperability among technical components, they remain insufficient to enable mutual understanding between humans and cyber-physical systems, since humans and machines process knowledge differently. The project therefore proposes extending interoperability from the semantic to the cognitive level by integrating models that represent concepts, cognitive processes and collaborative dynamics, with a human-centric orientation aligned with trustworthy [Artificial Intelligence \(AI\)](#). Within AI4C2PS, CDTs are conceived as a key mechanism to integrate and train cognitive abilities and to support collaboration in dynamic industrial environments, leveraging neuro-symbolic approaches that combine symbolic consistency with learning-based adaptation ([Lamb et al., 2020](#); [Marcus, 2020](#)).

0.4 Research questions and objectives

Building on the challenges highlighted in the AI4C2PS project, this thesis aims to address the specific challenge arising from the lack of a theoretical or operational basis for cognitive interoperability between humans and CPSs. Although semantic interoperability is now well understood, what constitutes cognitive interoperability remains to be detailed and how it can be formalised within CPE.

The overall objective of the thesis is therefore to *establish the conceptual, methodological and computational principles that enable cognitive interoperability between humans and CPSs*. To achieve this, several areas of research have been explored: the characterisation of cognition for industrial collaboration; the representation of shared cognitive structures; and the design of architectures capable of supporting dynamic cognitive alignment. We formulated accordingly four research questions:

RQ1: *How can cognitive interoperability within a CPE be formally defined and characterised, and how can a maturity model capture the transition from semantic to cognitive interoperability?*

Cognitive interoperability has been discussed in human contexts, but remains weakly formalised for human-machine and machine-machine collaboration in industrial ecosystems. This question aims to (i) propose a rigorous definition and characterisation adapted to CPE settings and (ii) derive measurable dimensions and criteria that distinguish cognitive interoperability from semantic interoperability that will provide the basis for a structured preliminary maturity model.

RQ2: *What constitutes a cognitive layer? How can a CCPS and a CDT be defined, and more generally, what is a cognitive system?*

While current industrial systems include sensors and data-driven adaptation, they lack an explicit cognitive layer with the capacity for reasoning, anticipation or explanation. In the AI4C2PS project, it is expected that DTs will provide cognition rather than physical machines, but the theoretical foundations of these entities remain underdeveloped. This question explores the conceptual elements required to define cognitive systems, CCPS and CDTs, and examines the cognitive functions necessary to enable it.

RQ3: *Which representation principles and knowledge structures are required to support cognitive interoperability in CDTs?*

Making cognition operational requires explicit representations of goals, roles, constraints, tasks, context and interaction state. This question investigates how such elements should be represented to support consistent interpretation and exchange between partners, and analyses the role of semantic models (e.g., ontologies/KGs) as a foundation for structuring and sharing the knowledge required for cognitive interoperability.

RQ4: *Which cognitive-architecture requirements enable a CDT to perform cognitive functions needed for collaboration (reasoning, adaptation, and explainability), and how can these architectures be integrated with the chosen knowledge representations?*

Beyond representation, cognitive interoperability requires computational mechanisms to exploit knowledge: reasoning about situations, updating beliefs, managing goals, selecting actions, and producing explanations. This question identifies the properties a cognitive architecture must satisfy to support these functions in a CDT, and studies how symbolic and learning-based components can be combined into a coherent architecture aligned with the objectives of the thesis.

0.5 Contributions and structure of the manuscript

This thesis makes significant contributions to our understanding of, and ability to implement, cognition within CPE. Firstly, it provides a formal characterisation of cognitive systems, cognitive CPSs and CDT. This establishes a conceptual basis for integrating cognition into industrial environments. Building on this, the thesis introduces the concept of cognitive interoperability and provides a formal definition of it. It also proposes a multi-level maturity model for assessing the progression from semantic to cognitive alignment in human-CPSs interactions. A second contribution is the design of an ontology that structures the knowledge required for collaborative tasks, creating a shared conceptual space between human and cognitive DT. This ontology facilitates symbolic reasoning within the CDT.

This thesis contributes to the understanding and operationalisation of cognition within a CPE through the following contributions:

- C1: Formal characterisation of cognitive entities.** The thesis provides a formal characterisation of cognitive systems and clarifies how cognition can be conceptualised for CPE, including the definition and positioning of CCPS and CDT. (*Addresses RQ2*).
- C2: Formal definition of cognitive interoperability.** The thesis introduces the concept of cognitive interoperability in HCC and provides a formal definition that distinguishes it from semantic interoperability by focusing on cognitive alignment mechanisms (e.g., intentions, roles, context and adaptive coordination). (*Addresses RQ1*).
- C3: Cognitive interoperability maturity model.** The thesis proposes a multi-level maturity model to assess the progression from semantic interoperability to cognitive interoperability in HCC, providing criteria to situate an interaction or system with respect to cognitive interoperability capabilities. (*Addresses RQ1*).
- C4: Ontology supporting cognitive interoperability in collaborative tasks.** The thesis designs an ontology that structures the knowledge required for collaboration (task context, roles, goals, constraints, interaction state), establishing a shared conceptual space for the CDT. This semantic layer supports consistent interpretation and enables symbolic reasoning within the CDT. (*Addresses RQ3; supports RQ4*).

Figure 1 summarises the organisation of the thesis and its alignment with the research questions.

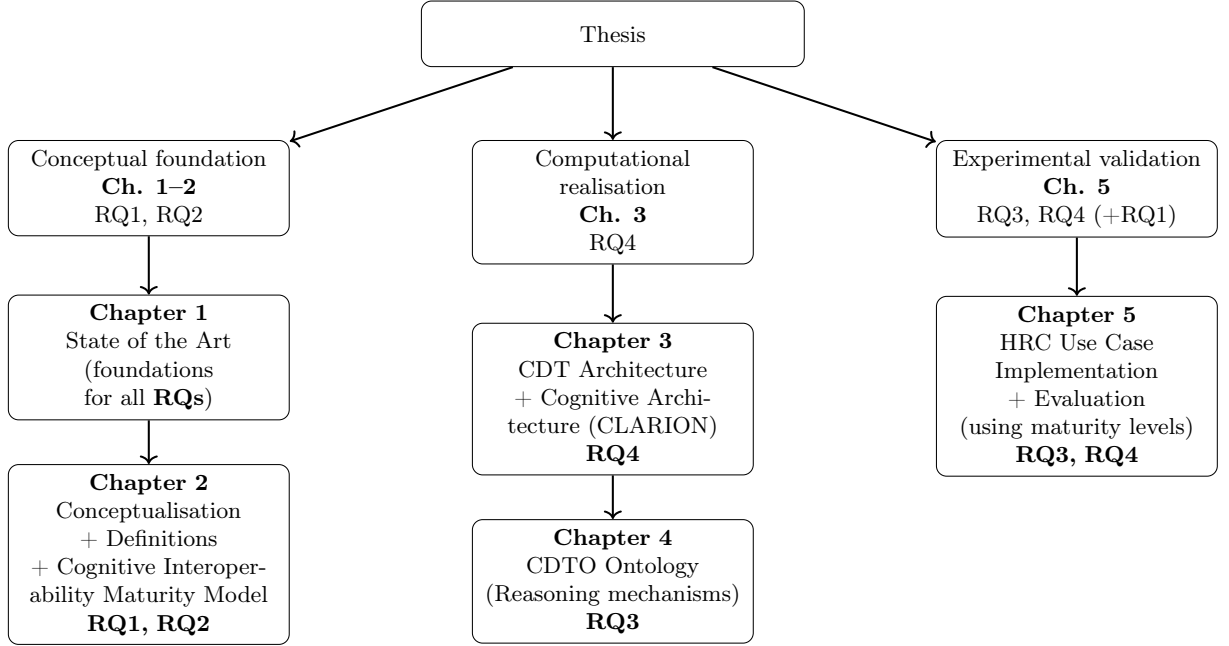


Figure 1: Structural map of the thesis and alignment with research questions.

The following sections provide an overview of how these contributions are developed, articulated and validated across the various chapters of the manuscript.

Chapter 1 – *State of the Art*

It provides the state of the art and theoretical background for the thesis [Al Haj Ali et al. \(2024a, 2025a\)](#). It reviews key notions of cognition from psychology, cognitive science and AI, and analyses existing work on cognitive capabilities in industrial systems, including CCPS and CDTs. It also surveys the current understanding of cognitive interoperability and the literature on cognitive architectures as computational models enabling reasoning, interpretation and adaptation in artificial agents. *(Supports all RQs by establishing foundations; in particular, it prepares RQ1 and RQ2.)*

Chapter 2 – *Synthesis and Conceptualisation of Cognitive Systems, Cognitive Interoperability, and Its Maturity Model*

It introduces the conceptual framework of the thesis [Al Haj Ali et al. \(2024a, 2025a,b\)](#). It formalises the notions of cognitive layer, cognitive system, CCPS and CDT, and identifies the cognitive functions that are relevant for HCC. Based on these elements, the chapter proposes a formal definition of cognitive interoperability and introduces a maturity model that characterises the progression from semantic interoperability to cognitive interoperability in a CPE. *(Addresses RQ1 and RQ2.)*

Chapter 3 – *Architecture of the Cognitive Digital Twin*

It focuses on the formalisation of a CDT as a computational entity and on the mechanisms that make cognition operational [Al Haj Ali et al. \(2025a, 2024b\)](#). The chapter analyses candidate

cognitive architectures (symbolic, sub-symbolic and hybrid), compares their properties with respect to the requirements derived from cognitive interoperability, and justifies the selection of an architecture suitable for a CDT in collaborative settings. (*Addresses RQ4.*)

Chapter 4 – *CDTO Ontology for Cognitive Digital Twins*

It presents the semantic model proposed in this thesis. Rather than aiming at a generic enterprise ontology, the chapter designs an ontology specifically targeted at the knowledge required to support cognitive interoperability in the collaborative activity (i.e., task context, roles, goals, constraints, interaction state, and collaboration-relevant concepts). This ontology establishes a shared conceptual space enabling consistent interpretation and exchange between the CDT and the HDT, and it provides the structured knowledge base required for reasoning and explanation in the collaborative scenario studied in the thesis. (*Addresses RQ3, and supports RQ4 by providing the knowledge layer used by the architecture.*)

Chapter 5 – *Use Case Implementation and Evaluation of Cognitive Interoperability*

It validates the proposed framework through an industrial use case in HRC, where the cobot is considered a CPS. The chapter instantiates the ontology on the collaborative assembly scenario, deploys inference mechanisms, operationalises the selected cognitive architecture within the CDT, and evaluates the ability of the overall CDT/HDT/CPS pipeline to support cognitive interoperability in practice. (*Primarily addresses RQ3 and RQ4, and provides evidence relevant to RQ1 by illustrating the maturity model levels.*)

Finally, **Chapter 6** concludes the manuscript by summarising the main scientific contributions, discussing the limitations of the current approach and outlining several directions for future research.

Chapter 1

State of the Art

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Introduction

This chapter provides the necessary scientific context to address the central objective of this thesis: defining, characterising and operationalising cognitive interoperability within Human-CPS collaboration. As mentioned in the introduction, semantic interoperability alone is insufficient in collaborative industrial environments, where humans and CPS must maintain mutual understanding, shared situational awareness and dynamic cognitive alignment. In order to achieve this, it is necessary to examine cognition itself in more detail, as well as how it is modelled in artificial systems and the technologies that can support cognitive functions within CDTs.

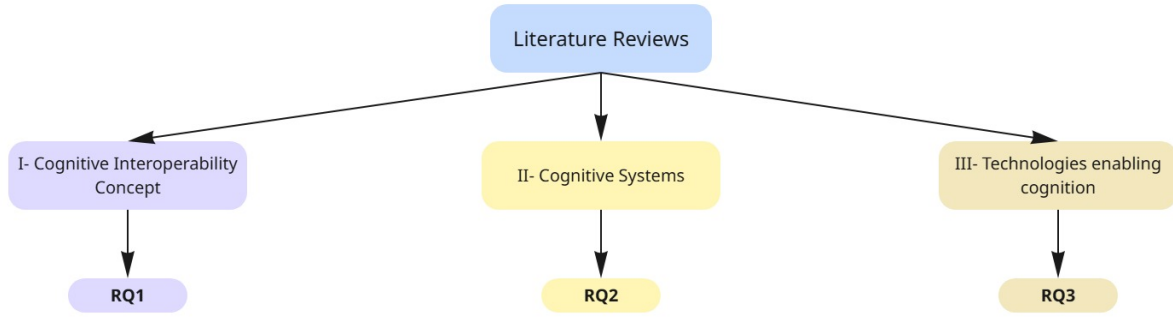


Figure 1.1: Literature review chapter overview

This chapter is structured around three complementary literature reviews, as illustrated in Figure 1.1. Each review corresponds to one of the research questions introduced in the introduction and contributes to establishing the conceptual foundations of this thesis. The reviews build in part on previously published work: Review (I) draws on (Al Haj Ali et al., 2025b) to examine definitions and related notions of cognitive interoperability and to contribute to **RQ1** by investigating how cognitive interoperability within the CPE can be formally defined and characterised. Review (II) builds on the systematic study in (Al Haj Ali et al., 2024a) and provides an [Systematic Literature Review \(SLR\)](#) of cognitive systems, drawing on cognitive science, psychology, AI, and literature on CCPs and CDTs; it contributes to **RQ2** by clarifying how a cognitive layer can be defined for CPS and by synthesising how cognitive systems, CCPs, and CDTs have been theoretically developed. Review (III) draws on (Al Haj Ali et al., 2025a) to survey cognition-enabling technologies such as ontological knowledge representation, ML approaches, hybrid neuro-symbolic methods, and cognitive architectures. It contributes to **RQ3** by identifying the representation principles and knowledge structures required to support cognitive interoperability in CDTs, and to **RQ4** by outlining the cognitive-architecture requirements and computational mechanisms (e.g., reasoning, adaptation, and explainability) needed to operationalise these representations in a CDT.

1.1 Review on the Cognitive Interoperability Concept

Collaboration between humans and machines goes beyond the simple exchange of data; it requires coordination of decisions and actions so that both agents respond appropriately to the operational context. The [European Interoperability Framework \(EIF\)](#) proposes a layered interoperability model (Figure 1.2) in which four interoperability layers are supported by an overarching *interoperability governance* context and coordinated through a cross-cutting integrated

public service governance function (European Commission, 2017).

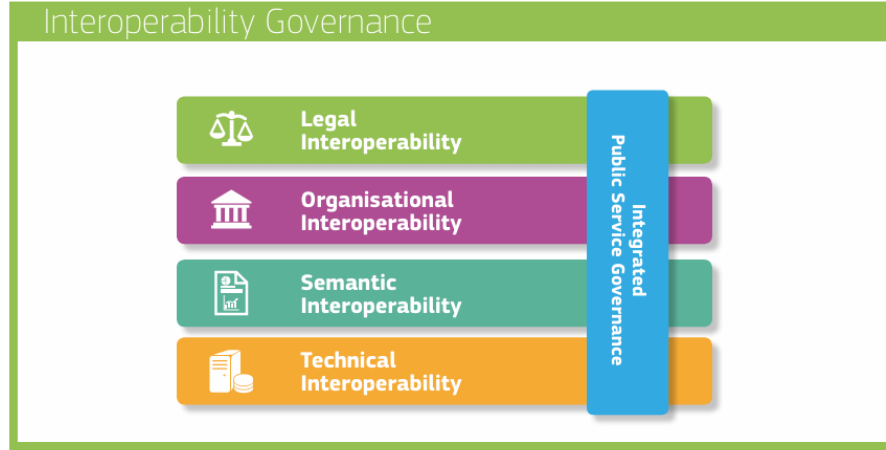


Figure 1.2: Interoperability layers in the EIF (European Commission, 2017)

Within this model, *legal interoperability* ensures that organisations operating under different legal frameworks, policies and strategies can work together and that legislation does not hinder cross-organisational service provision. *Organisational interoperability* concerns the alignment of business processes, responsibilities and expectations in order to achieve commonly agreed, mutually beneficial goals. *Semantic interoperability* ensures that the precise format and meaning of exchanged data and information are preserved and understood throughout exchanges then covering both semantic aspects (meaning and relationships of data elements) and syntactic aspects (the exact structure and format of information). Finally, *technical interoperability* covers the applications and infrastructures linking systems and services, including interface specifications, interconnection and data integration services, data exchange and presentation mechanisms, and secure communication protocols. Across these layers, *integrated public service governance* provides the coordination structures, roles and responsibilities, interoperability agreements (including service-level arrangements), change management, and continuity planning needed to sustain interoperability over time (European Commission, 2017).

This model of interoperability has been the basis for many advances in enterprise integration and CPS interoperability to date. However, when humans collaborate with machines, the limitations of semantic interoperability become apparent (Al Haj Ali et al., 2024a). Although ontology-based semantic solutions enable CPSs to share data and knowledge using a common vocabulary, ontologies that are readable by machines are not easily readable by humans. Humans may interpret formally defined terms differently, fail to understand logical formalisms or apply reasoning patterns that vary from person to person. In other words, even if two systems agree on the semantics of the data, a human operator might interpret the information differently to an artificial system. Therefore, simply having semantic interoperability “*is not always sufficient to ensure that CPSs and human agents understand each other well enough to cooperate or collaborate effectively*” (Naudet et al., 2023a). This finding highlights a fundamental gap: semantic alignment of information must be complemented by additional forms of alignment, particularly when bridging human and artificial systems..

To fill this gap, our earlier work has shown that we need to do more than just make sure that different systems can share information. We also need to introduce a new way of doing this, which we call “*cognitive interoperability*” (Naudet et al., 2023a; Ali et al., 2024). This concept is presented through a review of the state of the art that mentions and uses cognitive interoperability. This review is presented in Section 1.1.1, which collects and analyses the different definitions proposed in the literature. Section 1.1.2 describes the application areas in which cognitive interoperability has been addressed.

1.1.1 Review of Existing Definitions

In our previous work (Naudet et al., 2023a), cognitive interoperability was seen as *a mean to align the minds of entities interacting together and a prerequisite for efficient cooperation*. Where the mind alignment implies a *shared, mutual understanding and perception of situations, which is also a prerequisite for collaboration*. For CPSs and humans, this does not mean that CPS (in general) are cognitive or socio-cognitive agents. Instead, it leads to two more careful points:

1. We mainly refer to CPS that are designed with cognitive/social capabilities (e.g., CCPS or CPSS), rather than CPS in the broad, purely technical sense.
2. Cognitive coupling is a goal of the interaction: humans and such systems should be able to build and maintain enough mutual alignment (about the situation and what to do next) to cooperate. This can be supported by mechanisms inspired by *social cognition*.

In this section, we extend our previous review with a more systematic search that includes both the state of the art covered in (Naudet et al., 2023a) and additional, more recent publications. The aim is to provide a broader and more diverse picture of how cognitive interoperability is described in the literature, and to move towards a shared basis for a final definition. At the same time, the number of relevant publications remains limited, which reflects the novelty of the topic.

However the number of articles on the subject remains quite small, showing the novelty of the research field. The definitions found in the literature are listed together and divided into two categories: **explicit** definitions, which are clearly stated in the articles and highlighted in *italic*, and **implicit** definitions, which are formulated from the underlying concepts and the way in which each article considers cognitive interoperability. These definitions are presented in Table 1.1, where the first column indicates the reference and the second column provides the corresponding definition. Based on the analysis of this list, we propose our definition of cognition in industrial systems, which is presented in Section 2.3.

Reference	Definition
(Goldkuhl, 2008)	<i>Cognitive Interoperability is a part of organisational interoperability related to the congruence in thought and perceptions or the human actors’ way of thinking: aligning the way they conceptualise and understand information.</i>

Reference	Definition
(Berthier, 2006)	The term semiotico-cognitive interoperability is used especially for man-machine communication, where an artificial agent <i>appears to behave in the same way as a human agent would in the same situation and, in particular, that (to a predefined extent) some meanings seem to be shared between the user and the agent.</i> Cognitive interoperability is limited here to virtual agent-to-agent communication using standardised languages and normalised means to translate between different knowledge representations, including ontologies.
(Vogt, 2023)	<i>Cognitive interoperability is a critical characteristic of data structures and information technology systems that plays an essential role in facilitating efficient communication of data and metadata with human users. By providing intuitive tools and functions, these systems enable users to gain an overview of data, locate data they are interested in, and explore related data points in semantically meaningful and intuitive ways. The concept of cognitive interoperability encompasses not only how humans prefer to interact with technology (human-computer interaction) but also how they interact with information (human information interaction), considering their general cognitive conditions.</i>
(Blad and Potts, 2003)	A unity of mindsets, confidence/trust and mutual understanding based on shared education and values.
(Kwon et al., 2011)	Socio-Cognitive Interoperability is the ability of systems and users to share and understand information coherently by incorporating socio-cognitive aspects, enabling effective communication, shared understanding, and joint decision-making, particularly in complex situations such as emergencies.
(Raubal, 2005)	Cognitive semantic interoperability consists of aligning the data representation of systems with human conceptual structures. This highlights that when sharing knowledge, because the meaning of terms are in people's heads, the mental models of both the sender and the receiver have to be mapped for a complete understanding.
(Bytniewski et al., 2015)	Cognitive interoperability refers to the ability of different systems or agents to work together, exchanging and understanding knowledge in a way that promotes collaborative decision-making and adaptive behavior, drawing on cognitive processes. This means that cognitive agents, despite having different architectures or working on different subsystems, can interpret data in the same way, share knowledge and work together to carry out complex tasks in an integrated management system.
(Myneni et al., 2016)	Cognitive interoperability refers to the ability of a system to align itself with the cognitive processes of users, making it easier for them to interact with the system in accordance with their habitual thought patterns.

Reference	Definition
(Krinkin et al., 2023)	The ability of human and artificial intelligence to work together by synchronizing their knowledge and sharing a common understanding, even when they have different ontological models. It enables human and artificial agents to interact harmoniously by developing shared knowledge models (common ontologies), which facilitate their mutual understanding and collaboration.

Table 1.1: Definitions of Cognitive Interoperability from the Literature

Examining explicit and implicit definitions reveals that cognitive interoperability is a multifaceted concept that extends beyond traditional understandings of semantic interoperability, which, according to the EIF ([European Commission, 2017](#)), ensures that the meaning of exchanged data is preserved and consistently understood by all parties involved. Several key dimensions of this concept emerge from the literature:

- There is a recurring theme of aligning mental models, whether between humans (mostly in organisational or military contexts) or between humans and artificial agents. This alignment is often characterised by shared understanding, mutual perception and trust, which are fundamental to collaborative action.
- Many definitions highlight the integration of cognitive processes, such as intentionality, reasoning and adaptation, into systems, enabling artificial agents to act in ways that are understandable and meaningful to human users.
- The importance of contextualisation and user-centred interaction is highlighted, particularly in data mining and decision-making systems, demonstrating the practical application of cognitive interoperability in facilitating intuitive and effective collaboration between humans and systems.

1.1.2 Application Domains

The different applications of cognitive interoperability reveal its fundamental role across a wide range of domains, as illustrated in Figure 1.3 ([Al Haj Ali et al., 2025b](#)).

Cognitive interoperability has applications in a wide range of fields, each characterised by the need for common understanding between heterogeneous agents, whether human, artificial or system-based. In the military field, it promotes unity of mind and mutual understanding through joint training and shared values in order to enable coordinated joint operations ([Blad and Potts, 2003](#)).

In crisis management, it promotes real-time communication and joint decision-making by integrating socio-cognitive factors that ensure consistent understanding in a context of uncertainty ([Kwon et al., 2011](#)).

In e-government, cognitive interoperability facilitates convergence of thought among public institutions by harmonising the way information is perceived and conceptualised, enabling integrated services and more fluid interactions with users ([Goldkuhl, 2008](#)).

In navigation and geographic information systems, it adapts data representations to the mental models of users, thereby improving intuitive interaction and personalising spatial guidance ([Raubal, 2005](#)).

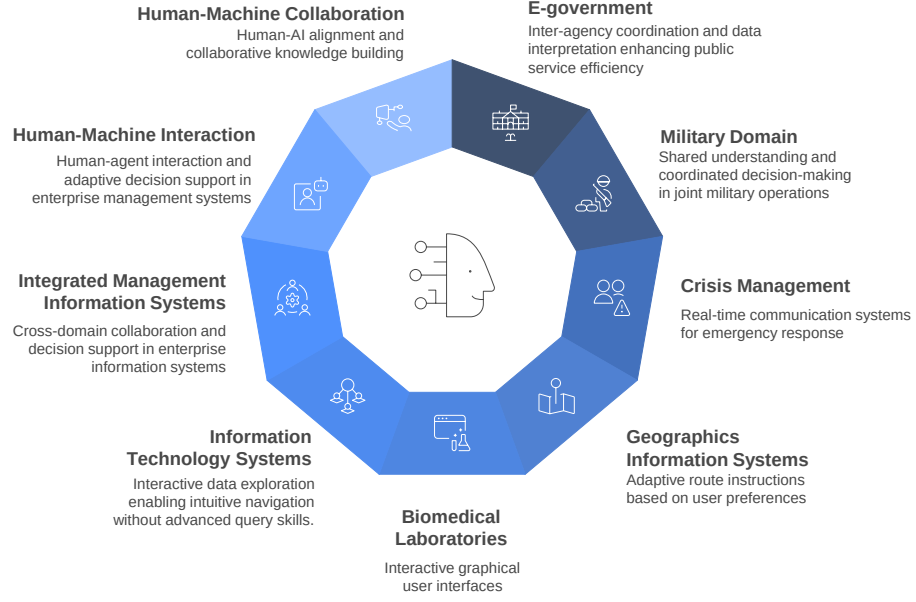


Figure 1.3: Application Domains of Cognitive Interoperability

In enterprise systems and human-machine interaction, it promotes semiotic-cognitive interoperability, where artificial agents simulate human-like reasoning and align with user intentions to improve decision-making and system acceptance (Berthier, 2006).

In the field of human-machine collaboration, it is used to continuously adapt and align human and artificial agents towards common goals; for example, AI systems can analyse data to detect complex patterns that humans then interpret and integrate into a common knowledge base, thereby promoting knowledge sharing and mutual understanding (Krinkin et al., 2023).

In integrated management systems, it enables various subsystems to share knowledge and collaborate through adaptive behaviour and collaborative interpretation of structured and unstructured data (Bytniewski et al., 2015).

In biomedical laboratories, it improves usability by aligning system interfaces with the usual thought patterns of users, allowing intuitive data exploration for non-experts (Myneni et al., 2016).

Finally, in information systems, it is applied in the creation of knowledge graphs, where users can explore data by following interaction strategies (e.g., zooming in on specific data points, filtering out irrelevant information) without the need for advanced query skills (Vogt, 2023).

Despite the diversity of these fields, one common theme emerges: **all involve situations where agents with different cognitive models must achieve mutual understanding in order to act coherently together.**

- heterogeneous agents,
- dynamic decision-making in uncertain contexts,
- variable user expertise,
- and the need for collaborative reasoning.

These characteristics highlight the limitations of traditional semantic interoperability and underscore the need for a more cognition-based approach.

Building on this perspective, the next section provides a detailed review of cognitive systems. It examines how cognition has been modelled in biological and artificial agents, and it provides the foundations required to conceptualise a cognitive layer for CPSs and CDTs.

1.2 Review on the Cognitive Systems

Having considered the concept of cognitive interoperability, this section turns to cognition and cognitive systems. Understanding these fundamentals is essential for defining a CCPS and a CDT. The objective is to identify how cognition is theorised, modelled and integrated into these industrial systems within a CPE. To ensure the relevance of the corpus, only studies that explicitly contribute to our understanding of the cognitive mechanisms in these systems were selected.

The review protocol implemented in this study follows Kitchenham guidelines ([Kitchenham et al., 2007](#)) to ensure a SLR, which structures the article around specific questions, aligned with the study objectives, to focus the analysis on the main topics of interest. Our study is based on our work [Al Haj Ali et al. \(2024a\)](#) and examines how cognitive functions are modelled and implemented in industrial environments, with a focus on CCPS and CDT systems. The aim is to better understand how cognitive aspects are understood, defined and implemented in such systems and how they are used to improve the overall industrial environment's performance and interoperability, particularly in the context of Industry 4.0/5.0.

This leads us to the following set of Sub-Research Questions (SRQs) for the SLR. These questions are used to structure the review and synthesise findings; they do not correspond to the main research questions of the thesis:

- SRQ1:** How is cognition characterised and defined in the literature of cognitive psychology and human theories, and to which extent cognitive CPS and DT refer to it in their own perspective?
- SRQ2:** Which cognitive functions are integrated into CPS and DT?
- SRQ3:** What particular methods, strategies or models are used to implement cognition in CPS and DT?
- SRQ4:** What are the theoretical foundations of cognition used in CPS and DT, and how do they help to implement the cognitive processes?
- SRQ5:** How are cognitive CPS and DT conceptualised, implemented, and applied within the context of Industry 4.0/5.0?
- SRQ6:** How can interoperability within industrial environments be enhanced by cognition, and is there a concept of cognitive interoperability?
- SRQ7:** What types of KRR approaches are commonly used in CPS and DT to structure and manage relevant knowledge for specific cognitive tasks, and facilitate cognitive interoperability?

This chapter provides an overview of the existing literature and the contextual basis needed for the thesis. The insights produced by the SLR are progressively consolidated and reused in the following chapters, where the thesis research questions are addressed in depth.

Regarding **RQ6**, this sub-question has already been addressed in detail in Section 1.1 and will not be repeated here in full. It was introduced earlier because cognitive interoperability is a central concept for the thesis and required an in-depth analysis from the outset. Moreover, since cognitive interoperability remains a relatively new topic in industrial systems, it was necessary to examine it separately before connecting it to other dimensions reviewed in this chapter.

Finally, **RQ4** and **RQ7** will be introduced and positioned in the next section (Section 3), where they are presented with the rest of the SLR protocol and the review scope.

1.2.1 SLR Methodology

This subsection provides an overview of our approach to selecting relevant articles, including the steps involved in choosing databases, defining keywords and search strings, and filtering articles.

The search process involves the following activities: selecting digital libraries; formulating the search string; conducting a pilot search; refining the search string; and retrieving an initial list of primary studies from digital libraries that match the search criteria.

1.2.1.1 Database selection

Before conducting a search, it is important to select the most appropriate databases to maximise the probability of finding highly relevant articles. To achieve the most comprehensive coverage possible, the most popular literature databases in the field should be explored. Taking a comprehensive and expansive approach is imperative to cover a diverse range of literature. For our study, we used the following digital databases:

- Web of Science (WOS),
- ACM Digital Library,
- IEEE Xplore,
- Scopus,
- PubMed.

We chose to search five databases, knowing that using at least four is typically considered sufficient for a thorough literature search (Kitchenham et al., 2010). Notably, Scopus stands out among these five because it is a comprehensive database that collects abstracts and citations from a variety of peer-reviewed journals. Articles sourced from Scopus and WOS are published by various well-known publishers, including Elsevier, Springer, Taylor & Francis Online and IEEE. This enhances the depth of our research. As is often the case, we anticipate finding duplicate articles in our results, which will be filtered out. It is important to note that each database requires a slightly different approach to query formulation.

1.2.1.2 Keywords and search strings definition

In order to find the articles allowing to answer our research questions, specific keywords were defined to compose a search string. This string was then split into search units and combined using Boolean operators. Acronyms, synonyms, alternative spellings, and plural forms were also included. This way, keywords related to *interoperability*, *digital twin (s)*, and *cyber-physical system (s)* were interconnected using the boolean operator **AND** with *cognition*, leading to the following query:

“Cognition” AND (“Interoperability” OR “Digital Twin” OR “Cyber-Physical Systems”)

To ensure a complete and systematic review of the literature, the search was first conducted using metadata from articles: titles, abstracts, authors and keywords. These elements provide a quick and reliable means of identifying studies focusing on cognitive aspects in different systems, since metadata is chosen and verified by authors and editors to represent the main content of each article.

However, to avoid missing relevant studies that may not be well described by their metadata, this method was complemented by manually checking the reference lists in the initially selected articles. This additional step enabled other important contributions, which were not captured by the metadata-based search, to be found.

Table 1.2 shows the number of articles retrieved from each database for each search string. These articles were then filtered according to the criteria described in the next section.

Publisher Databases	Number of Articles
Scopus	503
Web Of Science	314
IEEE Xplore	175
ACM	64
Pubmed	23
Total	1079

Table 1.2: Number of papers retrieved from each publisher database.

1.2.1.3 Papers Filtering

During the filtering process, selection criteria are employed to identify studies that address the research questions directly, in line with the recommendations of (Kitchenham et al., 2007). These criteria are defined according to the research objectives and documented from the protocol design stage to minimise the risk of bias.

After collecting all potentially relevant articles, filtering is carried out in two main steps. First, the title, abstract and keywords of each article are examined to assess their relevance. Based on this initial screening, articles that meet any exclusion criteria are removed. The second step involves reading the full text of the remaining articles to evaluate them in more detail. At this stage, articles are evaluated according to the defined inclusion criteria to confirm their relevance to the review.

The inclusion and exclusion criteria used in this selection process are presented in Table 1.3. The final set of selected articles, along with their descriptive analysis, is presented in the next section.

1.2.2 Descriptive Analysis of Papers

The initial search revealed 1079 references from the digital libraries, and 17 papers based on the snowball sampling. We then applied filtering on the total of 1079 papers guided by a series of inclusion and exclusion criteria as described in table 1.3. First we excluded those papers that are not accessible, not written in English and duplicates. Consequently the number of considered

E/I	Criteria
Exclusion	Duplicate papers
	Papers not accessible: a paper without full text to be assessed
	Papers written in other languages than English
	Entire conference proceedings
	Irrelevant primary studies
	Redundant paper from a same author: it refers to situations where an author publishes multiple papers on the same topic or with highly overlapping content
	Loosely related: articles that are not directly connected to the concept of cognition
Inclusion	Articles offering definitions
	Studies providing answers to the research questions
	Relevance papers: direct association with Cognition or featuring Cognition as a key element within systems

Table 1.3: Exclusion and Inclusion Criteria.

papers dropped to 607. Moving forward, we analysed the rest of the papers, considering their title, abstract and keywords. Thus, the number of considered papers dropped to 152. Moreover, after reading and analysing the full text of the remaining papers according to inclusion and exclusion criteria, we selected 90 of them. Once the papers are selected, we classify them by the year of publication, country and application areas. Table 1.4 shows the details of the selection phase.

Phase	Total N° of Papers
Total number of paper from digital libraries	1079
N° of papers after snowballing sampling	1096
N° of papers after exclusion based on the paper access, language and type of research	607
N° of papers after exclusion based on title, abstract and keywords	152
N° of papers after exclusion based on full text = N° of included papers	90

Table 1.4: Paper selection phase.

The study covers papers published up to July 2023. Figure 1.4 shows the temporal distribution of these publications. There was a period of relatively low activity from 1995 to 2007, after which there was a gradual increase in publications. The number of articles fluctuated between 2011 and 2018, with no clear constant growth trend. A significant peak in publications is observed in 2022, indicating renewed and robust interest in Cognitive Industrial Systems research that year and suggesting an increase in the importance of the field.

Our study targets industrial environments. However the analysed articles address cognitive

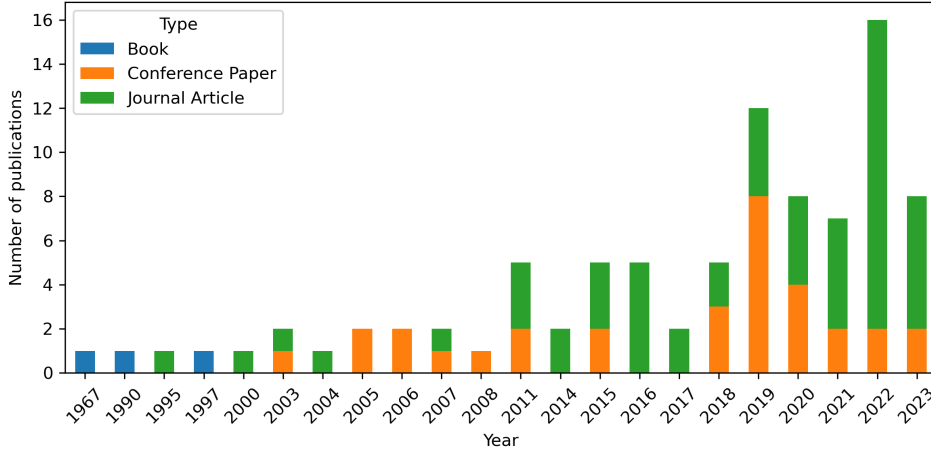


Figure 1.4: Number of papers published per year.

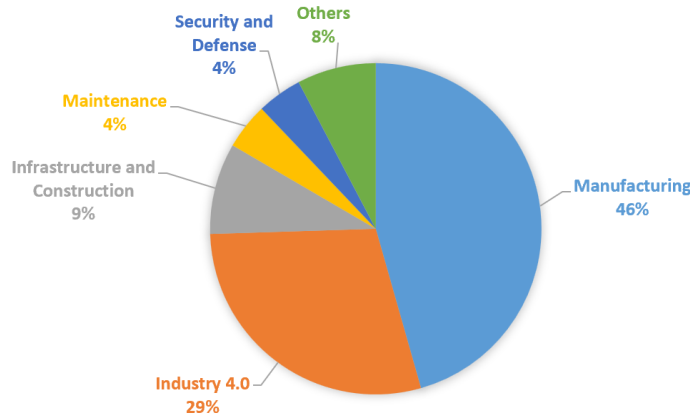


Figure 1.5: Application areas of cognitive systems.

systems for a diverse range of problems in multiple application areas. Figure 1.5 shows the distribution among these. Without surprise, a big majority of articles concern Manufacturing and Industry 4.0. However it is interesting to see scores of some other areas. Infrastructure, referred to in our study as *smart cities*, shows a significant application of cognitive systems in urban management and optimisation. *Others* refers to the process industry, the chemical industry, the automotive industry and aerospace, and the cognitive enterprise thus bringing together a diversity of sectors that also have an impact in cognitive systems. For example Aerospace, although being at the origins of the Digital twin concept, seem to have a few interest for cognition. A few articles also refer to the Cognitive Enterprise but the concept seems only emerging.

In the following sections, we detail the answers to our search questions, starting with the foundations of cognition exploited in articles, thus covering our first four questions.

1.2.3 Theoretical Foundations for Systems Cognition

This section introduces the theoretical foundations of systems cognition, providing a structured overview of how cognition is understood and applied in industrial systems. First, it clarifies the

concept of cognition as defined in cognitive psychology and human cognitive theories. It then examines how cognition is interpreted and implemented in industrial contexts, with a particular focus on CCPs and CDTs. The section also identifies the primary cognitive functions present in these systems and reviews the technologies typically employed to support these functions. Together, these elements provide a solid foundation for analysing how cognition can be modelled and operationalised in industrial systems.

1.2.3.1 Cognition

In this section, we address SRQ1: *How is cognition characterised and defined in the literature of cognitive psychology and human theories, and to which extent cognitive CPS and DT refer to it in their own perspective?*

Cognition is a concept deeply rooted in cognitive psychology. It has been the subject of numerous studies and interpretations over the years and several authors have drawn on human cognitive theory to explore this notion. (Neisser, 1967), often considered the father of cognitive psychology, was one of the first to lay the theoretical foundations of cognition. According to him (Neisser, 1967) the cognition refers to the mental processes by which individuals perceive, think, understand and interact with their environment. Central to these processes is the reception and interpretation of information (US, 1996), which, when effectively integrated through experiences, senses, and learning (Flower and Hayes, 1981), transforms into knowledge. This knowledge is not simply a set of isolated pieces of information; it is an integrated form of understanding that allows an agent to interpret situations, evaluate alternatives, and make appropriate decisions (US, 1996). This process involves not only the acquisition but also the storage and utilisation of knowledge (Niu et al., 2009). Then, human theories of cognition were proposed. (Newell, 1990) brought the notion of cognition as a symbolic information processing system, a complex network where symbols are manipulated to produce thoughts and decisions. (Minsky, 1988) proposed a modular view with his *Society of Mind*, where mental agents interact to create an overall cognitive experience.

Building on this foundational understanding of cognition, a wide range of paradigms has emerged (Pacaux-Lemoine et al., 2022), each offering unique insights into the nuances and complexities of cognitive processes. *Cognitive mimic* analyses human information processing in order to be mimicked by intelligent technologies (Karvonen and Saariluoma, 2023). *Cognitive science* is an interdisciplinary approach to understanding the mind that combines several different levels of analysis (Hozdić and Makovec, 2023; Ye et al., 2022). Derived from this field, cognitive heuristics have been identified. It can be considered a mental shortcut or a model inspired by the functioning of the human brain. Cognitive heuristics are designed to facilitate the distribution and interpretation of semantic data and associated data elements. Their design is inspired by the way humans store and process semantic information in their memory (Mordacchini et al., 2016).

Cognition refers to the set of mental processes that enable humans and other agents to interact effectively with their environment. It emerges from continuous and intentional interactions between an individual and the surrounding world, allowing tasks and goals to be achieved across different contexts (Anttila et al., 2022). At a fundamental level, cognition encompasses the ability to acquire, process, store, and interpret information. This includes both basic forms of learning and more advanced forms of understanding that allow concepts to be analysed and used

meaningfully (Angulo et al., 2023).

The human brain plays a central role in these processes due to its capacity to handle large amounts of information and adapt its behaviour accordingly. Through perception, memory, and reasoning, it enables individuals to interpret stimuli, learn from experience, and respond appropriately to changing situations (Zheng et al., 2017). Cognitive processes are strongly influenced by prior experience and acquired knowledge, including both *procedural knowledge* (how to perform actions) and *declarative knowledge* (facts and concepts) (Anderson, 1996; Juarez-Espinosa, 2003). Similar mechanisms can also be observed in other animals, which perceive their environment and act upon it based on learned patterns and sensory inputs (Castro and Andrade, 2018).

Cognition is therefore not a static capability but a continuous process of exploration and adaptation. It integrates sensory information with internal knowledge through processes such as perception, attention, learning, memory, and language (Jones et al., 2006). Through these processes, agents are able to build representations of their environment, reason about situations, and make decisions that guide their actions.

Within the context of the Cognitive Factory (Colangelo et al., 2019), cognition is commonly defined as the ability of a system to interpret data, learn from experience, and support decision-making (Niu et al., 2009; Sternberg and Sternberg, 2012; Baum, 2004). This interpretation aligns with industrial needs, where systems must adapt to dynamic environments, manage uncertainty, and interact with human operators.

In industrial research, including the articles analysed in this thesis, cognition is addressed from multiple perspectives, reflecting its multidimensional nature. In CCPS, cognition is mainly associated with perception of the environment, continuous interaction, and decision-making based on real-time data (Al-Turjman, 2017; Abie, 2019). This view emphasises learning, adaptation, and responsiveness. In contrast, CDT often focus more on cognition as a process of knowledge acquisition, organisation, and exploitation to support prediction and reasoning (Johansen et al., 2023). Despite these differences, both CCPS and CDT rely on common principles of information processing and reasoning, as highlighted in (Zheng et al., 2017; Kalaboukas et al., 2021).

These perspectives are rooted in foundational work in cognitive psychology, notably the definition proposed by (Neisser, 1967), which remains influential for understanding cognition in both humans and artificial systems.

Table 1.5 summarises the different definitions of cognition identified in this review. Overall, cognition is consistently described as a capability involving mental processes that support interaction with the environment. The definitions are grouped according to the primary objective assigned to cognition by different authors. The category *Awareness* emphasises perception and understanding of the environment to support decision-making. *Knowledge acquisition* and understanding focuses on how information is learned and structured. Finally, *Information processing* refers to the transformation and use of information or knowledge to guide reasoning and action.

In conclusion, cognition cannot be reduced to a purely theoretical notion or to a set of isolated functions. It is a broad range of processes that enable perception, understanding, learning, reasoning, and decision making. In modern industrial contexts, particularly with the emergence of Industry 4.0 (Faruque et al., 2021; Mortlock et al., 2022), cognition within industrial systems such as CCPSs and CDTs has become a key capability for handling and analysing large volumes of data (Fischbach et al., 2020). These capabilities support applications such as real-time monitoring, process optimisation, and anomaly detection.

Theme	Definition	CDT	CCPS
Awareness	Cognition is defined as the integrative capability to perceive and interpret the environment and its context, to learn from past actions, and to use this knowledge to reason, anticipate, and guide decisions toward specific goals. It involves real-time situation awareness and intentional understanding that support problem-solving and task execution.	(Johansen et al., 2023; Kalaboukas et al., 2021)	(Al-Turjman, 2017; Abie, 2019; Anttila et al., 2022)
Knowledge acquisition and understanding	Cognition refers to the mental processes through which knowledge is acquired, structured, and understood by combining experience, perception, and reasoning.	(Johansen et al., 2023)	(Angulo et al., 2023)
Information processing	Cognition is described as the set of processes that enable an agent to transform, organise, store, retrieve, and use information obtained from the environment. These processes include perception, data interpretation, memory management, and reasoning based on sensory and symbolic information.	(Neisser, 1967; Faruque et al., 2021; Mortlock et al., 2022; Kalaboukas et al., 2021; Yitmen et al., 2021)	(Zheng et al., 2017; Fischbach et al., 2020)

Table 1.5: Comparative Definitions

However, despite the increasing use of the term cognition in industrial research, it is often employed as a general concept without explicit reference to established definitions from cognitive psychology or neuroscience. Many contributions directly address CCPS or CDT architectures without clearly grounding their approaches in fundamental cognitive principles. For this reason, it is essential to clarify the underlying cognitive mechanisms before analysing higher-level systems such as CCPS and CDTs, which will be examined in greater detail in the following sections. The next section therefore focuses on the cognitive functions that support the integration of human cognition into these systems.

1.2.3.2 Cognitive functions

This section concerns SRQ2: *Which cognitive functions are integrated into CPS and DT?*

Understanding the cognitive functions embedded in these systems is essential for characterising them as cognitive systems and for analysing their role in cognitive interoperability.

Cognitive functions refer to the set of mental processes involved in acquiring knowledge, manipulating information, reasoning, and guiding action (Kiely, 2014). These functions typically include perception, attention, memory, learning, reasoning, decision making, problem solving, and language-related abilities (Vernon et al., 2007). In cognitive science, such functions can be

organised according to the type of process they involve, for example memory-related processes, attentional mechanisms, or reasoning capabilities (Harvey, 2019). Core cognitive skills⁶ often cited in the literature include short-term memory, logical reasoning, processing speed, attention, and spatial abilities.

Classical models of human cognition are largely grounded in the information-processing paradigm, which draws an analogy between mental processes and computational operations (Kiely, 2014). Within this paradigm, cognition is viewed as a sequence of processes that transform sensory inputs into internal representations, manipulate these representations through reasoning and memory, and produce decisions or actions.

When CPS or DT integrate such cognitive functions, they are commonly referred to as cognitive systems, which constitute a distinct area of research. Humans can naturally be considered cognitive systems, whereas CPS and DT acquire this property only when equipped with mechanisms that support perception, reasoning, learning, and decision making. In particular, cognitive systems aim to support human-like reasoning and learning capabilities, and, in some cases, higher-level features such as motivation or adaptive behaviour. Introducing such characteristics is expected to facilitate human-machine interaction by making system behaviour more interpretable and predictable, thereby supporting trust, collaboration, and partnership between humans and machines.

Research on CCPSs and DTs represents an initial step toward transforming CPS and DT into cognitive systems. The literature shows that various cognitive functions have been implemented in these systems to enable learning, reasoning, and decision making. However, the set of cognitive functions considered is not uniform across studies. Figure 1.6 presents the cognitive functions identified in the analysed articles and their frequency of occurrence in CCPS and CDT research.

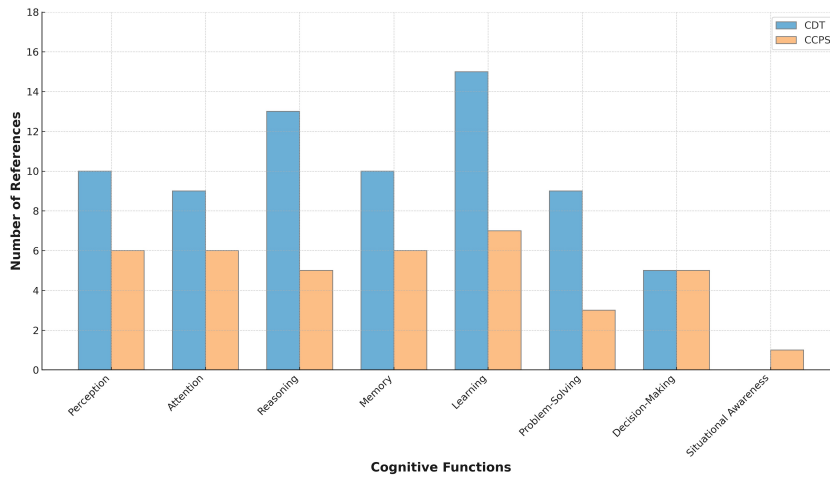


Figure 1.6: Comparison of cognitive functions in CDT and CCPS.

In the context of CCPS, several cognitive functions are commonly discussed in the literature. *Reasoning* supports information processing, logical inference, and decision support. *Attention* and *perception* are often described as mechanisms for selecting, grouping, and retrieving relevant data under limited processing capacity and large volumes of environmental inputs. *Situational*

⁶<https://www.verywellmind.com/what-is-cognition-2794982>

awareness is also emphasised, as it enables adaptation to environmental changes and contributes to safety and adaptability. In addition, *learning*, *decision-making*, *problem solving*, and *memory* are frequently integrated to support adaptive behaviour and long-term system improvement in CCPS (John et al., 2021; Angulo et al., 2023; Colangelo et al., 2019; Dumitrache et al., 2019; Jiang et al., 2022; Oliveira et al., 2019; Abie, 2019; Topal et al., 2020; Bocklisch et al., 2022; Mizanoor Rahman, 2019).

In the context of CDT, cognitive functions have been explored by many authors (Abburu et al., 2020b,a; Eirinakis et al., 2022; Faruque et al., 2021; Johansen et al., 2023; Mokhtari et al., 2022; Mortlock et al., 2022; D’Amico et al., 2022; Enrique and Soares, 2022; Intizar Ali et al., 2021; Jinzhi et al., 2022a; Kalaboukas et al., 2021; Yitmen et al., 2021; Zheng et al., 2022; Zhang et al., 2020). In particular, (Faruque et al., 2021) distinguish the main functions as follows:

- *Perception*: transforms data representations into useful information.
- *Attention*: selectively focuses on relevant tasks/data, filtering distracting elements.
- *Reasoning*: draws conclusions from a starting point (or perception), often relying on existing memory or assertions.
- *Learning*: converts experience into reusable knowledge, enabling adaptation and evolution.
- *Problem-solving* and *decision-making*: work together to find solutions and reach goals.
- *Memory*: stores and recalls information, from specific events to general knowledge about the environment.

The differences observed between CCPS and CDT with respect to cognitive functions can be explained by their distinct objectives and technological constraints. CCPS, which tightly couple digital and physical components, typically prioritise real-time interaction and situational awareness. In contrast, CDTs, as digital representations of physical systems, primarily focus on simulation, prediction, and reasoning over complex models. This distinction reveals a deeper epistemic difference: CCPS emphasise situated cognition and immediate action, whereas CDTs emphasise model-based understanding, anticipation, and explicit reasoning. This epistemic orientation makes CDTs particularly suitable as hosts for higher-level cognitive functions and interoperability mechanisms. These differences also reflect current research trends and application requirements. Enhancing cognitive functions in both types of systems is expected to improve the integration of domain knowledge and support more informed real-time decision making (Rožanec et al., 2022).

Finally, several studies emphasise that cognitive functions are essential for transforming data into actionable knowledge, particularly in uncertain environments (Abburu et al., 2020b). Cognition enables systems to incorporate new knowledge (Dhakal and Bobrin, 2023), learn improved models, enhance situational understanding, and plan actions accordingly. These capabilities raise classical challenges related to knowledge representation, acquisition, and updating, which are central topics in knowledge engineering.

1.2.3.3 Technologies enabling cognition in CCPS and CDT

This section addresses SRQ3, SRQ4, and SRQ7 by reviewing the main technologies used to implement cognitive functions in CCPS and CDT, and by clarifying how these technologies support knowledge management, reasoning, and cognitive interoperability.

AI-based technologies supporting cognitive functions (SRQ3): Across the analysed literature, cognition in CCPS and CDT is typically operationalised through a set of AI technologies that realise concrete cognitive functions (e.g., perception, attention, reasoning, learning, and interaction). In practice, these functions are implemented using: (i) *data-driven* approaches (ML and DL), which extract patterns from data and support prediction and classification; (ii) *language technologies* (NLP and, more recently, LLM-based components) to enable text understanding, knowledge extraction, and natural-language interaction; and (iii) *vision-based* approaches (computer vision) to interpret visual signals and support perception-like capabilities (Enrique and Soares, 2022; Elia and Margherita, 2022; Hozdić and Makovec, 2023).

In CCPS, these technologies are often selected according to operational needs: reasoning and analytics methods are used to interpret data in context and support decision-making; predictive models support anticipation of system states; and NLP/LLM-based components enable more flexible interaction and knowledge access from unstructured sources (Abie, 2019; Hozdić and Makovec, 2023). Computer vision methods (often DL-based) are used when perception relies on visual signals, and specialised sensing such as eye-tracking can support attention-aware interaction by capturing human visual behaviour (Bocklisch et al., 2022; Hozdić and Makovec, 2023).

In CDT, cognition is commonly implemented through a combination of data-driven and knowledge-oriented components: NLP methods (including sequential models such as LSTM in some works) support textual analysis and knowledge extraction, while DL-based vision models (e.g., VGG-style architectures) support visual recognition and interpretation (Lv et al., 2023). Additional approaches focus on memory and knowledge management (e.g., memory allocation mechanisms) and on learning strategies that improve behaviour over time, including graph-based learning in several CDT contributions (Asadi, 2022; Eirinakis et al., 2020; Mokhtari et al., 2022; Mortlock et al., 2022; Kalaboukas et al., 2021).

From tools to systems, cognitive computing and cognitive architectures (SRQ4): While the technologies above provide individual cognitive capabilities, building a *cognitive system* also requires integration: coordinating perception, reasoning, learning, memory, and interaction so that the system can select appropriate processes depending on the task and context. This motivation underpins the cognitive computing paradigm, often presented as a shift from deterministic software to systems that learn, reason, and adapt in changing environments (Zheng et al., 2017). In parallel, cognitive architectures aim to provide unified computational frameworks inspired by theories of cognition, offering reusable building blocks and control structures to combine multiple cognitive functions in a coherent way (Sathi, 2016).

In this chapter, we introduce these concepts here to position cognitive architectures as an integration mechanism rather than as a standalone technology. The reason is practical rather than conceptual: CDT implementations naturally rely on an explicit pipeline (data acquisition, synchronisation, modelling, simulation/emulation, and user interaction), which makes the need for a coordinating architecture more visible and easier to analyse. By contrast, CCPS contributions often embed cognitive capabilities in a more distributed and application-specific way (e.g., within control, analytics, or interaction modules), and the literature less frequently reports full, architecture-level designs in an explicit and comparable manner. Therefore, we introduce architectures here as an *integration mechanism*, but we detail their principles and state of the art later in Section 1.3.4 where the focus shifts from generic enabling technologies to CDT-specific

implementation choices.

Knowledge Representation and Reasoning for cognitive tasks and cognitive interoperability (SRQ7): A recurring requirement in both CCPS and CDT is the ability to structure, share, and exploit knowledge and not only to store information, but to support cognitive tasks such as interpretation, explanation, diagnosis, and decision support. This is where [knowledge representation and reasoning \(KRR\)](#) plays an important role: it provides formal languages and mechanisms to represent knowledge explicitly and to perform reasoning over it, which is particularly useful when systems must remain interpretable and when knowledge must be exchanged across heterogeneous components ([Johansen et al., 2023](#)).

In the analysed literature, two approaches of KRR types are dominant: ontologies and [knowledge graphs \(KGs\)](#). Ontologies provide shared vocabularies and formal conceptualisations that help maintain semantic consistency, enabling rule-based reasoning and making assumptions explicit. KGs represent entities and relations in a graph structure that supports contextual linking, querying, and inference, and can be extended over time as new knowledge becomes available. Table 1.6 summarises the studies applying ontologies and KGs in CCPS and CDT.

	CDT	CCPS
KG	(D’Amico et al., 2022 ; Eirinakis et al., 2020, 2022 ; Jinzhi et al., 2022a ; Johansen et al., 2023 ; Kalaboukas et al., 2021 ; Lv et al., 2023 ; Mokhtari et al., 2022 ; Intizar Ali et al., 2021 ; Asadi, 2022)	(Iarovyi et al., 2015 ; Liu et al., 2022a ; Dourlens and Ramdane-Cherif, 2011)
Ontologies	(D’Amico et al., 2022 ; Jinzhi et al., 2022a ; Kalaboukas et al., 2021 ; Li et al., 2022a ; Mokhtari et al., 2022 ; Intizar Ali et al., 2021 ; Eirinakis et al., 2020)	(Abie, 2019 ; Iarovyi et al., 2015 ; Liu et al., 2022a ; Castro and Andrade, 2018 ; Dourlens and Ramdane-Cherif, 2011)

Table 1.6: Number of references for each category and domain.

Furthermore, KRR contributes to cognitive interoperability by offering a common, machine-processable structure for meaning and context. Semantic web technologies (e.g., RDF-based representations) support consistent interpretation and exchange of knowledge between systems ([D’Amico et al., 2022](#)). Graph-based representations further help by making relations and context explicit; when combined with domain-specific information (including spatial or operational context), they can improve cross-system understanding and reasoning ([Krueger et al., 2016](#)). Beyond technical exchange, several works emphasise that socio-cognitive interoperability also requires aligning interaction and intent; in this view, KRR provides the structural layer on top of which higher-level alignment mechanisms can be built, by ensuring that what is exchanged is not only data, but explicit knowledge with stated relations and constraints ([Kwon et al., 2011](#); [Krinkin and Shichkina, 2023](#)).

Overall, the reviewed studies suggest a complementary pattern: data-driven AI supports perception and prediction, while KRR supports explicit knowledge, interpretability, and cross-

system alignment. The following section focuses on cognitive systems, CCPSs and CDTs.

1.2.4 Cognitive systems in CPE

Current industrial systems are increasingly integrating cognitive capabilities based on advances in cognitive systems. In the context of Industry 4.0, managing large volumes of data is a major challenge (Fischbach et al., 2020). However, the objective is not only to process data, but also to improve decision-making and enable intelligent behaviour and adaptation in manufacturing systems (Dumitrache et al., 2019). These developments contribute to the emergence of the cognitive enterprise, in which cognition is embedded directly within industrial systems, rather than being limited to offline data analysis (Elia and Margherita, 2022).

This evolution is reflected in the growing adoption of concepts such as CCPS and CDT within the context of Industry 4.0 and 5.0 research. These concepts indicate a clear shift towards systems that can perceive their environment, reason about situations and support decision-making.

In order to address the remaining research questions in this SLR, the following sections focus on two key aspects identified in the literature. The concepts and definitions of CCPS and CDTs are analysed to clarify how cognition is integrated into these systems.

1.2.4.1 Cognitive Cyber-Physical Systems (CCPS)

This section addresses SRQ5: *How are cognitive CPS conceptualised, implemented, and applied within the context of Industry 4.0/5.0?*

CPS are a key enabling technology of Industry 4.0, connecting the physical world with digital services. This connection relies on sensing, networking, computation and data storage to support interactive industrial environments and smart factories (Carvalho et al., 2018). CPS rely on a combination of sensing/communication infrastructures, data management, and intelligent decision functions, while facing recurring challenges related to heterogeneity, integration, and interoperability across components and enterprise layers (Dafflon et al., 2021).

The concept of CCPS builds on this by adding cognitive capabilities. Although the notion of CCPS is relatively recent, several complementary definitions exist. According to (Zhou et al., 2019), cognition in CCPSs aims to acquire and exploit the knowledge required to achieve system objectives, thereby directly supporting decision-making and control. Similarly, (Khargonekar, 2019) defines CCPS as CPS equipped with cognitive functions that can be designed explicitly or learned through interaction with humans and other CCPS.

Based on the analysed literature, CCPS are generally described by three closely related characteristics: autonomy, adaptability and awareness. These characteristics capture how cognition is used in CCPS to support decision-making, learning and interaction with the environment. *Adaptability* focuses on the ability of CCPS to modify and optimise their behaviour in response to changing environments and internal dynamics. *Autonomous* agents refer to the ability of CCPS to self-monitor and solve problems. Finally, *awareness* focuses on the system's perception of its environment and its own state. All CCPS systems have a detailed set of characteristics; however, in the literature, studies are generally organised according to the specific characteristic they examine, such as adaptability, autonomy, or awareness. Table 1.7 summarises the definitions

Characteristic	Definition	References
Autonomous agents	A CCPS is an integrated, decentralised system that combines cognitive capabilities with the structure of a CPS. It is designed to function as an autonomous agent, capable of behaving and making decisions independently, without human intervention.	(Fournaris et al., 2020; Mizanoor Rahman, 2019; Oliveira et al., 2019, 2021)
Adaptability	CCPS is a CPS with cognitive abilities, capable of adapting their actions according to sensory information and uncertainty management.	(Krueger et al., 2016)
Awareness	CCPS is a system that holds an advanced understanding of its environment and human interactions.	(Al-Turjman, 2017)

Table 1.7: Characterisation of CCPS in the Literature.

extracted from the selected publications and organises them according to the main characteristic highlighted in each study.

Across the reviewed studies, CCPS are described as systems that can operate autonomously, adapt to changes, and maintain awareness of their environment. For example, (Oliveira et al., 2019) presents CCPS able to detect and analyse environmental conditions and then select actions based on this analysis. This cognitive dimension is further explored in (Mizanoor Rahman, 2019), where cognition is used to support mutual trust in HRC scenarios.

From a system-of-systems perspective, (Fournaris et al., 2020) describes CCPS as holistic and decentralised systems enabling cyber-physical system-of-systems to operate without direct human intervention. In robotics-oriented work, cognitive capabilities support adaptation under uncertainty by combining sensory inputs with uncertainty management (Krueger et al., 2016). More recently, (Oliveira et al., 2021) defines CCPS as cooperative CPS equipped with AI-enabled cognitive architectures, allowing deeper interaction with the physical system.

In several contributions, cognition is explicitly linked to learning from past actions and using stored knowledge to improve future decisions and system performance (Al-Turjman, 2017). This is consistent with (Zhou et al., 2019), which highlights that cognition in CCPS is primarily a mechanism for acquiring and using knowledge to support decision-making and control.

Finally, some work addresses CCPS through specific enabling technologies, such as the integration of cognitive radio into CPS. In this case, cognitive mechanisms are used to analyse transmission constraints and spectrum availability in order to optimise communication performance (Topal et al., 2020).

This section has summarised how CCPS are conceptualised and characterised in the literature. The resulting interpretation and the definition adopted in this thesis are discussed in Section 2.1.2.

1.2.4.2 Cognitive Digital Twin (CDT)

This section deals with SRQ6: *How are CDTs conceptualised, implemented, and applied within the context of Industry 4.0/5.0?*

A DT is a digital representation of a physical product or system. It is typically implemented as a virtual engineering model that reflects the structure, behaviour, and state of the real-world object or system. High-fidelity DTs are developed using a combination of modelling and simulation technologies (Schleich et al., 2017). These include simulation, which predicts system behaviour under predefined conditions (Law and Kelton, 2000), and emulation, which reproduces system behaviour in real time (Ayani et al., 2018). While simulation provides an predictive, offline view of system behaviour, emulation enables dynamic interaction with live data, closely mirroring the operation of the physical system (Semeraro et al., 2021).

In many industrial applications, DTs are used primarily as accurate digital replicas of physical assets, without embedded cognitive capabilities (Intizar Ali et al., 2021). To overcome this limitation, the concept of the CDT was introduced. A CDT extends the traditional DT by integrating AI and cognitive functions, enabling capabilities such as reasoning, decision-making, learning, and autonomous behaviour. As observed in the literature, the definition of a CDT varies across different studies, reflecting various application objectives and technological approaches.

Based on the analysed publications, the CDT can be characterised by four closely related aspects combining digital twinning and intelligence. These aspects are summarised in Table 1.8. *Semantic modelling* is the ability of CDTs to create structured, interpretable representations of physical systems, which supports an understanding of system dynamics and relationships. *Autonomous agents* emphasise the capacity of CDTs to operate independently, interpreting information and making decisions without direct human intervention. *Real-time reasoning and decision-making* focus on the ability of CDTs to process incoming data, reason about current states and trigger actions with minimal delay. Finally, *adaptability* highlights the capability of CDTs to evolve over time by learning from new data and adjusting their behaviour in parallel with the physical system.

The literature highlights the importance of semantic modelling for understanding system evolution and supporting informed decision-making, particularly in complex and interconnected systems (Kalaboukas et al., 2021; Lu et al., 2020; Li et al., 2022a; D’Amico et al., 2022). The integration of autonomous agents demonstrates that CDTs are active entities capable of real-time reasoning and autonomous problem-solving, rather than passive models. This is essential in applications requiring operational independence (Asadi, 2022; Mokhtari et al., 2022; Yitmen et al., 2021). Real-time reasoning and decision-making are frequently identified as advanced features that enable CDTs to react immediately to changes based on contextual information and accumulated knowledge (Eirinakis et al., 2020; Zhang et al., 2020; Intizar Ali et al., 2021; Mokhtari et al., 2022). Finally, adaptability reflects the ability of CDTs to evolve continuously with their physical counterparts throughout the system lifecycle, ensuring long-term relevance and flexibility (Abburu et al., 2020a; Jinzhi et al., 2022a; Zheng et al., 2022).

Although a single CDT may integrate all these characteristics, studies are typically organised around one or more specific aspects depending on the application context. This highlights both the diversity of CDT use cases and the interdependence of these characteristics in achieving fully cognitive behaviour.

From a cognitive perspective, CDTs are designed to understand context, reason over existing information, predict system behaviour, and optimise decisions. To support these functions, CDTs integrate services for sensing, reasoning, learning, and optimisation. As a result, CDTs are increasingly recognised as key enablers for agile and intelligent supply chains, supporting

Characteristic	Definition	References
Semantic modelling	A DT with augmented semantic capabilities for identifying the dynamics of virtual model evolution, promoting the understanding of interrelationships between virtual models and enhancing the decision-making based on DT.	(Kalaboukas et al., 2021; Lu et al., 2020; Li et al., 2022a; D’Amico et al., 2022)
Autonomous agents	A DT that uses real-time reasoning and problem-solving skills to interpret information and make autonomous decisions. This system is adept at using its cognitive abilities to control processes, react to unexpected events, and handle unknown situations independently, without human intervention.	(Asadi, 2022; Mokhtari et al., 2022; Yitmen et al., 2021)
Real-Time Reasoning and Decision	A DT with the ability to reason, interpret and act in real time. It integrates enhanced communication, analytical and cognitive capabilities, enabling decision-making and process control based on accurate observations, in-depth knowledge and experience.	(Eirinakis et al., 2020; Zhang et al., 2020; Intizar Ali et al., 2021; Mokhtari et al., 2022)
Adaptability	A DT composed of interlinked digital models that reflect various lifecycle stages of the physical counterpart, including all subsystems and components. Notably, it has the distinctive ability to continuously evolve in parallel with the physical system throughout its entire life-cycle.	(Abburu et al., 2020a; Jinzhi et al., 2022a; Zheng et al., 2022)

Table 1.8: Characterisation of CDT in the Literature.

synchronisation, knowledge sharing, responsiveness, and optimisation across complex networks of actors.

Thanks to their cognitive capabilities, CDTs can detect patterns in the behaviour of their physical counterparts, anticipate deviations and predict the impact of different operating conditions. These properties mean that CDTs can be considered autonomous intelligent agents, as defined in the fields of AI (Sabharwal and Selman, 2011; Maes, 1995) and agent-based computing (Luck et al., 2004). For example, (Kalaboukas et al., 2021) implements a CDT as an ontological knowledge graph enriched with system states, behaviours, processes, and optimisation services that support cognitive reasoning. Another approach is proposed by (Abburu et al., 2020a), who define three levels of cognitive augmentation: a basic Digital Twin, a Hybrid DT with interconnected models and predictive capabilities, and a Cognitive DT capable of knowledge manipulation and problem-solving in unknown situations.

Overall, the cognitive capabilities of CDTs include sensing, reasoning, learning, and self-adaptation, enabling continuous evolution of both structure and behaviour. This section has reviewed how cognition is analysed and implemented in DTs. The synthesis of these findings and the CDT definition adopted in this thesis are discussed in chapter 2.

1.3 Technologies Enabling Cognitive Digital Twin

The results of the systematic literature review presented in the previous section conclude the state-of-the-art analysis. Section 1.2.4.2 introduced the cognitive functions of the CDT, alongside the two classical DT functions: simulation and emulation, and the additional cognitive functions that distinguish a CDT from a conventional DT.

In the remainder of this chapter, we focus specifically on the CDT. This choice is motivated by both methodological and practical reasons. Methodologically, the CDT provides a clear and operational framework to study how cognitive functions can be integrated into an industrial digital representation: it combines the traditional DT pipeline (data acquisition, synchronisation, modelling, simulation/emulation, and interaction) with additional cognitive capabilities. Practically, this focus also reflects the positioning adopted in the *AI4C2PS* project, where the CDT is used as the main vehicle to introduce and operationalise cognition within the CPS. In other words, rather than attempting to make the entire CPS “cognitive” end-to-end, the project approach is to embed and orchestrate cognitive functions through the CDT, which then supports decision-making and interaction at system level.

Building on this foundation, the subsequent sections review the technologies employed to implement the three functions in CDTs. Each subsection organises the state of the art according to the CDT functions introduced previously, and discusses representative technological approaches used in the literature. Particular attention is given to cognitive architectures, as these provide a structured way to integrate multiple cognitive functions within a coherent system. While full architecture-driven CDT implementations are still relatively uncommon, they remain important to understand how cognition can be modelled systematically and operationalised in DTs.

1.3.1 Technologies for Emulation function

The emulation function of a DT aims to reproduce the behaviour of the real system *in synchrony with its operation*. In line with early work on component-based and interoperable emulation for industrial systems, emulation can be understood as a paradigm where the virtual system remains compatible with the real system’s inputs/outputs and is continuously aligned with the physical execution (Iung et al., 2000). In a CDT, this corresponds to maintaining a live mirror of the physical asset by ingesting sensor signals and control data so that the digital representation reflects what is happening now (or with minimal delay). Table 1.9 shows the key technologies and approaches that enable the emulation function.

Based on Table 1.9, industrial IoT provides the acquisition and low-latency transport required by emulation by continuously streaming sensor and controller data (e.g., via MQTT, OPC UA, or 5G gateways) from the shop floor to the CDT (Mikołajewska et al., 2025). This connectivity layer is what keeps the digital representation “in phase” with its physical counterpart.

On top of data acquisition, *real-time 3D models and virtual environments* provide a synchronised visual representation of machines, conveyors, and parts, which can support live monitoring and operator understanding of the current state (Redelinghuys, 2020). Virtual commissioning and controller emulation extend this principle by running real controller logic against an emulated plant, enabling safe validation of behaviour while preserving I/O consistency with the real system (Redelinghuys, 2020). Similarly, *augmented reality* interfaces can project the current

Technologies	Key Tools & Platforms	Description
Industrial IoT Connectivity	MQTT, OPC UA, 5G gateways, IoT sensors	Provides real-time “perception” of machine state (Mikołajewska et al., 2025).
Real-Time 3D Models & Virtual Environments	Siemens Tecnomatix Plant Simulation, game engines	Visualizes live shop-floor conditions for anomaly detection (Redelinghuys, 2020).
Virtual Commissioning & Controller Emulation	PLC simulators (hardware-in-the-loop)	Enables risk-free testing of control logic in sync with real hardware (Redelinghuys, 2020).
Augmented Reality for Live Monitoring	AR headsets, tablet apps	Overlays current metrics on physical assets to enhance situational awareness (Ullah et al., 2025).
Supervised Machine Learning (ML)	Random Forest, Decision Trees	Real-time sensor-to-state mapping and anomaly detection (Hamel et al., 2024; Garg et al., 2021).
Knowledge-Based Systems (Ontologies) and Semantic Graphs	Protégé, RDF stores, Neo4j + ML pipelines	Integration of expert rules and contextual knowledge and semantic labelling of sensor/data streams for contextual perception (Madani et al., 2022; Mohammed et al., 2022; Zappa et al., 2024).

Table 1.9: Emulation Function Technologies in CDTs

emulated state (e.g., measurements, tool positions, alarms) onto the physical asset, improving situational awareness during operation (Ullah et al., 2025).

Beyond mirroring, emulation often requires online state interpretation: mapping raw streams to operational states, events, and anomalies. Supervised ML models (e.g., decision trees, random forests) are frequently used to classify system states or detect deviations in real time (Hamel et al., 2024; Garg et al., 2021). In parallel, ontology-based and semantic-graph approaches support emulation by providing *contextual labelling* of live data streams: they represent domain concepts (components, modes, alarms, conditions) and attach semantics to measurements so that the CDT can maintain a structured, interpretable view of the current situation (Madani et al., 2022; Mohammed et al., 2022; Zappa et al., 2024). Importantly, in this emulation-focused context, these knowledge models are used primarily to *organise and contextualise the current state* (e.g., what a signal refers to, which component it concerns, which condition it suggests), rather than to perform high-level maintenance decision-making, which is discussed later under the cognition function.

Overall, the emulation function relies on (i) connectivity and synchronisation mechanisms to keep the CDT aligned with the physical system, and (ii) online interpretation mechanisms (ML and semantic labelling) to turn streamed signals into meaningful states and events. When ML is combined with semantic graphs or RDF-based knowledge structures, the emulated state can become easier to interpret and reuse across tools because signals are linked to explicit domain concepts and relationships (Ayani et al., 2018). This strengthens the CDT’s ability to maintain

a coherent and explainable representation of the current situation within the emulation loop.

1.3.2 Technologies for Cognition function

The cognitive function of a CDT enables it to interpret information, reason about system states, learn from experience and support decision-making. While simulation and emulation provide predictive and real-time representations of the physical system, cognition enables the CDT to analyse and act upon these representations in context.

Implementing cognition requires a combination of technologies that support reasoning, learning, memory and planning. These include symbolic, data-driven and hybrid approaches and cognitive architectures. Table 1.10 summarises the main technologies used to implement cognitive functions in CDTs.

Technologies	Key Tools & Platforms	Description
Cognitive Architectures	SOAR engine, ACT-R framework	Symbolic reasoning and human-like planning via rules, chunking, modular memory and RL (Balaji et al., 2023).
ML Models (DL, RL)	TensorFlow, PyTorch, scikit-learn, RL toolkits (e.g. OpenAI Gym)	Data-driven perception, anomaly detection and policy optimization via RL (El Kalach et al., 2024).
KGs and Ontologies	Neo4j Aura, Amazon Neptune, RDF4J, OWL APIs	Semantic modeling of domain knowledge (long-term memory) and context-aware inference (Lam et al., 2024 ; Jinzhi et al., 2022b).
Rule-Based Reasoning Engines	Expert systems, logic solvers, Prolog engines	Deductive if-then reasoning for diagnosis, planning, and anomaly explanation, high transparency of decision logic (El Kalach et al., 2024 ; Johansen et al., 2023).
Hybrid Approaches (cognitive multi-agent, codelet-based)	JADE (agents), custom codelet frameworks, MPI for parallelism	Neuro-symbolic AI, and distributed agents (e.g. many small “codelets” for perception, memory, action) aligned with cognitive theories (global workspace, attention) (Johansen et al., 2023).

Table 1.10: Cognitive Function Technologies in CDTs

According to Table 1.10, each technology category brings distinct cognitive capabilities to CDTs. Cognitive architectures are theoretical frameworks that formalise perception, working and long-term memory, decision making and learning in an integrated processing cycle; SOAR and

ACT-R use production rules and modular memory constructs to support goal-directed planning, episodic retrieval and adaptive rule adaptation (Laird, 2022; Balaji et al., 2023).

Next, ML frameworks (TensorFlow, PyTorch, scikit-learn, OpenAI Gym) provide data-driven perception and reinforcement control policies for real-time anomaly detection and process optimisation, but their black-box nature and large data requirements usually require coupling with symbolic methods to ensure interpretability and robustness (El Kalach et al., 2024).

Then, KGs and ontologies (Neo4j Aura, Amazon Neptune, RDF4J/OWL) provide structured semantic stores that enable contextual inference and provenance tracking across heterogeneous data sources, although building and synchronising large domain models with high-speed sensor feeds is labour intensive (Lam et al., 2024; Jinzhi et al., 2022b).

In addition, rule-based engines (expert systems, Prolog) provide transparent deductive reasoning for fault diagnosis and planning with clear audit trails, essential for safety-critical scenarios, but require continuous updating of the rule base and specialised middleware to integrate noisy real-time sensor streams (El Kalach et al., 2024; Johansen et al., 2023); and hybrid, multi-agent frameworks (JADE, MPI) aim to unify symbolic reasoning and sub-symbolic learning within the framework of cognitive theories such as the global workspace to produce adaptive and emergent behaviours, although they introduce considerable complexity in terms of integration and validation (Johansen et al., 2023).

Among these approaches, cognitive architectures can be seen as a useful integration option for CDTs, because they offer a structured way to coordinate several cognitive functions within a single control loop. Unlike isolated AI components, cognitive architectures explicitly model the organisation of cognition, including perception, memory, learning, reasoning, and decision-making. This systemic organisation is particularly relevant for CDTs, where cognitive capabilities must not only be connected to the DT pipeline, but also remain coherent, interpretable, and aligned with human cognitive processes in collaborative contexts (continuous state updates from the physical system, internal state representation, and interaction with users and services). In this sense, cognitive architectures are not presented here as the only or necessarily central solution; rather, they represent one promising family of approaches to organise and combine heterogeneous mechanisms (including symbolic and data-driven components) in a coherent and explainable manner (Al Haj Ali et al., 2024b). Their potential and limitations, as reported in the state of the art, are examined in detail in Section 1.3.4.

1.3.3 Technologies for Simulation function

The simulation function of a CDT focuses on predictive modelling and scenario exploration. In this context, simulation means that the CDT can perform virtual experiments of the manufacturing process or product, predicting future states or testing hypothetical conditions without affecting the actual system. Table 1.11 shows the key technologies and approaches that enable the emulation function.

Various technologies are presented in Table 1.11. First, discrete-event process simulators replicate production workflows and, when coupled with optimisation algorithms such as genetic algorithms and recurrent neural networks such as bi-GRU with attention, generate synthetic data on flow and failure rates to identify optimal operating parameters (Zhu and Ji, 2022).

In parallel, high-fidelity physics engines are often implemented in the form of Simulink models derived from SysML system designs that reproduce equipment dynamics, including robot trajectories, fluid and thermal flows, and sensor outputs, which are then fed into machine learning

Technologies	Key Tools & Platforms	Description
Discrete-Event & Process Simulation	Plant Simulation, custom DES engines	Enables “what-if” analyses and AI-driven process optimization (Zhu and Ji, 2022)
Physics-Based Simulation Engines	MATLAB/Simulink, multibody & CFD solvers	Generates high-fidelity time-series data for AI training (Jinzhi et al., 2022b)
Co-Simulation & Multi-Model Integration	Functional Mock-up Interface (FMI), HLA middleware	Captures system-of-systems behavior for holistic scenario planning (Li et al., 2022b)
AI and Reinforcement Learning (RL) in Simulation	A2C, PPO, SAC agents in discrete-event simulators	Provides a virtual “playground” for learning and refining control policies via sim-to-real transfer (Liu et al., 2022b ; Mikołajewska et al., 2025)

Table 1.11: Simulation Function Technologies in CDTs

pipelines to build predictive models ([Jinzhi et al., 2022b](#)). To capture the interactions between systems, co-simulation frameworks based on standards such as the Functional Mock-Up Interface or High-Level Architecture coordinate multiple domain-specific simulators (mechanical, electrical, control) in real time, enabling holistic simulation analysis of entire production cells ([Li et al., 2022b](#)).

Finally, AI and RL enable large-scale virtual experimentation and safe transfer: a DRL agent interacts continuously within a high-fidelity simulated robotic cell with precise kinematics, sensor models and control logic, exploring millions of action sequences to optimise goals such as capture success or cycle time without risk to the physical hardware ([Liu et al., 2022b](#)). After initial training, the CDT uses domain randomisation (varying lighting, friction, payload, etc.) and live feedback to bridge the reality gap, iteratively refining the policy so that the behaviours learned *in silico* remain robust at the time of deployment.

Beyond RL, generative models such as GANs synthesise rare or faulty sensor data, such as bearing wear signatures or thermal excursions, to enrich the sparse real-world examples and enable the twin to train more accurate fault detection and prognostic models entirely from synthetic datasets ([Mikołajewska et al., 2025](#)).

At this stage, DRL appears as one promising option for simulation-based optimisation when the control problem is complex and hard to specify analytically, because it enables safe exploration and policy learning in a virtual environment ([Liu et al., 2022b](#)). However, DRL is not universally preferable: training can be long and costly, and classical predictive and optimisation models are often faster and easier to validate when system dynamics are well understood. Consequently, simulation technologies should be selected according to the objective and constraints, rather than assuming a single best-performing method.

1.3.4 Cognitive Architectures

Cognitive architecture is a theory of the fixed mechanism and structures that underlie human cognition (Lehman et al., 2006). In essence, a cognitive architecture is a map of how cognitive processes (perception, memory, reasoning, etc.) are organised and integrated to produce behaviour. Researchers in cognitive science and AI use this concept to develop unified theories of cognition and to design intelligent agents. Historically, cognitive architectures have been explored both in biological contexts to explain how the human brain gives rise to mind and in artificial contexts to build AI systems with general intelligence. This exploration covers several paradigms, from classical symbolic models of the mind to connectionist models (neural networks), as well as hybrid approaches that aim to combine the two. In recent years, researchers have worked to extract the common core of many architectures into a single consensus framework represented in Figure 1.7 and known as the Common Model of Cognition (Rosenbloom et al., 2024) (previously the Standard Model of Mind (Laird et al., 2017b)).

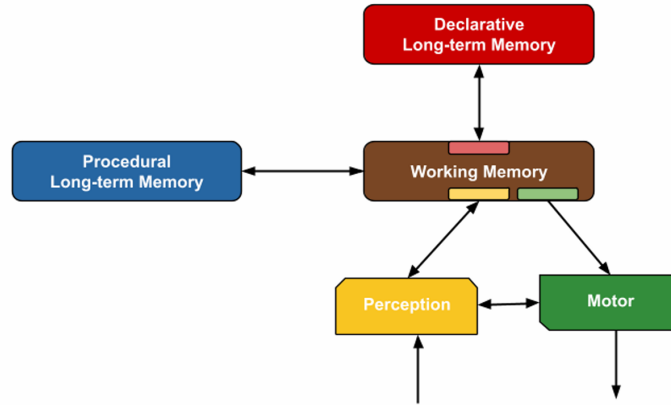


Figure 1.7: The Common Model of Cognition (Rosenbloom et al., 2024).

The diagram shows how the common model of cognition works as a simple, interconnected system. At the centre is working memory, which holds everything you are thinking about at any given time. On one side is declarative memory, which provides facts and experiences when you need them, and on the other is procedural memory, which uses “if-then” rules to decide what to do next. Below that, the perceptual module turns raw sensory information into useful information, and the motor module turns decisions into actions. Each part communicates with working memory through its own channel, allowing the system to perceive the world, think, remember and act in a coordinated way. In the following sections, we first examine cognitive architectures from the perspective of neuroscience and psychology, then review the main architectures of artificial intelligence, before discussing applications in human-machine interaction, robotics and manufacturing.

1.3.4.1 Cognitive Architectures in Biological Systems

Early theories of cognitive architecture in psychology adopted an information processing perspective, largely independent of biology. The mind was seen as “software” running on the “hardware” of the brain, with researchers focusing on abstract cognitive structures (e.g. memories, rule-based processors) without reference to neuronal anatomy (Petersen and Sporns, 2015). In this symbolic or *cognitivist* paradigm, cognition is seen as the manipulation of symbols according to

formal rules, as in Allen Newell and Herbert Simon’s physical symbol system hypothesis (Card and Newell, 1990). Classical symbolic architectures assume that complex thought emerges from sequences of discrete symbolic operations, like a digital computer. However, in the early 1980s, *Connectionism* or subsymbolic paradigm proposed a different approach: rather than symbols and explicit rules, cognition would emerge from networks of simple units (similar to neurons). These distributed parallel processing models automatically learn activation patterns between units, which more closely matches the biological functioning of the brain. Throughout the late XX^e century, the debate opposed:

1. Symbolists (cognitivist), for whom thought is based on the manipulation of formal symbols.
2. subsymbolists (Connectionists), for whom thought arises from the collective interaction of artificial neurons.

Some critics have questioned the ability of connectionist models to account for the complex structures and rules that seem to drive human thought (Card and Newell, 1990).

To balance these points of view, hybrid models have emerged that incorporate both symbolic and subsymbolic representations. Cognitive architects have recognised that human cognition can involve several levels of representation: high-level symbolic reasoning and low-level associative learning. An important example is the Adaptive Control of Thought-Rational (ACT-R) architecture developed by John R. Anderson, which, although often considered a symbolic architecture, has explicit links to neuroscience (Ritter et al., 2019). ACT-R is organised into modules supporting different cognitive functions (e.g., declarative memory, goals, and visual perception), and these modules are associated with specific brain regions based on neuroimaging evidence. This mapping is illustrated in Figure 1.8, which shows how ACT-R components relate to brain activity patterns. The architecture uses rule-based production systems for cognition (“if-then” rules to model stages of thought), each memory chunk in ACT-R has an activation level that determines how easily it can be retrieved, and these activation levels spread between chunks much like neural signals in the brain. In fact, much of the research that went into developing ACT-R involved analysing the neurological evidence for the existence of different modules and their interconnections.

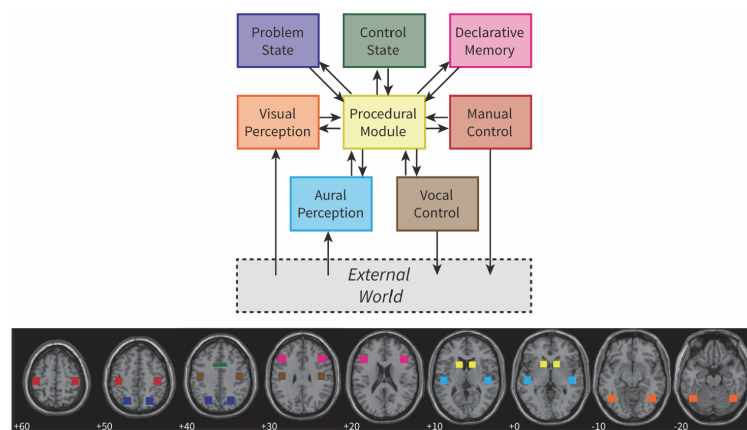


Figure 1.8: The cognitive architecture ACT-R and its mapping on brain regions (Borst et al., 2015).

Another influential biological model is Global Workspace Theory (GWT), which provides a

cognitive architecture for higher-order consciousness and cognition (Signa et al., 2021). Originally proposed by Bernard Baars, GWT compares the brain to a theatre: many unconscious processes (backstage actors) compete for attention, and the “stage” of the global workspace is a limited-capacity memory where the winning information is broadcast to the whole system as conscious content. In this model, attention is the projector that brings certain information into the global workspace, making it globally available to otherwise separate brain modules. Cognitive processes such as decision making, working memory and problem solving are supported by this global diffusion, while once an element enters the workspace it can influence system-wide processes (Signa et al., 2021). As a cognitive architecture, GWT provides a functional explanation of consciousness, which arises from the interaction of specialised and unconscious components in the brain. Its influence is evident in modern neurally plausible cognitive models. For example, Franklin’s LIDA architecture (Franklin et al., 2013) implements GWT in software agents and remains a reference point for consciousness research in both neuroscience and AI.

1.3.4.2 Cognitive Architectures in Artificial Intelligence

In the field of AI, cognitive architectures serve as blueprints for the construction of AI agents that exhibit human-like intelligence or cognition. AI researchers have developed numerous architectures, generally classified into three categories: symbolic, subsymbolic (connectionist) and hybrid. These AI architectures are inspired by psychological models, but are implemented as computational systems that can be run on a computer. Below, we present each category and some key examples, detailing the theoretical and technical aspects of each.

Symbolic Architectures

Symbolic cognitive architectures, as explained in the previous section, come from the classical tradition of AI in symbol manipulation (Kotseruba and Tsotsos, 2020). They are based on the principle that intelligence results from the execution of logical operations on symbolic representations (like a programme operating on data structures). Symbolic systems represent concepts using symbols that can be manipulated using a set of predefined instructions (Kotseruba and Tsotsos, 2020). A canonical example is the Soar architecture, originally developed by Allen Newell, John Laird and Paul Rosenbloom as a unified theory of cognition and problem solving (Laird et al., 1987; Laird, 2019). Figure 1.9 illustrates the Soar cognitive architecture, which is organised around a single, global buffer memory (the central box) in which perceptual inputs are stored and from which actions are sent back to the environment (Chong et al., 2007). In Soar, long-term knowledge is divided into three separate memories: procedural memory (productions), semantic memory (general facts) and episodic memory (event records) alongside the unified working memory, rather than being encoded only as productions:

- The procedural memory contains all the *if-then* rules (productions) that correspond to the working memory in order to propose actions or new problem states.
- The semantic memory provides declarative facts when requested by the productions.
- The episodic memory provides traces of past experiences that serve as reference points for problem solving.

Whenever a production matches the contents of the working memory, it triggers to suggest an action or modify the state of the problem. If no production applies, i.e., in the event of a deadlock, Soar automatically creates a subgoal, represented by a new state in the working memory, and invokes its learning mechanism (illustrated on the left) to resolve it. Once successfully

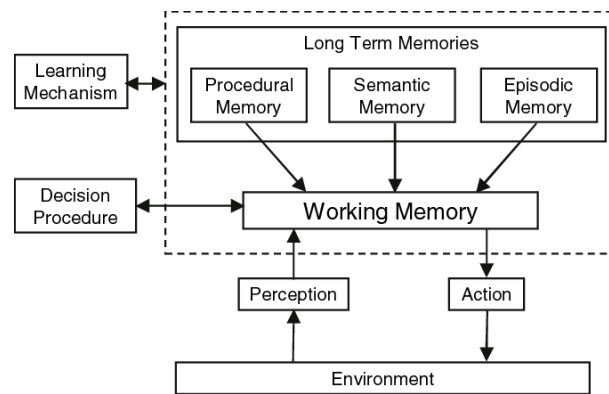


Figure 1.9: The Soar cognitive architecture (Chong et al., 2007).

resolved, the fragmentation process compiles the reasoning trace into a new production, which is stored in the procedural memory for future use.

Soar has demonstrated its ability to simulate a wide range of cognitive tasks (Newell, 1994), from simple puzzle solving to complex decision making and can even approach human response times for certain tasks. (Newell, 1994) positioned Soar as the basis for a unified theory of cognition, demonstrating that with the appropriate knowledge, the same architecture could model behaviour in various domains: language use, reasoning, spatial navigation, etc. Technically, Soar’s rule-based engine and its use of universal subgoals make it a powerful problem-solving tool that searches through possible actions until the goals are achieved. Soar continues to be extended to include RL, semantic memory, mental imagery modules, etc., but it remains essentially a symbolic problem space architecture.

Subsymbolic Architectures (Neural Networks and Emergent Models)

Unlike symbolic approaches, sub-symbolic architectures don’t use explicit rules and symbols, instead, they use neural-like units and weighted connections as the basis for cognition (Kotseruba and Tsotsos, 2020). These architectures are built on Artificial Neural Networks (ANN), which learn and compute by activating and transforming patterns. The terms sub-symbolic, neural, and emergent architectures are often used interchangeably, as they all refer to systems where cognitive abilities arise from the interaction of numerous simple processing units rather than from explicit symbolic manipulation. They are called emergent because cognition is not pre-programmed but emerges from the distributed dynamics and adaptive learning processes of neural networks. Traditional AI neural networks (such as multilayer perceptrons or recurrent neural networks) typically focused on narrow tasks, but some researchers have designed large-scale neural architectures aimed at broader cognitive functions.

Internally, Spaun’s design incorporates principles from neuroscience: for example, it uses a spiking neural model where neurons communicate via discrete pulses and employs the Neural Engineering Framework to encode and decode information into neural discharge patterns. The key to Spaun lies in the concept of semantic pointers, a neural representation that allows complex information to be compressed. Technically, Spaun demonstrates how a single neural architecture can exhibit a variety of cognitive behaviours by changing context: without any modification to the code, the same network can handle perception, working memory, decision-making and motor control in different tasks.

A notable example is the Semantic Pointer Architecture Unified Network (Spaun), a large-scale architecture that goes beyond a conventional ANN by combining (i) a biologically grounded spiking-neuron implementation, (ii) an explicit functional organisation into interacting brain-like subsystems, and (iii) a shared internal representational scheme that supports multiple cognitive functions within one integrated model (Choo, 2018; Stewart et al., 2012). In contrast to typical task-specific ANNs that are trained end-to-end for a single objective, Spaun is designed as a unified system: it couples perception, working memory, action selection, and motor control through structured interactions between modules, coordinated by a central action-selection mechanism. In addition, Spaun relies on semantic pointers which are compressed neural representations that allow the same representational format to be reused across tasks (e.g., representing symbols, goals, and intermediate results in a way that remains compatible with neural computation). This combination makes Spaun an example of how a neural system can support more than pattern recognition, by integrating memory, control, and task structure within a single architecture.

Spaun, developed by Chris Eliasmith and colleagues (Stewart et al., 2012), consists of approximately 2.5 million spiking neurons organised into modules reflecting brain regions (visual cortex, prefrontal cortex, basal ganglia, etc.). As shown in Figure 1.10, these modules interact via information projections (solid black lines) and flow control projections (dashed blue lines), coordinated by a central action selection network. Spaun is able to perform several cognitive tasks, including handwritten digit recognition, sequence memorisation, and simple reasoning, by receiving only visual inputs (images of handwritten digits) and producing outputs via a physically modelled arm. Internally, Spaun incorporates principles from neuroscience: it uses spiking neural communication and employs the Neural Engineering Framework to encode and decode information into neural activity patterns. Technically, Spaun illustrates how a single neural architecture can exhibit a range of cognitive behaviours by reconfiguring context and internal state, rather than changing the code between tasks (Stewart et al., 2012; Choo, 2018).

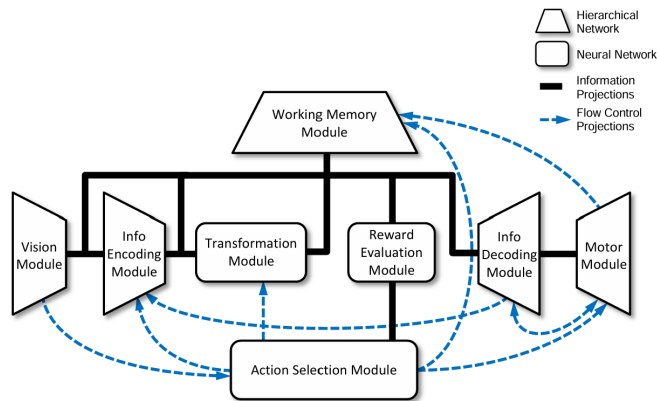


Figure 3.20: High-level architectural diagram of the Spaun model.

Figure 1.10: The Spaun cognitive architecture (Stewart et al., 2012).

This is a major achievement for subsymbolic AI, demonstrating the viability of brain-inspired architecture in large-scale simulation. More generally, modern DL architectures (such as deep convolutional and recurrent networks) can be considered subsymbolic cognitive architectures when designed to handle multiple cognitive functions. For example, a DRL agent that processes vision and produces actions, such as DeepMind’s Atari game network (Mnih et al., 2013), follows

a cognitive architecture model: a pipeline of neural networks transforming sensory inputs into motor outputs, modulated by learned value signals. However, most DL systems lack the cognitive scope and modularity of classical cognitive architectures.

Despite their successes, purely subsymbolic architectures have limitations in abstract reasoning, structured problem solving, and integration of prior knowledge. This has led to the emergence of hybrid cognitive architectures in the field of AI, which integrate symbolic and subsymbolic techniques.

Hybrid Architectures (Combining Symbolic and Neural)

Hybrid architectures attempt to achieve the best of both worlds by combining explicit symbolic reasoning with the learning and robustness of neural networks (Kotseruba and Tsotsos, 2020). A notable hybrid architecture is CLARION by Ron Sun (Sun, 2006), which is explicitly designed around the distinction between explicit and implicit cognition. In CLARION, cognitive processes are divided into two levels: a higher level that uses symbolic representations via symbolic rules and chunks, and a lower level that uses subsymbolic representations via distributed neural networks. For example, the Action-Centred Subsystem (ACS) of CLARION illustrated in Figure 1.11 may include a lower-level neural network that learns skills through trial and error, and a higher-level rule system that contains explicit strategies; the two levels interact and can learn from each other's experience. This dual-process structure is inspired by psychological theories that humans possess both intuitive, unconscious skills and conscious, declarative knowledge. CLARION also includes a Non-Action-Centric Subsystem (NACS) for semantic knowledge that includes a motivational subsystem for drives and rewards and a metacognitive subsystem for higher-level control, making it very comprehensive.

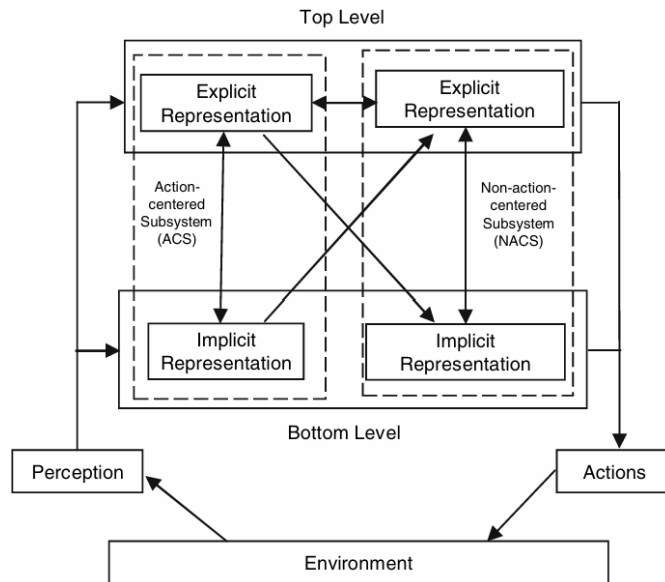


Figure 1.11: The CLARION cognitive architecture (Chong et al., 2007).

The technical aspect of CLARION lies in its ability to reason, for example by following rules, while at the same time improving these rules or acquiring new ones through lower-level neural learning. This makes it possible to model how a person or an agent can progressively automate a

skill: initially their performance may follow explicit instructions (higher level), but with practice the knowledge becomes implicit (encoded in lower level networks). CLARION has been used to simulate cognitive tasks and even social situations, demonstrating the flexibility of hybrid architectures.

Sigma, developed by Paul Rosenbloom and colleagues, is another hybrid architecture ([Rosenbloom, 2013](#)). Sigma is designed to unify several AI paradigms using a graphical model-based ([Koller and Friedman, 2009](#)) approach. In Sigma, all cognitive operations, from symbolic reasoning and probabilistic inference to continuous signal processing, are formulated as message passing in a factorial graph. In essence, Sigma’s engine generalises logic and probability into a single mathematical framework, allowing the architecture to deal natively with uncertainty, learning and discrete structures. Sigma’s design goals include *grand unification*, covering cognition, perception, and motor control in a single architecture, and *functional elegance*, meaning that it strives to derive many capabilities from a small set of fundamental principles. Specifically, Sigma represents knowledge using predicates and conditionals ([Rosenbloom, 2010](#)), like a symbolic AI system, but implements them in the background as factorial graphs that can also encode probabilities and utilities. A single inference algorithm, a synthesis of logical and Bayesian updating, drives the whole system. This allows Sigma, for example, to perform probabilistic reasoning about rules or to integrate perceptual signals into symbolic memory. Although still under development, Sigma has already demonstrated examples such as visual scene understanding combined with cognitive queries, and aims to address aspects such as emotions and personality by integrating them as additional factors in the model.

Beyond CLARION and Sigma, many other architectures adopt a hybrid stance. In practice, this often means combining symbolic structures (rules, goals, explicit knowledge) with subsymbolic mechanisms (activation, learning dynamics, statistical selection). For instance, Soar has been extended with neural learning modules in several lines of work, and ACT-R is frequently described as hybrid because its symbolic production system is modulated by subsymbolic activation and utility equations that influence memory retrieval and rule selection.

ACT-R is also widely used as a programming framework to model human reasoning and memory in applied settings ([Chong et al., 2007](#)). In particular, ACT-R models have been used to reproduce human-like performance patterns in tasks such as multitasking, driving, or problem solving, including characteristic timing and error profiles. A key aspect of ACT-R is that it connects functional modules (e.g., declarative memory, goals, and visual processing) to empirically grounded components; this link is illustrated in Figure 1.8. At the same time, purely symbolic approaches such as Soar and ACT-R are valued for interpretability: one can inspect rules and knowledge structures to understand the agent’s behaviour, and for the ability to encode prior knowledge explicitly. However, they are also criticised for being brittle when facing noisy, high-dimensional, or rapidly changing real-world conditions, which motivates the integration of subsymbolic learning mechanisms.

Hybrid approaches are now common in cognitive systems research, reflecting the view that human intelligence probably needs both robust learning from data (a strength of neural networks) and abstract reasoning with structure (a strength of symbolic AI). Technically, designing a hybrid architecture involves specifying how the symbolic and sub-symbolic parts interact. For example, does the neural component provide suggestions from which the symbolic component chooses, or does the symbolic component provide goals for the neural network to learn? Different architectures give different answers. But all share the idea that a single paradigm is insufficient

to cover the breadth of cognition. As (Sun, 2025) argues, the integration of neural and symbolic processes can also shed light on psychological theories of dual processes, providing a computational embodiment of System 1: fast, intuitive thinking, and System 2: slow, deliberative thinking, within the same cognitive system.

These architectures are essentially cognitive operating systems to which specific skills or knowledge can be added, making them attractive for practical applications, as we will see below.

1.3.4.3 Practical Applications of Cognitive Architectures

Cognitive architectures are not just theoretical constructs: they have been applied in various fields to stimulate intelligent behaviour in machines and model human performance. Below, we highlight applications in the fields of Human-Machine Interaction (HMI), robotics and industrial manufacturing.

Human-Machine Interaction

Cognitive architectures are used in the field of HMI mainly to model user cognition and improve interface design in order to understand how humans interact with automated systems. Researchers have built an ACT-R cognitive model to simulate a driver taking control of an autonomous vehicle (Scharfe and Russwinkel, 2019). This model reproduces the sequence of mental steps required to regain situational awareness during a control transfer and has helped explain the wide variability in driver response times. Such cognitive simulations are valuable for identifying factors that influence HMI performance, such as interface alerts and user distractions, and for designing safer embedded interfaces.

Another application is the development of HDT for industrial operators. In 2023, an HDT of a chemical plant control room operator was developed using the ACT-R architecture (Balaji et al., 2023). This cognitive twin can mimic an operator's monitoring and decision-making behaviour in abnormal situations such as process disturbances. In experiments with 426 simulated scenarios, the ACT-R-based twin showed attention patterns similar to those of humans and validated against eye-tracking data from real operators. The result shows that ACT-R can be used successfully to simulate human-like operator behaviour. However, more work is needed to enable emulating the behaviour and actions of a specific individual.

These models demonstrate the potential of cognitive architectures to predict human responses to interface alerts and to aid training and evaluation. For example, the HDT can generate a large database of human responses to rare emergency situations, helping designers identify interface flaws and train novice operators in multiple alarm management.

Robotics

Recent work in autonomous robotics increasingly investigates brain-inspired cognitive architectures in connection with the principles of developmental robotics. Developmental robotics is an interdisciplinary field at the intersection of robotics, developmental psychology, and neuroscience, which studies how embodied agents can progressively acquire sensorimotor and cognitive skills through exploration and social interaction (Metta et al., 1998; Schmidhuber, 2006; Min et al., 2016). In contrast to many industrial manipulators, typically designed for predefined tasks in controlled settings, developmental approaches emphasise open-ended learning, goal formation, and lifelong adaptation. The focus is therefore less on reproducing the optimised performance of an adult brain, and more on modelling the processes of skill acquisition and self-organisation that can lead to autonomous decision-making (Sandini et al., 2024).

Several robotic systems illustrate how cognitive architectures are used to structure these capabilities. The Gentlebots project, for instance, reports a cognitive architecture for social robots performing domestic tasks (Martín et al., 2020). The architecture evolved from behaviour-based control toward the integration of symbolic memory and planning components to address more complex human-robot interaction scenarios. The most recent version combines behaviour trees for low-level control with a distributed memory graph for high-level knowledge, and it has been evaluated in service robotics benchmarks such as RoboCup@Home and the European Robotics League (Martín et al., 2020). These evaluations highlight typical requirements for domestic service: perceiving the environment, planning actions (e.g., object manipulation), and interacting with humans in a natural and robust way.

In industrial HRC, architectures are also used to support coordination and adaptation during teamwork. The DAC-HRC architecture (Freire et al., 2024), grounded in distributed adaptive control theory, combines modules for perception, goal management, and learning to support interaction features such as joint intention and interdependence. It has been applied in an intelligent recycling plant scenario to coordinate humans and robots in e-waste dismantling tasks, with reported behaviours including turn-taking, partner-specific adaptation, and multimodal interaction (e.g., gesture-based communication) (Freire et al., 2024). This line of work illustrates how architecture-level integration can support social awareness and real-time adjustment in dynamic environments.

Beyond these examples, other cognitive architectures have been extended and connected to robotics middleware to address navigation and planning in realistic conditions. For example, LIDA has been augmented with hierarchical spatial memory and integrated with ROS, with demonstrations in a simulated 3D environment (Gazebo) that includes physical effects and sensor noise (Madl et al., 2016; Quigley et al., 2009). Similarly, Soar has been used for robot control and planning; for instance, Soar-based approaches have been proposed for autonomous exploration and mapping in unknown environments (Lindes, 2022). Overall, the robotics literature suggests that cognitive architectures are often leveraged as integration frameworks to combine perception, memory, planning, and interaction, especially in settings where robots must operate under uncertainty, interact with humans, and adapt over time.

Manufacturing

In manufacturing, the application of cognitive architectures is part of the broader trend of cognitive manufacturing or Industry 4.0/5.0, where production systems are becoming more intelligent and adaptive. Modern factories generate huge amounts of data from machine sensors, production logs and more, requiring complex real-time decision making. (El Kalach et al., 2024) defines cognitive manufacturing as the use of AI systems capable of reasoning from multimodal data and domain knowledge to manage the complexity of modern production. To achieve this, manufacturing research has turned to cognitive architectures to equip machines and software with advanced planning, diagnostic and learning capabilities similar to human cognition.

One area of application is logistics and autonomous operations in factories. The TROCA project (2019) (Gudwin et al., 2020) developed an autonomous transport robot for warehouses, controlled by a cognitive architecture called MECA. The robot’s “artificial mind” was responsible for several tasks: navigating the facility, transporting packages, managing its own battery level, interacting with human operators and coordinating its work with other robots. The cognitive architecture enabled the robot to perform these multiple tasks. For example, it could explore the

inventory, then move on to a delivery task, all the while monitoring its power and responding to human instructions. By integrating all these capabilities into a single cognitive framework, the TROCA robot illustrates how cognitive architectures can create flexible AGVs (Automated Guided Vehicles) or mobile robots that adapt to the changing needs of a production environment.

Another example of an application is the cognitive orchestration of manufacturing processes. (Fischbach et al., 2020) presented CAAI, a cognitive architecture for cyber-physical production systems. CAAI acts as an intelligent decision layer in a factory’s software stack: it takes high-level goals from human engineers (e.g., the goal of optimising the throughput of a production line), selects the appropriate AI models/algorithms, and configures a data processing pipeline to achieve the goal. The architecture continuously monitors performance and can automatically adjust or change algorithms, for example changing a predictive maintenance model if it is underperforming. Due to its modular design using containers and streaming platforms, CAAI has been deployed on a big data platform and evaluated in a real use case in production, demonstrating its ability to reduce the effort of integrating AI into manufacturing.

Cutting-edge research is also exploring neuro-symbolic AI for manufacturing decision support by combining cognitive architectures with advanced machine learning. (Wu et al., 2024) presented LLM-ACTR, which integrates the ACT-R cognitive architecture with a Large Language Model (LLM) to support design-for-manufacturing decisions (e.g., optimising an assembly process). In their experiments, the authors report that LLM-ACTR produced more consistent and better-structured solution rationales than a baseline LLM, by constraining the generation process with an explicit ACT-R-based task model. More specifically, ACT-R is used to encode structured decision processes for the manufacturing task, and these structures are then leveraged to guide the LLM’s responses. This illustrates how combining data-driven models with architecture-level task structure can improve transparency and reliability in complex manufacturing scenarios (Wu et al., 2024).

1.4 Chapter conclusion

This chapter provides the conceptual foundations required to address the central objective of this thesis: defining, characterising, and operationalising cognitive interoperability within the CPE. Through three complementary literature reviews, it clarifies how cognition is understood, how it is implemented in industrial systems, and how it can support a new form of interoperability between humans and CPS.

The first review contributes to **RQ1**, it analyses the concept of cognitive interoperability and highlights its emergence as a response to the limitations of traditional interoperability models. While technical and semantic interoperability enable systems to exchange data and preserve meaning, they remain insufficient in collaborative environments where humans and industrial agents must share understanding, interpret intentions, and reason coherently about situations. Although the literature suggests that cognitive interoperability goes beyond data alignment, it rarely specifies which cognitive mechanisms must be aligned, how such alignment can be operationalised, or how it can be evaluated in industrial settings. As a result, cognitive interoperability remains an emerging and fragmented concept, particularly in environments characterised by heterogeneity, uncertainty, and human-machine collaboration. It also shows that the concept remains emerging and fragmented, which reinforces the need for a clearer conceptual framework.

The second review contributes to **RQ2**, focuses on cognition and cognitive systems, grounding the thesis in cognitive psychology, cognitive science, and AI. Cognition is not treated as a single function, but as a set of interrelated processes including perception, attention, memory, learning, reasoning, and decision-making, which together enable agents to interpret their environment, learn from experience, and act purposefully. The review shows that, although industrial research increasingly refers to cognition, it often lacks explicit grounding in these theoretical foundations. By systematically analysing how cognitive functions are modelled and integrated into CPS and DT, this chapter clarifies how cognitive capabilities transform traditional CPS into CCPS and DTs into CDTs. CCPS are characterised as systems that exhibit autonomy, adaptability, and awareness, while CDTs extend classical DTs through semantic modelling, reasoning, learning, and self-adaptation.

The third review contributes to **RQ3, RQ4**, it examines the technologies that enable cognition in industrial systems. It highlights the role of AI and ML methods, as well as integration mechanisms such as cognitive architectures, for implementing cognitive functions. In particular, KRR approaches such as ontologies and KGs stand out as key enablers for structuring knowledge, supporting reasoning, and facilitating interoperability. These technologies support a shift from raw data processing toward knowledge-based interpretation, anticipation, and decision support. They also contribute to the semantic and structural foundations that can help align machine representations with human conceptual models, which is relevant for cognitive interoperability.

Taken together, the three reviews show that cognition is an important capability for future industrial systems, not only to enhance autonomy and adaptability, but also to support meaningful collaboration between humans and machines. Cognitive interoperability emerges as a key missing link that connects cognitive systems to collaborative performance by supporting shared understanding, coordinated reasoning, and mutual adaptation.

At the same time, this chapter also shows that existing research remains fragmented: definitions of cognition, CCPS, CDTs, and cognitive interoperability vary across disciplines, and there is no unified conceptual framework that integrates these elements coherently. This justifies the need for the conceptual work developed in the remainder of the thesis.

Consequently, this chapter establishes the necessity of:

1. refining the definition of cognition in an industrial context,
2. clarifying the concepts of CCPS and CDT based on cognitive foundations, and
3. formally defining cognitive interoperability and identifying how it can be operationalised and evaluated.

These objectives are addressed in the following chapters. Chapter 2 builds on these foundations to develop a conceptual model of cognition in industrial systems, propose consolidated definitions of CCPS and CDTs, and introduce a formal definition of cognitive interoperability. This progression moves from literature synthesis toward conceptual construction, enabling the subsequent development of architectures, metrics, and evaluation frameworks for cognitive interoperability between humans and CPS.

Chapter 2

Conceptualisation of Cognitive Systems, Cognitive Interoperability, and its Maturity Model

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Introduction

This chapter develops the conceptual foundations of the thesis and contributes primarily to **RQ1** and **RQ2**. First, it provides a structured analysis of cognition in industrial systems, leading to a refined definition of cognitive systems in the context of Industry 4.0 and 5.0. Building on this analysis, the chapter then proposes consolidated definitions of Cognitive Cyber-Physical Systems (CCPS) and Cognitive Digital Twins (CDTs), clarifying their respective roles, characteristics and cognitive capabilities within industrial environments (**RQ2**).

The second part of the chapter focuses on cognitive interoperability and contributes mainly to **RQ1**. Building on the theoretical foundations introduced earlier, cognitive interoperability is formally defined. A conceptual architecture is then proposed to support cognitive interoperability between humans and CPS. Finally, the chapter introduces a set of evaluation levels and metrics to assess cognitive interoperability maturity. This framework starts from Level 0, where systems only achieve semantic interoperability, and progresses through Levels 1–4, where cognitive interoperability develops from shared situation awareness and shared mental models to intent and reasoning alignment, and ultimately to joint decision-making with metacognitive regulation.

2.1 Cognitive Systems in Industry

Following the synthesis of the state of the art on cognition obtained through the SLR presented in section 1.2.3, this section focuses on how cognition is interpreted and integrated in industrial contexts. The limitations observed in traditional CPS and DTs show that purely data-driven or rule-based approaches are often insufficient in complex environments. Industrial settings require systems that can understand context and adapt to change, especially under variability or unexpected events. When human involvement is present (e.g., supervision, maintenance, commissioning, or recovery), systems also benefit from mechanisms that support effective human–system coordination. Addressing these needs requires a clear understanding of cognition and of the mechanisms that allow cognitive behaviour to emerge in industrial and artificial systems.

In cognitive psychology, cognition is fundamentally concerned with how an agent, whether human or artificial, makes sense of its environment in order to act with purpose. It is described as a set of mental processes through which sensory information is transformed into meaningful knowledge, supporting understanding, judgement, decision-making and action (Neisser, 1967). This interpretation highlights cognition as an active, continuous and adaptive process rather than a static function.

This interpretation of cognition is particularly relevant in industrial environments. Industrial systems operate in contexts characterised by complex processes, dynamic conditions, uncertainty and human interaction. In such settings, cognition cannot be reduced to simple data processing or automation. Rather, it refers to the system’s ability to interpret situations, understand context, and respond appropriately to evolving conditions. In this perspective, cognition in industrial systems can be described through the following key characteristics:

- A key aspect of cognition is the transformation of raw data into **knowledge**. Sensory inputs or measurements alone do not carry meaning. Cognition enables an agent to organise information, relate it to prior experience and interpret it within a given context.

This process leads to the formation of internal representations that support reasoning and decision-making. In industrial contexts, this is essential for tasks such as diagnosing anomalous situations, anticipating system behaviour or adapting to changes in production conditions.

- Cognition is also inherently **goal-oriented**. Cognitive processes are driven by objectives, constraints, and intentions. In humans, these may relate to task completion, safety, or efficiency. Similarly, in industrial systems, cognitive behaviour is oriented towards operational goals such as maintaining performance, ensuring safety, and supporting collaboration. This goal-oriented nature distinguishes cognition from passive monitoring or reactive control.
- Another defining feature of cognition is its reliance on **experience**. Cognitive agents do not operate based only on predefined rules; they also rely on knowledge acquired through past interactions. This includes both procedural knowledge (how actions are performed) and declarative knowledge (facts, concepts, and relationships) (Anderson, 1996). In industrial settings, this enables agents to refine their behaviour over time, improve the quality of their decisions, and handle previously unseen situations more effectively.
- Cognition is also **context-dependent**. The same information may lead to different interpretations depending on the situation, environment or agent's internal state. Industrial environments often involve contextual factors such as machine states, human actions, operational constraints and external disturbances. Cognitive behaviour therefore requires the ability to integrate these factors and adapt decisions accordingly rather than applying fixed responses.
- Importantly, cognition is not a single function, but rather a **system-level** phenomenon that emerges from the interaction of multiple cognitive functions. Perception enables awareness of the environment, memory supports the retention of knowledge, reasoning allows inference and judgement, and learning enables adaptation. These functions operate together and continuously influence one another. Section 2.1.4 briefly revisits the main cognitive functions relevant to cognitive systems and clarifies how they contribute to the overall behaviour of a CDT.

The state of art reveals that, while cognition is frequently referenced in industrial research, it is often employed implicitly or inconsistently. Many contributions refer to cognitive systems without explicitly grounding them in established cognitive theories. This makes it difficult to compare approaches or assess cognitive maturity. A clear conceptual understanding of cognition is therefore necessary when defining cognitive systems, CCPS and CDTs.

2.1.1 Defining a Cognitive System

The term “*cognitive system*” is frequently used in industrial research, but with heterogeneous scopes: it may refer to (i) a dedicated cognitive *subsystem* embedded in a technological artefact, (ii) a *system* (human or artificial) capable of cognitive behaviour, or (iii) a broader *socio-technical* setting where cognition emerges from interactions between humans and technological components. To make the concept usable in this thesis, we adopt an explicit systemic framing

and formalise the common characteristics that appear across definitions.

The state of art shows that the term “*cognitive system*” is frequently used in industrial research, with different scopes and levels of precision. In order to make the concept usable in this thesis, the next step is to identify and formalise the common characteristics that appear across definitions.

Cognitive systems in industry are complex, comprising numerous interacting elements. The frequent use of the term “*system*” in definitions found in the literature therefore suggests the need to adopt a systemic perspective. In line with General Systems Theory, (Von Bertalanffy, 1972) defines a system as *a complex set of interacting elements, whose properties are richer than the sum of its parts*. When applied to cognitive industrial systems, this implies that cognitive behaviour does not arise from a single function, but from the coordinated interaction of multiple elements supporting observation, internal representation, reasoning, learning and action.

Beyond this systemic dimension, current research describes cognitive systems as systems that emulate certain aspects of human cognition in order to facilitate autonomous and adaptive behaviour. Rather than replacing human operators, their objective is to operate in symbiotic collaboration with them, in line with the human-computer symbiosis paradigm (Vermesan et al., 2022). In industrial environments, cognitive systems are expected to observe operational contexts, interpret heterogeneous information, learn from experience and continuously improve their behaviour (Angulo et al., 2023). Some studies also suggest that integrating human-like characteristics, could enhance Human-CPS collaboration by making system behaviour more intuitive and predictable (Abie, 2019).

Across the reviewed studies, cognitive systems are associated with the integration of fundamental cognitive processes, including perception, attention, memory, learning, reasoning, decision-making and problem-solving (Kiely, 2014; Vernon et al., 2007). These functions enable the system to construct internal representations of situations, reason about possible actions and adapt decisions to dynamic conditions. This capacity for learning and adaptation is particularly important in industrial contexts, which are characterised by uncertainty, variability and human interaction.

Based on this synthesis, the following definition is proposed to capture the essential characteristics of cognitive industrial systems while preserving the systemic nature of the concept:

Definition: Cognitive System

A cognitive system is a system that integrates cognitive functions inspired by the functioning of the human brain, capable of learning, reasoning and decision-making. They can interact naturally with humans and continuously improve through their ability to adapt. These systems process and understand complex information, making it easier to solve a variety of problems and carry out complex tasks.

2.1.2 Defining a Cognitive Cyber-Physical System (CCPS)

A CCPS is an industrial extension of a CPS designed to operate under conditions of variability, uncertainty, and human interaction. The SLR shows a clear convergence. In addition to sensing and control, a CCPS integrates cognitive functions that enable the system to interpret its environment, adjust its behaviour, and support interaction with human operators in real time.

Unlike Cognitive Digital Twins, CCPSs emphasise situated cognition and direct coupling with the physical system (Zhou et al., 2019; Khargonekar, 2019). Unlike Cognitive Digital Twins, CCPSs emphasise situated cognition and direct coupling with the physical system.

Three system-level characteristics emerge from the synthesis of the reviewed definitions as the most stable descriptors of CCPS: autonomy, adaptability, and awareness. Autonomy captures the capacity to self-monitor and make decisions without continuous human supervision. Adaptability captures the ability to adjust behaviour in response to changes and uncertainty, rather than following fixed strategies. Awareness captures the ability to maintain an understanding of the environment and internal state, including the interaction context when humans are involved. The SLR indicates that, depending on the application, studies often emphasise one of these characteristics, but in practice they remain interdependent: awareness supports decision quality, adaptability supports robustness, and autonomy enables continuous operation.

Overall, the SLR positions CCPS as a CPS with an additional cognitive layer that enables contextual interpretation, learning from interactions, and autonomous decision-making. Based on this synthesis, the following definition is proposed, extending the foundational CPS concept to include the properties of a cognitive system:

Definition: Cognitive Cyber-Physical System (CCPS)

A CCPS is a CPS and a cognitive system. As such it has cognitive functions, with abilities enabling not only to interact with its environment, but also to understand and learn from it, to make autonomous decisions and adapt dynamically to the context.

2.1.3 Defining a Cognitive Digital Twin (CDT)

A CDT is an evolution of the DT when cognitive capabilities are required. While conventional DTs primarily provide digital representations of physical systems through simulation and emulation, the SLR shows that this is often insufficient in industrial environments where interpretation, anticipation and autonomous decision-making are required. CDTs emerge precisely to address this limitation.

Our review of the literature indicates that a CDT is not just a DT with greater detail or real-time capabilities, but rather a DT that has been augmented with the properties of a cognitive system. In this sense, cognition enables the DT to do more than mirror behaviour; it allows it to actively exploit knowledge in order to understand situations, reason about system states and guide decisions. Therefore, CDTs combine the core purpose of digital twinning, maintaining a consistent link with a physical system, with cognitive mechanisms that support interpretation, learning, and adaptation.

Across the analysed studies, four characteristics consistently emerged as defining dimensions of CDTs: semantic modelling, autonomy, real-time reasoning and adaptability. Semantic modelling enables the CDT to structure and interpret information about the physical system in a meaningful way rather than relying on raw data alone. Autonomy refers to the CDT's ability to operate as an active entity, capable of making decisions without continuous human intervention. Real-time reasoning captures the capacity to analyse incoming data and trigger actions with minimal delay, which is essential for operational relevance. Finally, adaptability reflects the CDT's ability to evolve over time by learning from new data and maintaining alignment with

the physical system throughout its lifecycle.

Taken together, these characteristics mean that the CDT is positioned as an active, evolving digital counterpart rather than a passive model. It continuously interacts with its physical twin, refining its internal representations and updating its behaviour based on experience and context. The SLR also shows that CDTs can be interpreted as autonomous intelligent agents capable of sensing, reasoning, learning and acting within an industrial ecosystem.

Based on this synthesis, a CDT can be defined as a combination of the properties of a digital twin and a cognitive system.

Definition: Cognitive Digital Twin (CDT)

A CDT is a DT and a cognitive system. As a DT it emulates a physical system that can be or not itself cognitive. As a cognitive system it possesses cognitive functions, bringing it in particular the ability to semantically model, process and interpret information autonomously and actively learning from its interactions.

While the above definitions clarify what constitutes a cognitive system, a CCPS and a CDT in industrial contexts, they also highlight an issue that still needs to be addressed. Industrial environments rarely involve a single cognitive entity operating in isolation. Instead, multiple cognitive agents, including humans, CCPS and CDTs, must interact, coordinate and collaborate within shared processes and environments. Therefore, defining cognitive capabilities at the system level is a necessary but insufficient step. The question of how these heterogeneous cognitive systems can effectively understand each other, align their internal representations, and cooperate to achieve common goals remains to be determined.

2.1.4 The Cognitive Functions

Cognition in industrial systems is typically operationalised through a limited set of functions that support adaptive behaviour and effective interaction. Across the reviewed works, we retain the following six functions as the most recurrent and relevant for industrial cognitive engineering: *perception, attention, memory, learning, reasoning, and decision-making*.

In parallel, the cognitive interoperability literature introduces notions such as *mutual understanding, shared mental models, or joint decision-making*. Although these terms are expressed at the interaction level, they implicitly refer to the same families of functions, but considered in a coordinated or aligned form between agents. This motivates the need for a unified reference, defined once, that can be reused consistently both for industrial cognition and for cognitive interoperability.

Different scientific disciplines nevertheless use different decompositions and terminologies. From a neuropsychological perspective, for example, the cognitive sphere is divided into domains that group related functions, processes, and abilities (Harvey, 2019). In such works, the terms *function, process, and ability* may be used interchangeably. From a systemic perspective, however, it is useful to distinguish between the **cognitive system** (the subsystem that generates cognitive behaviour) and **cognitive abilities** (the observable capacities resulting from its internal processes). This distinction is particularly relevant for industrial systems, where cognition

must be analysed independently of any biological substrate.

In (Harvey, 2019), eight main cognitive domains are identified: *sensation, perception, motor skills and construction, memory, attention and concentration, executive functioning, processing speed, and language-verbal skills*. These domains structure human cognitive assessment and provide a functional view of cognition that can be abstracted beyond the human brain. When artificial systems are intended to collaborate with humans and reach comparable levels of task understanding, evaluating humans and artificial agents along similar functional dimensions becomes meaningful (Naudet et al., 2023b). This supports the idea of a shared cognitive reference framework as a basis for cognitive interoperability.

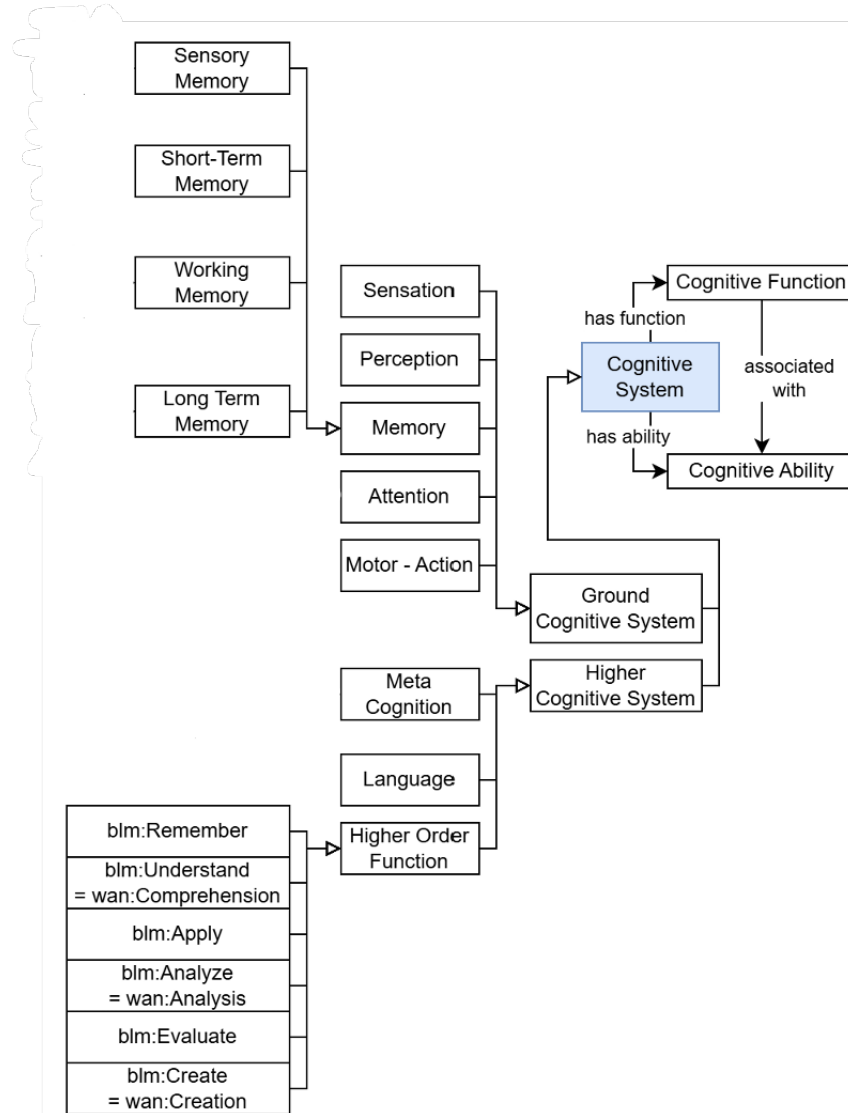


Figure 2.1: Cognition model: hierarchical meta-model of the different components of a cognitive system and their relations to cognitive functions and abilities (Adapted from Naudet et al. (2025)).

Other perspectives provide complementary structure. Cognitive informatics (Yingxu Wang et al., 2006) conceptualises cognition as interacting processes organised in layers (*sensational, subconscious, meta-cognitive, higher cognitive*). While this layered view provides useful insight, we do not retain a strict separation between conscious and unconscious processes, as the literature does not converge on a stable dichotomy and many processes can operate at both levels. Cognitive architectures adopt yet another viewpoint by modelling cognition through interacting functional components, typically *perception, working memory, long-term memory (procedural and declarative), and action*, emphasising information flow and functional coupling (Laird et al., 2017a).

To unify these perspectives and avoid redefining functions multiple times in later sections, we rely on a study that explicitly integrates several viewpoints on cognitive functions and synthesises them into a single conceptual model (Naudet et al., 2025; Zehnder and Naudet, 2026). Here we focus exclusively on the cognitive domain and formalise it as a **Cognitive System**. This model is a functional abstraction of cognition: it does not aim to represent the human nor any physiological or psychological aspects beyond cognition, and it is independent of its biological or artificial realisation. It also does not imply a strict sequential processing order or any anatomical localisation; rather, it structures the modelling and analysis of cognition in both human and artificial systems.

Figure 2.1 presents the resulting model. The **Cognitive System** is associated with **Cognitive Functions** (internal processes) and **Cognitive Abilities** (observable capacities), and is hierarchically decomposed into two main subsystems: the **Ground Cognitive System** and the **Higher Cognitive System**. The figure uses prefixes (namespaces) to indicate the origin of some sub-classifications. In particular:

- **blm** refers to Bloom/Anderson’s revised taxonomy, used here as a first-level structure for **Higher Cognitive Functions** (e.g., **blm:Remember**, **blm:Understand**, **blm:Apply**, **blm:Analyze**, **blm:Evaluate**, **blm:Create**) (Anderson and Krathwohl, 2021).
- **wan** refers to Wang’s Cognitive Informatics perspective (Layered Reference Model of the Brain), which is used in the source study to map and relate higher-order functions across perspectives (e.g., correspondences such as comprehension/analysis/creation in Wang’s terminology aligned with Bloom/Anderson levels in the figure) (Yingxu Wang et al., 2006).

This explicit provenance is important for our purpose: the model is not a new discipline-specific taxonomy, but a unifying meta-model that maintains correspondences between established decompositions.

The **Ground Cognitive System** comprises the core functions required for basic interaction with the environment and internal information processing, namely **Sensation**, **Perception**, **Memory**, **Attention**, and **Motor/Action**. These functions enable the acquisition, transformation, storage, selection, and use of information.

The **Higher Cognitive System** comprises functions that operate on the outputs of the ground system to support abstraction, reasoning, regulation, and complex interaction. It includes **Metacognition**, **Language**, and **Higher Cognitive Functions**.

Each component of the model, as defined in Naudet et al. (2025) and Zehnder and Naudet (2026), is described in the following paragraphs:

- **Sensation and Perception.** **Sensation** refers to the acquisition of raw signals from the external world (via biological senses or engineered sensors), whereas **Perception** refers to the interpretation and organisation of these acquired signals into meaningful representations of objects, events, and situations. Although some approaches merge sensation into perception, the meta-model distinguishes them to make the functional chain explicit. In our work, we keep this conceptual distinction for clarity; however, when focusing on industrial cognitive systems, **Sensation** is realised by sensing hardware and data acquisition mechanisms and is therefore treated as part of the overall sensing–perception pipeline rather than as a standalone cognitive function.
- **Memory:** supports the encoding, storage, and retrieval of information and provides continuity across time. It enables the preservation of past states and experiences and underpins learning, reasoning, anticipation, and consistent behaviour. In engineered systems, it corresponds to mechanisms that maintain and update internal representations (e.g., models of system state, context, and history) at multiple time scales.
- **Attention:** regulates the allocation of limited cognitive and computational resources. It supports the selection and prioritisation of information streams, events, or hypotheses, and it enables the system to focus processing on what is relevant with respect to goals, risks, or anomalies. Attention can be stimulus-driven (reactive to salient events) or goal-driven (directed by internal objectives).
- **Motor/Action:** includes the mechanisms by which a cognitive system produces actions, both physical and internal. It includes selecting, initiating, and adapting actions based on perceived context and internal state. In industrial systems, it covers the execution of control actions on the physical process, as well as internal actions such as updating plans, triggering alarms, or reconfiguring models.
- **Metacognition:** refers to the monitoring and regulation of the system’s own cognitive processes. It includes assessing confidence and uncertainty, detecting inconsistencies or errors, and adapting strategies, models, or resource allocation when needed. This function supports self-regulation and robustness, especially under uncertainty and change.
- **Language:** supports symbolic representation and communication. It enables the externalisation of internal representations (e.g., explanations, instructions, reports) and the interpretation of communicated information from other agents. In industrial contexts, this function may be realised through natural language interaction, structured messages, or semantic representations that support shared meaning.
- **Higher Cognitive Functions:** include reasoning, planning, evaluation, problem-solving, and decision-making. They operate on the outputs of the ground cognitive system to support abstraction, goal management, and complex regulation of behaviour. In the model, Bloom/Anderson’s revised taxonomy provides a first-level structuring of these functions (e.g., remembering, understanding, analysing, evaluating, creating), offering a hierarchical view of increasing cognitive complexity.

2.2 Cognitive interoperability: Concept and Vision

This section introduces the concept of cognitive interoperability and outlines the approach taken in this thesis. The state of the art on cognitive interoperability is first introduced in Section

1.1, and this section builds upon that baseline to refine and operationalise the concept. First, it explicitly discusses the limitations of semantic interoperability, showing why semantic alignment alone is not sufficient to ensure robust collaboration between humans and CPS. Second, it analyses the terminological field using a bag-of-words approach to identify the key cognitive notions underlying existing definitions. It then adopts an architectural perspective to identify the core elements of cognitive interoperability. Based on these insights, the section concludes with a definition of cognitive interoperability, which will be used throughout the rest of the thesis.

2.2.1 Why semantic interoperability is not sufficient

Semantic interoperability is a necessary foundation for collaboration because it enables agents to exchange *actionable knowledge* through shared vocabularies, ontologies, and formal constraints, so that information is interpreted consistently and without ambiguity across systems (European Commission, 2017; Tolk and Muguira, 2003). However, even under a strong semantic regime, collaboration between humans and CPS can fail because agents still need to align cognitive states and processes that are *not guaranteed* by semantic alignment alone.

First, semantic interoperability does not guarantee that agents align their **perception and interpretation of the current situation**. Situation awareness requires selecting relevant cues, integrating them into a coherent interpretation, and projecting near-future evolution (Endsley, 1995; Stanton et al., 2001). Two agents may perfectly agree on the meaning of terms (e.g., what “temperature” and “overheating threshold” denote) while still diverging on what is currently happening, what is relevant, and what is likely to happen next, especially in dynamic settings where attention and resource limits play a role (Endsley and Jones, 2012).

Second, semantic interoperability does not guarantee alignment of **intentions, goal priorities, and commitments**. Shared representations enable knowledge exchange, but they do not guarantee that agents will pursue compatible goals or make the same choice when goals are conflicting (i.e., when improving one goal detracts from another), for example, by prioritising safety over throughput. In multi-agent cooperation, this gap is precisely why the literature introduces explicit models of intentions, commitments, and joint activity (Cohen and Levesque, 1991; Rao and Georgeff, 1995; Jennings, 1995). These mechanisms go beyond shared meaning to establish how agents coordinate goals and responsibilities over time.

Third, semantic interoperability does not ensure alignment of **reasoning processes**. Even when agents share the same ontologies, they may rely on different internal models, assumptions, inference strategies, or constraint-handling methods, leading to different conclusions from the same knowledge base. This motivates the need for mechanisms that make reasoning *interpretable and alignable* across agents in collaborative settings. More broadly, collaboration requires building and maintaining common ground through interaction, not only conforming to static standards (Coiera, 2023).

Finally, semantic interoperability does not provide alignment of **confidence, uncertainty handling, and error monitoring**. Effective collaboration requires knowing *when* an agent is uncertain, when a recommendation is fragile, and when verification or delegation should occur. These metacognitive aspects (confidence calibration, uncertainty awareness, error detection) are key to safe and adaptive coordination, yet they are not implied by shared semantics (Fleming and Daw, 2017; Endsley and Jones, 2012).

In summary, semantic interoperability is a necessary foundation for exchanging knowledge consistently, but it does not ensure alignment of situation interpretation, intentions, reasoning processes, or confidence regulation during collaboration. This is why, in the next section, we turn to **cognitive interoperability**.

2.2.2 Terminological Landscape: A Bag-of-Words Analysis

This section aims to extract the main key elements of each definition in the form of two-word expressions, called ‘bag of words’ and analyse them to build a common generic definition of Cognitive Interoperability. Table 2.1 shows the corresponding references and the bag of words associated with each definition. These keywords are often linked to concepts such as thinking, understanding, mental models, human factors and state of mind, highlighting a common conceptual space.

References	Bag-of-words
(Goldkuhl, 2008)	Congruence of thoughts, human actors, understand information, aligning perceptions.
(Berthier, 2006)	Shared meanings, predefined extent, same situation.
(Vogt, 2023)	Human information, general cognitive, interact technology, interact information.
(Blad and Potts, 2003)	Unity of mindsets, mutual understanding, shared education, shared values.
(Kwon et al., 2011)	Share information, understand information, socio-cognitive aspects, joint decision-making.
(Raubal, 2005)	Sharing knowledge, mental models, sender receiver, complete understanding.
(Bytniewski et al., 2015)	Shared knowledge, adaptive behavior, collaborative decision-making.
(Myneni et al., 2016)	Cognitive processes, habitual thought, align system.
(Krinkin et al., 2023)	Shared knowledge, common ontologies, mutual understanding.

Table 2.1: Bag-of-Words Extracted from Definitions of Cognitive Interoperability with References

In this section, we provide a detailed explanation of each bag-of-words, which we have organised into five groups: understanding, perception, mental model, thought, and decision-making, as represented in Figure 2.2. In the following section, we will select the key functions from each group and compare them with those addressed by semantic interoperability, highlighting the additional capabilities required to achieve cognitive interoperability.

2.2.2.1 Group 1: Understanding

Understanding means the acquisition, organization, and appropriate use of knowledge to produce a response directed toward a goal, when that action is taken with awareness of its perceived

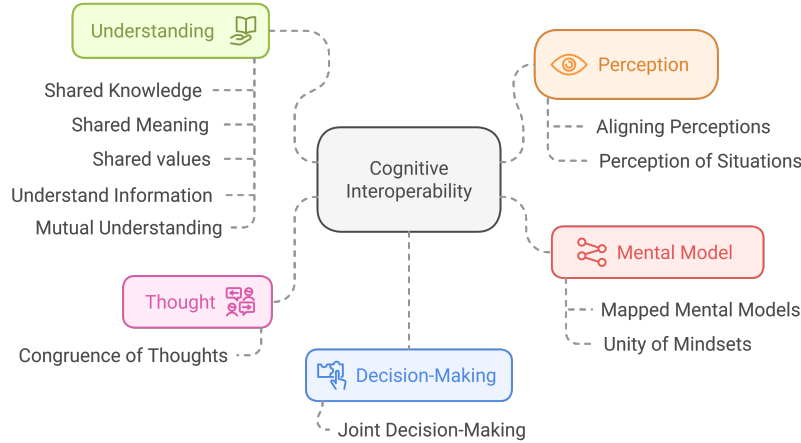


Figure 2.2: Key elements Contributing to Cognitive Interoperability

purpose (Hough and Gluck, 2019). In the context of agents representing digital systems, understanding can be defined as the system’s ability to reason about relationships between events or elements, to interpret and integrate multimodal information, and to generalise knowledge across diverse tasks and domains (Maetschke et al., 2021). This involves not only recognising patterns in data, but also building and applying composite and hierarchical knowledge representations to new scenarios, while effectively managing uncertainty and incomplete information.

The literature operationalises this group through more specific notions that capture how knowledge, meaning, and awareness are established and aligned between agents (Horton, 2012):

- **Shared knowledge** refers to the fact that two or more agents possess the same information, without requiring that they know that the others also possess it (Horton, 2012). In an industrial context, this can correspond to agents having access to the same facts about an object or situation (e.g., observations, measurements, specifications), even if they do not explicitly establish this commonality.
- **Shared meaning** refers to the alignment of interpretation, ensuring that the agents involved interpret a piece of information (or a term, signal, or observation) in a consistent way within the operational context. This is particularly critical when agents use different *views* of the same system: the same object may be conceptualised differently depending on the role, the level of abstraction, or the model used.
- **Shared values** correspond to common priorities or evaluation criteria that may be identified or established once meaning is sufficiently aligned. They provide a basis for consistent judgments and coordinated choices (e.g., safety-first vs. performance-first, minimal down-time vs. minimal cost).
- **Common understanding** refers to the fact that agents reach a compatible understanding of the same object or situation after processing and interpreting the available information (i.e., they converge toward the same meaning or conclusion), without necessarily being aware that they agree. For example, a driver and a maintenance engineer may refer to the same *engine* but rely on different conceptualisations: the driver may view the engine through observable effects (noise, vibrations, loss of power), while the engineer views it through subsystems and causal mechanisms (cooling circuit, combustion, lubrication, fault

modes). Common understanding is reached when they align these perspectives sufficiently to support the same interpretation (e.g., “this behaviour indicates a cooling failure”).

- **Mutual understanding** extends common understanding by adding *reciprocal awareness*: each agent recognises that the other has reached the same understanding and recognises that this agreement is itself recognised (Horton, 2012). In this sense, mutual understanding can be seen as *shared knowledge* combined with *common understanding* and a shared awareness of this alignment. It is defined as the process by which different minds mutually infer that they agree in their understanding of an object, person, place, event or idea (Stolk et al., 2016). Continuing the engine example, mutual understanding emerges when the driver and the engineer not only converge on the same interpretation, but also establish that they are indeed talking about the *same aspect* of the engine and that their perspectives are aligned (e.g., the driver’s symptoms correspond to the engineer’s diagnosed subsystem), enabling coordinated action such as deciding to stop, selecting a diagnostic procedure, or planning the maintenance intervention.

In summary, while **shared knowledge** concerns co-possessed information, **shared meaning** and **common understanding** concern aligned *interpretation*, and **mutual understanding** additionally requires *shared awareness* of this alignment, which is essential for coordinated action.

2.2.2.2 Group 2: Thoughts

In cognitive science, **thoughts** are generally understood to be mental representations or processes that arise, develop and evolve over time (Andrews-Hanna et al., 2022). They are often rooted in conceptual associations within the mind. Thoughts can be spontaneous, such as dreaming, or more deliberate, involving conscious reasoning and problem-solving. These conceptual associations constitute the basic elements of thought, enabling individuals to organise their experiences, make sense of the world, and recognise patterns. Indeed, **shared thoughts** refers to a collaborative intellectual process where two or more individuals work together to solve a problem, clarify a concept, evaluate an activity, or extend understanding. Shared thoughts represent a stock of common knowledge about a social object that is shared by members of a community (Epstein and Street, 2011).

In digital systems, the concept of thoughts is often mapped to discrete units of information, data, or computational processes that mimic or support human cognitive functions (Oosthuizen and Manzini, 2022). Furthermore, sharing thoughts and ideas refers to the exchange and distribution of information, ideas or content between users or systems, usually via online platforms or collaborative tools. This sharing is not only transactional, but often motivated by social and cognitive reasons, such as the desire for connection, personal expression or collective problem solving.

2.2.2.3 Group 3: Perception

Perception is the process by which an agent (biological or artificial) gathers raw sensory information from its environment in various forms (modalities), and processes them in order to derive meaningful information (features) (Efron, 1969) and understand the environment (Agrawal et al., 2023). In studies of perception, **congruence in perception** refers to the alignment, harmony or correspondence between the meanings or interpretations of various stimuli or components during the perceptual process (Laurienti et al., 2004). Within this framework, semantic congruence plays an essential role, as it determines the extent to which the meanings of different sensory

inputs align to form a coherent and unified representation by allowing the perceptual system to decide whether stimuli from different modalities should be linked or processed separately (Laurienti et al., 2004). This process ensures that the perception of an environment is not only unified, but also logical and consistent with previous experiences.

In digital systems, aligning perception in digital systems involves synchronizing the system's interpretation of sensory or user data with human expectations or with other systems for effective interaction (Chen et al., 2024). This can be achieved by developing unified frameworks (e.g., SYNERGAI) that enable both the robot and the human to share and align their perceptual models, enhancing collaboration and reducing misunderstandings (Chen et al., 2024).

2.2.2.4 Group 4: Decision-Making

In cognitive science, human decision-making is generally defined as the cognitive process of evaluating information, considering possible alternatives and selecting a course of action based on goals, preferences and constraints. This process is influenced by memory, emotions, heuristics and contextual awareness, and occurs in both routine scenarios and complex ones requiring reflection and judgement (Kahneman, 2011). By contrast, decision-making in artificial agents refers to the mechanism by which an autonomous system selects from available actions to achieve its goals based on its perception of the environment, coded policies, or reasoning procedures. Russell and Norvig define it as “the process by which an agent selects a course of action from among several alternatives based on its internal goals, knowledge, and perceived state of the environment” (Russell and Norvig, 2021). For joint decision-making, in humans, is a socially embedded process in which two or more people coordinate their reasoning to reach a collective decision or agreement. This coordination involves negotiation, the exchange of views and the inference of intentions, often resulting in compromise. These processes are shaped by mutual understanding and a shared context (Sebanz et al., 2006). Participants must not only reason individually, but also integrate the views and preferences of others. In artificial agents or digital systems, joint decision-making refers to the collaborative deliberation of multiple agents, whether human or artificial, through mechanisms such as consensus algorithms, argumentation frameworks, or belief fusion. The aim of these systems is to align the beliefs, goals or strategies of the agents in order to produce a coordinated plan of action (Wooldridge, 2009).

2.2.2.5 Group 5: Mental Models

Mental model mapping, often referred to as cognitive mapping or mental mapping, is the process by which individuals create internal representations of external reality. These mental models help people perceive, navigate, and interact with their environment, integrating both subjective experiences and objective facts. Cognitive maps are dynamic, flexible, and unique to each person, shaped by prior experiences, motivations, and abilities (Smelser and Baltes, 2001). They serve as schemas or frameworks for organizing spatial and conceptual information, supporting problem-solving, and decision-making. In cognitive science, mapping these mental models allows researchers to understand how individuals process, store, and recall information about their surroundings and experiences. It is a technique for analysing and resolving difficulties in decision-making processes (Kolkman et al., 2005). These difficulties are often due to differences in perspectives, knowledge, assumptions and subjective beliefs between different stakeholders.

In digital systems, mental model mapping involves designing interfaces, information architectures and user experiences which align with how users mentally represent how systems should work.

Unity of mind refers to a shared cognitive orientation among individuals in a group, characterized by aligned thought processes, perspectives, and attitudes (Ratchford et al., 2021). This unity emerges when people communicate, negotiate, and adapt their internal representations to create a common framework or shared cognitive map.

In digital systems, unity of minds refers to the alignment of mental models between users and systems, or among multiple users collaborating within a digital environment. This is achieved by creating shared digital representations, collaborative tools, and synchronized interfaces that support common understanding and coordinated action. This concept, which indicates the alignment of mental frameworks, is directly linked to these key words: shared knowledge, mutual understanding, congruence of perceptions and congruence of thoughts.

2.3 Architecture for Cognitive Interoperability

Based on the previous analysis of existing definitions, this section explains *why* we retain a reduced set of keywords to characterise cognitive interoperability, and *how* these selected components interact in the framework proposed in Fig. 2.3. While the previous section provided detailed definitions for the concepts found in the literature (grouped in Fig. 2.2), the objective here is to justify the final selection and organise it into a coherent operational flow for human-CPS collaboration.

The bag-of-words in Table 2.1 and the five groups introduced previously (understanding, perception, mental model, thought, and decision-making) contain terms that are partially overlapping, and sometimes refer to similar mechanisms at different levels of granularity. To avoid redundancy and to make cognitive interoperability operationally assessable, we retain one *representative* component per essential cognitive mechanism that (i) appears recurrently across definitions, (ii) brings an additional requirement beyond what is addressed by semantic interoperability, and (iii) can be positioned within a functional interaction flow between agents.

(1) Alignment of perceptions and thoughts is retained because it captures the *dynamic* coordination required for cooperation in real time. Beyond agreeing on the meaning of exchanged messages, agents must converge on what to focus on in the current situation (e.g., which signals, events, objects, or changes are relevant) and on how these elements are organised into an interpretable situation *for reasoning and action*. This idea is central to alignment-based accounts of interaction, where coordination is achieved through progressive convergence not only in linguistic choices but also in conceptualisations and interpretations during communication and joint action (Rasenberg et al., 2020). In collaborative settings, such alignment is described as a mechanism that supports efficient coordination, reduces misunderstandings, and enables agents to adapt their interpretations as the task and context evolve (Aho et al., 2022).

(2) Mutual understanding is retained because it operationalises an interaction-level requirement that goes beyond common interpretation. Communication and coordination research distinguishes between merely having *shared knowledge* and achieving *mutual understanding* (or common ground), where agents additionally establish reciprocal awareness that they agree (Horton, 2012). In other words, agents may interpret information similarly without being able to rely on that alignment for prediction and coordination; mutual understanding is precisely the mechanism that makes alignment *actionable* in interaction. Conceptually, it can be described as the process by which agents mutually infer that they agree in their understanding of an object,

event, or idea (Stolk et al., 2016). This keyword is essential because the reciprocal-awareness property is directly linked to trust, predictability, and the ability to coordinate beyond literal statements in dynamic contexts.

(3) **Joint decision-making** is retained because collaboration ultimately requires convergence toward coordinated choices and actions. Even when agents exchange well-structured and interpretable knowledge, shared decisions typically require an additional collaborative process: synthesising distributed evidence, resolving ambiguities or conflicting interpretations, and negotiating priorities and constraints as circumstances evolve. The literature on human-system collaboration highlights that effective coordination depends not only on shared meaning but also on shared reasoning about the decision context (including uncertainties, intent, and justification), which supports agreement on *what to do next* rather than only agreement on *what information means* (Myneni et al., 2016). Joint decision-making therefore represents the functional outcome of the previous components: once perceptions/thought are aligned and mental models are mapped, agents can deliberate collectively and adapt decisions under change, providing a direct bridge to action and control in industrial settings.

(4) **Mapped mental models** are retained because they provide the structural basis for sustained collaboration, especially when agents rely on different viewpoints, levels of abstraction, or internal formalisms. The literature on team cognition and collaborative systems shows that coordination improves when agents maintain compatible internal representations of the task, the environment, and each other’s roles and expectations (Aho et al., 2022). However, cognitive interoperability does not require that agents hold *identical* mental models; rather, it requires that their models can be *related* (i.e., translated or matched) sufficiently to support prediction, explanation, and coordinated adaptation. This is why we use the term *mapped*: it emphasises correspondences between representations rather than strict equivalence. The same requirement is captured across the mental-model-related vocabulary identified in the literature (e.g., shared situation model, shared conceptual space, unity of minds), which stresses that a common conceptual structure supports transparency of assumptions and smoother coordination.

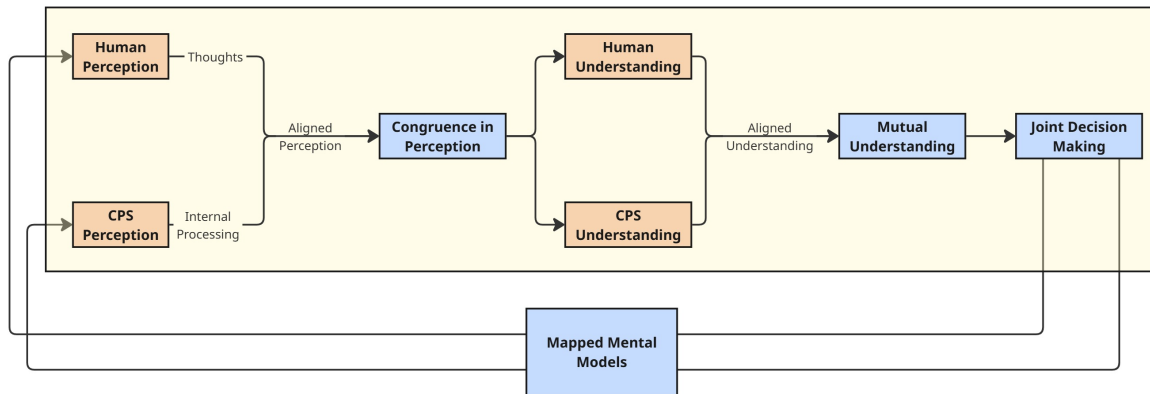


Figure 2.3: Functional Architecture for Achieving Cognitive Interoperability between Human and CPS Agents

Figure 2.3 organises these four components into a coherent flow. First, *alignment of per-*

ceptions and thoughts supports compatible interpretation of the ongoing situation and reduces divergences in salience and reasoning. In parallel and over time, *mapped mental models* provide the stable correspondences required to relate viewpoints and maintain consistent expectations across agents. These two mechanisms enable *mutual understanding*, where agents not only converge in interpretation but establish shared awareness that they agree. On this basis, agents can engage in *joint decision-making*, i.e., collectively reason, arbitrate between alternatives, and adapt choices under uncertainty and change.

Building on this analysis, we propose the following updated definition:

Definition: Cognitive Interoperability

Cognitive Interoperability refers to the ability of different agents (human or artificial) to align their thoughts and perception of information, allowing for mutual understanding and shared intentions. It involves creating shared mental models, aligning knowledge, and ensuring a common way to use, interpret and reason over knowledge.

Several aspects of cognitive interoperability such as mutual understanding have been explored in multi-agent systems through models of beliefs, desires, intentions (BDI) agents, teamwork, and joint intentions, which formalise goal coordination, commitments, and cooperative planning (Rao and Georgeff, 1995; Cohen and Levesque, 1991; Jennings, 1995). Here, we build on these foundations but focus on heterogeneous (human and/or artificial) industrial collaboration, where alignment must also cover perception, mapped mental models, and mutual understanding.

2.4 Maturity Model of Cognitive Interoperability (MMCI)

This section introduces a preliminary maturity model that operationalises cognitive interoperability as a progression from semantic alignment toward cognitive alignment in industrial collaboration. Building on the definition established in the previous section, the model organises the required capabilities into successive levels that are applicable to heterogeneous agents (human and/or CPS) interacting together.

2.4.1 Measurement foundations and design principles of the MMCI

The MMCI is proposed as an assessment-oriented maturity model: each level is defined by capabilities that are observable, testable, and measurable in principle. The goal is to characterise the degree of interoperability capacity between agents (human or artificial) and, more generally, any cognitive agent for which representative behavioural or operational data can be collected. Since this model is not yet experimentally validated, it should be considered as a *proposal to be tested* and refined through studies in targeted application contexts.

In psychology and neurocognitive research, constructs such as situation awareness, executive control, and metacognition are evaluated through behavioural measures and well-established task families, because internal cognitive states are not directly observable (Endsley, 1995; Diamond, 2013). Importantly, this measurement logic is also used in comparative cognition: researchers assess cognitive capacities in non-verbal agents (infants, primates, animals) by designing paradigms where behaviour reveals what the agent represents or predicts. For example, studies on great apes have used gaze-anticipation paradigms to test whether agents predict the actions of others

based on their beliefs (Krupenye et al., 2016). This illustrates that cognitive levels can be operationalised using observable indicators that remain meaningful even when agents differ in terms of embodiment, language, or internal architecture.

For artificial agents, the same principle applies: we assess maturity through operational evidence that agents can align state beliefs, task models, goals/intentions, and uncertainty regulation during interaction. The MMCI therefore uses indicators that can be computed from interaction traces (messages, state estimates, plan revisions, uncertainty outputs, decision logs). This makes the model applicable to heterogeneous collaborations (human-CPS, CPS-CPS, or broader multi-agent systems), while remaining agnostic to whether an agent implements a cognitive architecture, a multi-agent coordination protocol, or a learned policy.

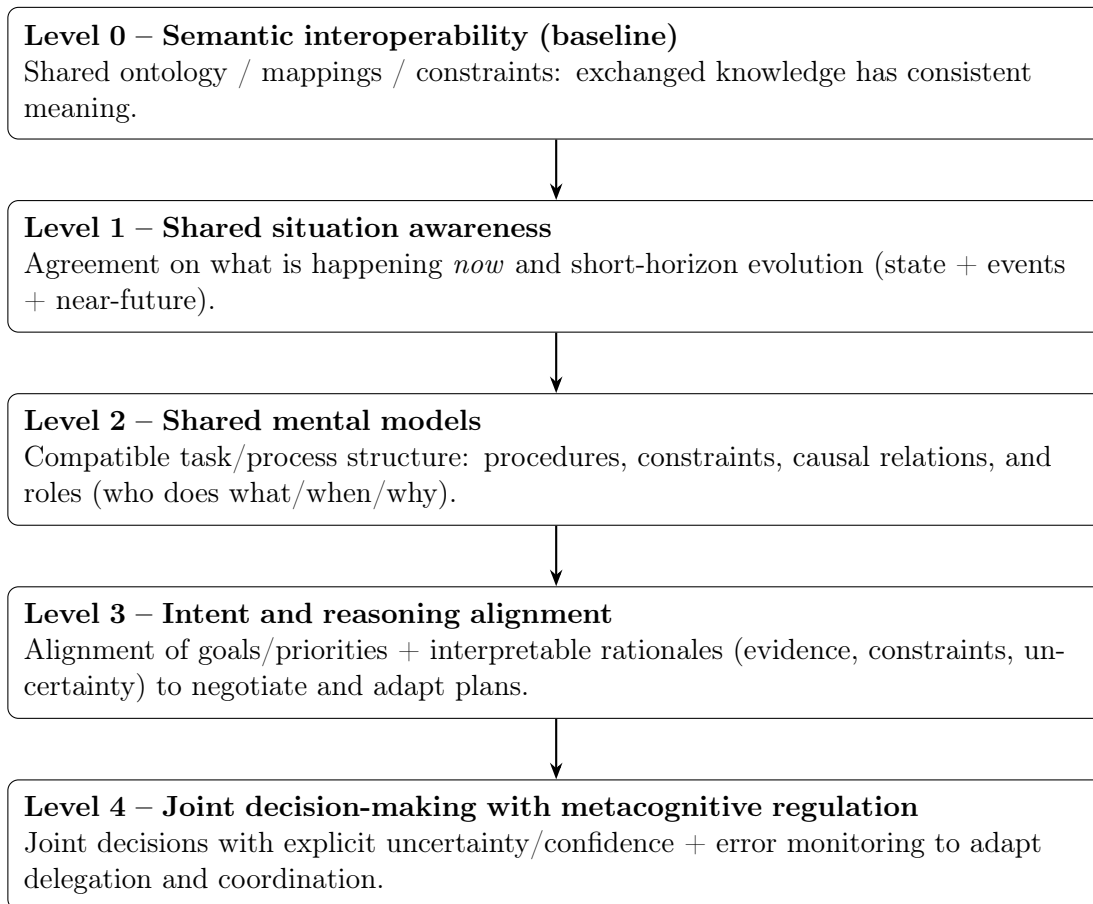


Figure 2.4: MMCI schema: progression from semantic interoperability (Level 0) to cognitive interoperability levels (Levels 1–4).

Level 0 is defined as the *maximum semantic interoperability* achievable (shared ontologies, mappings, constraints). Levels 1–4 progressively add cognitive alignment capabilities that semantic interoperability does not guarantee (see Fig. 2.4): (i) shared situation awareness, (ii) shared/mapped mental models, (iii) intent and reasoning alignment, and (iv) metacognitive regulation for joint decision-making. Each level is specified in terms of measurable characteristics, and examples are provided only as illustrations.

2.4.2 Level 0: Semantic Interoperability

At Level 0, we assume the strongest feasible semantic interoperability in an industrial ecosystem: agents share (or can reliably map to) a reference information model and shared ontology, including units, constraints, identifiers, and validation rules. Meaning is preserved end-to-end for exchanged information (European Commission, 2017; Tolk and Muguira, 2003).

What Level 0 *maximises*:

- **Ontology-aligned meaning:** consistent interpretation of terms, relations, and constraints across agents (European Commission, 2017).
- **Semantic mapping and governance:** harmonisation of synonyms/homonyms, schema mappings, and unit conversions (Tolk and Muguira, 2003).
- **Machine-processable semantics:** rules and constraints enabling automated conformance checking on exchanged knowledge (European Commission, 2017).

What Level 0 *still does not guarantee*: Even with perfect semantic agreement, agents may still diverge on: (i) how they interpret the *current* situation (situation awareness requires perception-comprehension-rojection, not only meaning) (Endsley, 1995); (ii) their intentions and priorities (semantic alignment does not impose joint goals or commitments) (Cohen and Levesque, 1991); (iii) their reasoning pathways (different inference/assumptions can yield different conclusions); and (iv) their confidence and error monitoring (uncertainty handling and calibration are not implied by semantics) (Fleming and Daw, 2017).

2.4.3 Level 1: Shared Situation Awareness

Level 1 is reached when agents **agree on the current situation** and on what will likely happen in the next moments. In practice, they align: (i) what they detect as relevant in the environment, (ii) what this means for the task, and (iii) a short-term prediction of the next state (Endsley, 1995). This is the first level of cognitive interoperability because it goes beyond shared meaning: agents must share a time-dependent view of what is happening now (Endsley and Jones, 2012).

Why this level is needed? Semantic interoperability can ensure that agents interpret terms consistently, but it does not ensure that they interpret the *current state* in the same way. Level 1 adds a measurable requirement: alignment of state interpretation and short-horizon prediction.

Measurable characteristics.

- **State agreement:** agents converge on key state variables within a defined tolerance.
- **Event agreement:** agents classify important events in the same way (e.g., normal vs anomaly).
- **Short-term prediction agreement:** agents produce similar near-future predictions within a defined threshold.

As an example, during a collaborative insertion task, both the operator and the robot agree that the part is misaligned *now* and that continued operation will likely cause a blockage; both pause and re-align before proceeding.

2.4.4 Level 2: Shared Mental Models

Level 2 is reached when agents share (or can map) a **common structured view of the task and system**. This includes: task steps, constraints, cause-effect relations, and role distribution (who is responsible for what and when). This level supports smoother coordination because agents can anticipate each other's actions based on the shared structure (DeChurch and Mesmer-Magnus, 2010).

Why this level is needed? Even if agents agree on the current situation (Level 1), they may still fail if they do not share the same understanding of the procedure, constraints, or responsibilities. Level 2 adds alignment of the *task structure* that guides coordination (DeChurch and Mesmer-Magnus, 2010).

Measurable characteristics.

- **Procedure consistency:** agents produce compatible expected next steps from the same task state.
- **Role consistency:** agents agree on responsibilities for checks and handoffs.
- **Constraint/causal consistency:** agents agree on which constraints apply and what actions will cause.

As an example, in a shared screwing task, both agents follow the same plan: the robot holds the part steady, the operator performs torque verification, and both agree that the next step is allowed only after verification.

2.4.5 Level 3: Intent and Reasoning Alignment

Level 3 is reached when agents align **goals and priorities** and can also understand each other's **reasoning**. This means: they agree on the active goal (and sub-goals), and they can exchange explanations about decisions using evidence, constraints, and uncertainty. This helps them negotiate when there is ambiguity or when the situation changes (Diamond, 2013).

Why this level is needed? With shared mental models (Level 2), agents can coordinate routine tasks. But when several actions are possible, or when priorities change, they need to align *why* they choose an action. Level 3 adds interpretable reasoning and goal/priorities alignment, so adaptations remain consistent.

Measurable characteristics.

- **Goal alignment:** agents converge on the active goal/sub-goal and priority policy.
- **Explanation usability:** explanations are clear enough that the partner can update its plan accordingly.
- **Consistent adaptation:** when conditions change, agents revise plans in compatible ways (low conflict rate, bounded repair time).

As an example, the operator starts preparing a tool change (goal shift). The robot detects this and adapts by retracting and presenting the tool tray. The robot also explains its change (workspace constraint and safety distance), so the operator understands the new sequence (Kekana et al., 2025; Zhou et al., 2022).

2.4.6 Level 4: Joint Decision-Making with Metacognitive Regulation

Level 4 is reached when agents make decisions together **and** regulate collaboration using explicit **uncertainty, confidence, and error monitoring**. Agents can decide when to act, when to ask for confirmation, and when to transfer control based on risk and confidence (Fleming and Dolan, 2012; Fleming and Daw, 2017).

Why this level is needed. Even with goal and reasoning alignment (Level 3), collaboration may fail under high uncertainty, rare events, or conflicting evidence. Level 4 adds a measurable capability: agents expose uncertainty, calibrate confidence, and adapt delegation to avoid overconfident or unsafe actions (Fleming and Daw, 2017).

Measurable characteristics.

- **Explicit confidence/uncertainty:** decisions include externally available confidence or uncertainty values.
- **Calibration:** confidence is consistent with real performance (bounded calibration error).
- **Error handling:** agents detect likely errors and trigger correction/verification actions.
- **Adaptive delegation:** role switching and confirmation policies follow clear risk/uncertainty rules.

Example (HRC). *Example:* during insertion, the robot detects abnormal force signals but reports low confidence in its diagnosis; it pauses and requests operator verification. If the operator is overloaded, the robot temporarily handles low-risk subtasks while keeping explicit confidence reporting and clear rollback conditions (Fleming and Dolan, 2012; Fleming and Daw, 2017).

2.5 Chapter Conclusion

This chapter contributes to **RQ1** and **RQ2**. For **RQ1**, we develop the conceptual foundations required to study and engineer cognitive interoperability in industrial environments. For **RQ2**, we clarify what is meant by cognition in industrial systems, defines the associated cognitive capabilities, and selects the subset of functions retained as essential for industrial cognitive systems. Cognition is formalised as a system-level phenomenon emerging from the interaction of multiple functions, including perception, memory, attention, reasoning, learning, and metacognition, and this abstraction is used to define cognitive systems, CCPS, and CDTs in a consistent and operational manner.

We then introduces cognitive interoperability as a necessary extension of technical and semantic interoperability for collaboration between humans and CPS. By analysing existing definitions and terminology, it identifies key cognitive dimensions, including perception alignment, mutual understanding, shared mental models, joint decision-making, and shared intentions. These elements are integrated into a functional architecture describing how cognitive alignment emerges from interactions between human and artificial agents.

Based on these foundations, we proposes a formal definition of cognitive interoperability. This definition explicitly positions cognitive interoperability as a multi-agent phenomenon that

supports coordinated, adaptive, and goal-directed collaboration.

Finally, we introduce the Cognitive Interoperability Maturity Model (MMCI), which structures the progression from semantic interoperability (Level 0) to increasingly advanced forms of cognitive alignment. Each level in the model is characterised by specific measurable capabilities, implementation mechanisms, and operational indicators. This provides both a theoretical basis and an operational framework for assessing and designing cognitive interoperability in industrial contexts.

A key perspective of this thesis is to validate and refine this maturity model through empirical evaluation and additional industrial scenarios, and to extend it by specifying additional metrics, measurable indicators, and assessment procedures that enable a robust and repeatable evaluation of cognitive interoperability at each maturity level.

Next, Chapter 3 moves from concepts to design. It presents the architecture of the CDT and explains how the selected cognitive functions are operationalised through representation mechanisms and computational components (e.g., knowledge structures, reasoning and learning modules) to support cognitive interoperability in HCC.

Chapter 3

Architecture for the Cognitive Digital Twin

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Introduction

This chapter supports **RQ2** by focusing on the detailed design of the Cognitive Digital Twin (CDT), paying particular attention to the implementation of its cognitive capabilities. First, it revisits the three core functions of a CDT: simulation, emulation and cognition, and clarifies their respective roles within an integrated CDT architecture. While simulation and emulation provide predictive and real-time representations of the physical system, cognition enables the

CDT to analyse and act upon these representations in context. In our work, cognition is not treated as an auxiliary component, but as a core architectural function that organises perception, interpretation, learning, and decision-making within the CDT. This design choice ensures that cognition directly mediates between system representations and action, rather than being applied as a post-processing layer.

The chapter then contributes to **RQ4**, focusing on the cognitive function of the CDT, A comparative analysis of existing cognitive architectures is presented to identify their respective strengths and limitations in relation to CDT applications. Based on this analysis, one cognitive architecture is selected and described in detail. Finally, the chapter demonstrates how this architecture is integrated into the CDT by explaining its components, information flows, and interactions with the simulation, emulation, and knowledge representation mechanisms.

3.1 Cognitive Digital Twin Framework

This section builds on the state of the art presented in Section 1.2.4.2 and its analysis. It also relies on the definition of a CDT introduced in Section 2.1.3: a “*CDT is a DT and a cognitive system. As a DT it emulates a physical system that can be or not itself cognitive. As a cognitive system it possesses cognitive functions, bringing it in particular the ability to semantically model, process and interpret information autonomously and actively learning from its interactions*” (Al Haj Ali et al., 2024a). Here, we clarify how cognition in CDTs complements the classical DT functions of emulation and simulation. The approach is presented here from the CDT perspective, the same principles are general and can also be applied to other types of cognitive twins (e.g., HDTs).

This dual characterisation of a CDT as both a digital replica and a cognitive system raises a central question for CDT design: **where and how should cognition be placed within the Digital Twin so that it effectively supports the physical system?**

Although many recent works refer to “cognitive” extensions of Digital Twins, they often do not specify clearly which components implement cognition, how cognition is separated from other DT components, and how the different modules should interact to produce coherent behaviour. This lack of a precise structure makes it difficult to compare approaches and to guide implementation choices.

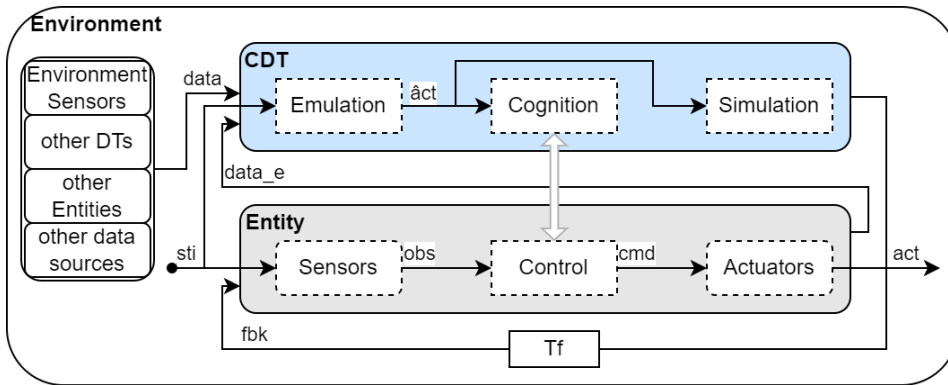


Figure 3.1: Entity-CDT Architecture. Entity here can be a CPS or a CCPS (Naudet et al., 2024).

This issue is addressed in our recent work (Naudet et al., 2024), where we propose the Entity CDT architecture to provide a clear and implementable organisation of a CDT. As shown in Figure 3.1, the architecture defines explicit processing layers and information flows, making it possible to locate cognitive mechanisms within the CDT and to specify how they interact with the other CDT components.

An entity is conceptualised as an agent composed of sensors, a control system and actuators, which enable it to perceive its environment (**obs**), transform observations into control decisions (**cmd**) and execute physical actions (**act**). These actions are responses to external stimuli (**sti**) and can in turn influence other entities or environmental conditions. The CDT receives structured data (**data_e**) from its physical counterpart, including sensor outputs, command notifications and internal state data, and can also access larger environmental data (**data**), such as that generated by other entities, DTs or information systems. The three essential functions of a CDT are detailed in the following subsections.

3.1.1 Emulation Function

The emulation function is introduced in Section 1.3.1; this subsection details its role and implementation within the CDT. The CDT emulation function computes an estimate of the response (action) of the entity **act** to a stimulus **sti**. This input may be supplemented by data provided by the entity **data_e**, which may include, in particular, its own observations **obs** and notification of commands sent **cmd**, as explained above. In the case of a CPS, it can be assumed that the observations of its sensors can be automatically transmitted to the DT. In the case of a human, the transmission would be indirect via a Human-Machine Interface (HMI), subject to specific interpretation by the human: it cannot be raw data from the human senses, except via a dedicated external sensor (e.g. ECG, EEG, oculometer, etc.). The DT then retrieves all useful data from its or the entity's environment: from sensors and other entities in the entity's environment, from other DTs, and from any other useful source of information. From this data, the DT's emulation function estimates the state of the entity and the resulting action **act** in response to the stimulus **sti**, using entity models built from the acquired, a priori known or learned data. This function enables synchronisation between the physical and digital layers.

3.1.2 Cognition Function

The cognition function is introduced in Section 1.3.2; this subsection details its role and implementation within the CDT. The CDT cognition function takes as input **act**, the estimated response of the entity by the emulation function, and data from the environment (**data**) and the entity (**data_e**) to perform any cognitive processing that is useful to the system. This function is essential to enable autonomous behaviour, context-sensitive reasoning and adaptation to unexpected situations. Any form of cognition can be implemented here, including learning, anomaly detection, symbolic reasoning, or goal-based decision making. A common objective is to detect deviations between the observed reality and the estimated state or action of the entity, and to reason about these gaps in order to derive possible explanations or corrective actions.

Cognition in CDTs can be implemented using two main strategies, as illustrated in Figure 3.2. The first approach (**1**) aims to *provide cognitive capabilities to a non-cognitive physical entity* (e.g. a standard CPS), thereby transforming it into a CCPS. In this configuration, a **strong coupling** is established via a bidirectional link between the cognition function of the CDT and

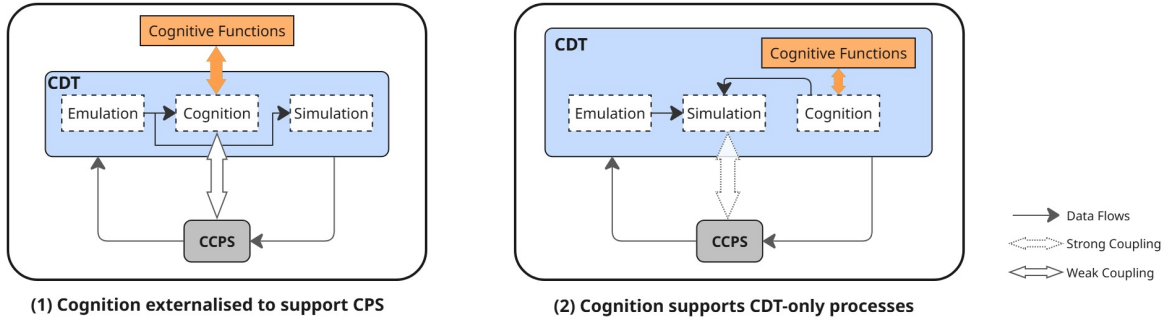


Figure 3.2: Two implementation strategies for cognitive functions within CDTs. **(1)**: Cognition is externalized to support a CCPS via strong coupling. **(2)**: Cognition is internal to the CDT and supports only its own functions, with weak coupling to the physical system.

the controller of the entity, effectively transforming the CDT into a *cognitive controller*. This allows the CPS to benefit directly from the reasoning and decision-making capabilities housed in the CDT. The second strategy **(2)**, more commonly found in the literature, involves implementing cognition as an *internal service* that supports the CDT's own operations, in particular, model simulation, analysis and optimisation. Here, the coupling between the CDT and the physical system is **weak**, as the cognitive functions remain within the CDT and do not directly control or enhance the CPS. Nevertheless, this configuration enhances the ability of the CDT to generate refined predictions, contextual recommendations or adaptive feedback, thus contributing to the overall intelligence of the system without changing the behaviour of the physical entity itself.

3.1.3 Simulation Function

The emulation function is introduced in Section 1.3.3; this subsection details its role and implementation within the CDT. The simulation function allows the system to explore possible future situations and test how the entity might behave in those situations. This function is particularly useful when the system is confronted with situations that have not been planned or programmed in advance. It can be activated during operation, for example when something unusual happens, or in parallel with operation to prepare for situations that may occur in the future. While the emulation function shows how the system is currently behaving, the simulation function helps the CDT to anticipate and prepare for what might happen. To do this, the CDT needs a model not only of the entity to which it is connected, but also of the environment around it, i.e. the other systems, variables and constraints that are likely to affect its behaviour. In a CPE, this might mean simulating a complete production cell, with all the machines and conditions that interact. Because each entity has its own CDT, it is easier to simulate how all parts of the system work together. Each CDT uses its own models to respond to a common situation and achieve its own goals.

The simulation starts with a defined situation, such as a problem or scenario to be solved. The simulation engine sends a version of this situation to all the CDTs involved. Each CDT first uses its emulation function to estimate how the real system would react. Then the simulation function adjusts or improves this action using the CDT's internal logic, such as optimisation rules or learning-based methods. In our work, we use RL to help the CDT explore better ways of acting (Section 5.4.3). This method allows the CDT to try different strategies and learn the one that

works best over time, using the same behavioural models already used in the emulation function (this approach has been selected on the basis of a comparison detailed in section 1.3.3). This approach offers more flexibility than other simpler models such as BDI (Belief-Desire-Intention). At each stage of the simulation, the simulation function generates an improved action based on the system's knowledge and objectives. The simulation engine collects the actions of all the CDTs, checks whether the global objectives and constraints are respected and helps the system to evolve towards a common solution.

3.2 Cognitive Architectures for Cognitive Digital Twins

The focus is on the cognition function of CDTs, given that the emulation and simulation functions are already extensively covered in the existing literature on DTs. Therefore, given the state of the art in technologies that enable cognitive functions in Section 1.3.2, we are focusing on cognitive architectures as a necessary approach to implementing cognition in CDTs when the objective is cognitive interoperability. Unlike isolated AI components, cognitive architectures provide an explicit organisational model of cognition, enabling coherent integration of perception, memory, learning, reasoning, and decision-making within a single system. The following sections compare several cognitive architectures with respect to the cognitive functions they support; this analysis leads to the selection of CLARION, which is then presented and detailed to support its integration within the proposed CDT framework.

3.2.1 Overview of Cognitive Architectures

As outlined in the state-of-the-art section (Section 1.3.4), cognitive architectures have been the subject of numerous studies in the fields of AI and cognitive science, leading to the identification of several key categories based on their unique representation methods and learning processes.

The integration of a cognitive architecture for CDTs becomes important for systems that display cognitive behaviour themselves (e.g. human operators). Indeed, if a real-world entity involves complex decision making (a human operator, an intelligent machine, etc.), its CDT must possess cognitive characteristics to faithfully mimic its behaviour. Several categories of cognitive architectures can fulfil this role, depending on the expected cognitive functions, the required level of explainability, and the type of learning and adaptation needed, and each offering different advantages:

- Symbolic cognitive architectures are preferred when a CDT needs to follow human-like problem-solving steps or explain its reasoning, as this type of architecture models cognition using symbolic representations and rule-based processing, closely mirroring human logical reasoning and declarative knowledge structures.
- Subsymbolic architectures, often based on neural networks or deep learning, abandon explicit symbolic reasoning and instead learn cognitive behaviours from data. These emerging models perform well at pattern recognition and adaptation from large amounts of data, which is well suited for CDTs that need to process high-volume sensor flows or learn complex patterns.
- Hybrid architectures that integrate both symbolic and sub-symbolic components, with the aim of capturing low-level learning and high-level reasoning in a single framework, are particularly interesting for CDTs, as they allow abstract reasoning while continuing to

learn from raw data streams, thus balancing the interpretability of symbolic models with the adaptability of neural networks.

Table 3.1 summarises the main differences between cognitive architectures and sub-symbolic, data-driven AI approaches, highlighting when each is more suitable depending on the type of reasoning required in the CDT.

Criterion	Cognitive Architectures	Data-driven AI
Main purpose	Integrates multiple cognitive functions (perception, memory, reasoning, learning, decision-making) into one coherent agent.	Performs a specific cognitive task (e.g., prediction, classification, control) with high accuracy.
Nature of cognition	Human-like, interpretable, and context-aware cognition.	Task-oriented, data-driven cognition focused on optimization.
Learning mechanism	Combines symbolic reasoning and adaptive learning (hybrid) or emergent learning (neural).	Primarily statistical or deep learning from large datasets.
Scope of application	Broad: needed when several cognitive capabilities must work together or when human-like cognition is required, such as in HRC or HDT–CDT interaction.	Narrow: suited for isolated functions (e.g., prediction, control, or classification) where human-like reasoning is not required.
Explainability	High (especially in symbolic or hybrid architectures).	Often limited (black-box models).
Adaptability	Learns and adapts continuously across multiple tasks.	Learns per task; retraining often needed for new conditions.
Complexity and development effort	Higher: requires modelling multiple cognitive processes and integration.	Lower: faster to deploy for a well-defined task.
Example use case	HRC, decision support, autonomous reasoning, HDT–CDT interaction.	Fault prediction, anomaly detection, optimization.

Table 3.1: Comparison between Cognitive Architectures and AI methods for CDTs.

Regarding table 3.1, the choice between a full cognitive architecture and another AI approach in a CDT depends on the requirements of the scenario. Based on the characteristics of cognitive architectures, we can conclude that they perform well in scenarios where a wide range of cognitive functions need to work together. They provide an integrated agent with perception, memory, decision making and learning subsystems, which is ideal when the CDT needs to mimic complex problem solving processes or human-like responses, as is the case with an HDT. If we tried to recreate this cognitive scope without an architecture, we would end up assembling many disparate models: one for anomaly detection, one for decision logic, one for planning learning updates, etc., which can be fragile and difficult to manage.

On the other hand, other cognitive models and techniques may be more appropriate when the cognitive capabilities required from the twin is limited or when predictive performance on a specific task is the primary objective. For example, if the primary function of a CDT is predictive analytics to predict the remaining lifetime of a machine, a DL model or a statistical ML model could perform this specific function with greater accuracy and less development effort than implementing a full cognitive architecture. In such cases, the twin can still be considered “cognitive” in the broad sense, as it has an acquired capability, but it does not require a comprehensive architecture coordinating multiple cognitive capabilities. The CDT can also be based on commonly understood decision rules. In a scenario where domain knowledge can be explicitly encoded, a rule-based engine may be sufficient to guide the actions of the CDT without the overload of a cognitive architecture.

3.2.2 Comparison of Cognitive Architectures and Selection of CLARION

The CDT supports a collaborative interaction between a human and a CCPS (for example, a collaborative robot), which imposes specific constraints in terms of perception, interpretation, and decision-making. Listed below are the key requirements for our application case:

- Ability to perceive and understand the environment: The architecture must enable CCPS to analyze sensory data to understand their environment in real time.
- Interpretation of intentions: It must provide the tools to interpret the intentions and actions of collaboration partners, whether human or CCPS, to enable fluid interaction.
- Intelligent decision-making: It should facilitate autonomous decision-making based on perceived and interpreted data, enabling CCPS to choose the most appropriate actions to achieve the common goal.
- Computational efficiency: It must offer sufficient computational efficiency to enable rapid information processing, essential for real-time interactions in a dynamic environments.
- Flexibility: It should be flexible enough to adapt to different scenarios.

To identify the architecture that satisfies these requirements, the following section proposes a detailed comparison between different cognitive architectures.

3.2.2.1 Comparative Analysis of Cognitive Architectures

Based on (Kotseruba and Tsotsos, 2020), which provides a large-scale review of cognitive architectures (84 architectures) and compares their coverage of fundamental cognitive functions,

we limit our study to four candidates: **SOAR**, **ACT-R**, **LIDA**, and **CLARION**. We select a small subset that (i) offers broad coverage of cognitive functions, (ii) is representative of different architectural paradigms (symbolic and hybrid), and (iii) has been documented for use or discussion in collaborative contexts.

In what follows, we present a detailed comparison of these four architectures, focusing on their performance in each cognitive function selected and defined in section 2.1.4.

Perception: We compare the selected cognitive architectures according to two main sensory modalities: vision and multimodal perception.

- **Vision:**

Table 3.2 represents a comparison of the visual processing capabilities of selected cognitive architectures, providing an overview of their performance in different sensory modalities. *Real Vision* refers to the architecture’s ability to process visual data from physical sensors. *Simulated Vision* refers to how the architecture handles visual data in a simulated environment. *Stages of Visual Processing* represents the number of visual processing steps the architecture can perform: 1. Feature detection. 2. Grouping into objects. 3. Object labeling. 4. Building spatial relationships. 5. Contextualization. 6. Deep scene interpretation.

Architecture	Real Vision	Simulated Vision	Stages of visual processing
SOAR	Partial	Basic	1-2
ACT-R	Partial	Advanced	1-4
LIDA	Advanced	Advanced	1-6
CLARION	Partial	Basic	1-3

Table 3.2: Comparison of Visual Processing Capabilities in Different Architectures

For ACT-R, advanced visual processing covers from initial detection of visual features to contextual interpretation, spanning four process steps. This indicates a thorough integration of visual perception, from the recognition of edges and colors to the contextualization of objects in their spatial environment. SOAR and CLARION, while also processing visual information, are limited to initial stages, focusing more on identifying and grouping features into objects without reaching the complexity of spatial relationships or contextual interpretation. LIDA stands out for its ability to perform complete visual processing, including additional steps such as the construction of spatial relationships and advanced contextualization, enabling a rich and nuanced understanding of the visual scene.

- **Multi-modal perception:**

Table 3.3 highlights the multi-modal perceptual processing capabilities of different cognitive architectures. *Integration of sensory modalities* measures the architecture’s ability to combine information from different sensory modalities, such as sight and hearing, to

create a unified perception. *Cross-modal interaction* evaluates whether the architecture can handle interactions between different sensory modalities.

Architecture	Integration of sensory modalities	Cross-modal interaction
SOAR	Basic	Limited
ACT-R	Advanced	Limited
LIDA	Advanced	Yes
CLARION	Partial	Limited

Table 3.3: Comparison of multi-modal perception capabilities in different architectures

Table 3.3 shows that ACT-R and LIDA are advanced in the integration of sensory modalities, with LIDA also distinguished by its ability to enable cross-modal interactions. SOAR and CLARION offer basic sensory integration with limited cross-modal interactions.

Attention: ACT-R and LIDA stand out for their advanced suppression, selection and restriction mechanisms, indicating a superior ability to handle information overload and focus on relevant elements in complex visual scenes. These systems can implement these attention mechanisms in both real and simulated vision contexts. Conversely, SOAR and CLARION have basic capabilities in these same areas.

Action selection: ACT-R takes a moderate approach to decision flexibility through rule-based assessment. SOAR, on the other hand, shows great flexibility thanks to its hierarchical learning. LIDA and CLARION stand out for their very high flexibility and the integration of emotions in decision-making, with CLARION adding mixed symbolic-connectionist learning.

Memory: Memory is described in terms of duration (short and long term) and type (procedural, declarative, semantic, etc.). Long-term storage is subdivided into semantic, procedural and episodic types, which respectively store factual knowledge, information about actions to be taken under certain conditions, and episodes of the system’s personal experience. Short-term storage is divided into sensory and working memory. Sensory or perceptual memory is a very short-term buffer that stores many recent percepts. Working memory is a temporary storage of percepts that also contains other elements related to the current task, and is often associated with the current focus of attention. ACT-R and SOAR share a similar approach, relying on active working memory and long-term memory that combines declarative and procedural elements. LIDA stands out with perceptual sensory memory and episodic working memory, indicating an ability to maintain recent sensory information for more complex processing. CLARION, however, presents no sensory memory but focuses on a working memory model integrating both bottom-up and top-down processes, and a long-term memory that clearly distinguishes between implicit and explicit components.

Learning: Learning is divided into two categories: declarative and non-declarative knowledge acquisition, which includes perceptual, procedural, associative and non-associative types of learning (Squire, 1992). ACT-R stands out for its multi-functionality, with capabilities in declarative

learning via chunking and in procedural learning through the use of Explanation-Based Learning (EBL) (Minton et al., 1989), as well as in associative learning for action adjustment. SOAR, on the other hand, focuses primarily on procedural learning by creating and managing rules via EBL, with no specific indications for perceptual, declarative, associative or priming learning. LIDA is unique in asserting a perceptual learning capability and using an activation model for priming, suggesting a facility for preparing the system for efficient information processing. CLARION appears to focus on declarative and procedural learning, with the ability to extract and induce rules and knowledge, while recognizing the importance of priming in the form of positive and negative priming.

Reasoning: ACT-R relies on probabilistic and inductive reasoning, using rule-based inference to simulate long-term declarative memory that mirrors human behavior (Kotseruba and Tsotsos, 2020). This approach enables ACT-R to produce reasoning behaviors that closely match the results observed in human experimental studies (Nyamsuren and Taatgen, 2014). In contrast, CLARION encompasses a broader spectrum of reasoning processes by integrating symbolic and sub-symbolic mechanisms that enable it to model deductive, inductive, analogical and heuristic reasoning. This gives it the ability to capture a diversity of psychological phenomena beyond traditional reasoning frameworks (Hélie and Sun, 2014).

Metacognition: ACT-R applies a theory of mind model to a mobile robot, enabling it to recognize and adapt its behavior to changes in human activities. SOAR focuses on improving learning by building on previous experience and eliminating ineffective strategies. CLARION, on the other hand, implements metacognition by assessing confidence in task-related knowledge.

3.2.3 Criteria for Selecting a Cognitive Architecture for CDT

The objective is not to choose the most comprehensive architecture in general, but the one that best meets our constraints: integration into a CDT and support for HCC in an industrial context. Therefore, the selection is based on criteria that reflect both the functional needs of the CDT and the practical requirements of implementation.

1. **Coverage of core cognitive functions:** A CDT must be able to interpret information, reason about system behaviour, select actions, and learn from experience. These functions are necessary for supporting diagnosis, adaptation, and decision-making in dynamic environments. Architectures that lack one of these core capabilities cannot fully support the objectives of a CDT.
2. **Continuous and online operation:** A CDT operates in permanent interaction with a physical or simulated system. It receives continuous data flows and must update its internal state accordingly. Therefore, the architecture must support reasoning, online learning, and continuous perception-decision-action loops rather than only offline or episodic processing.
3. **Transparency and explainability:** In industrial contexts, system behaviour and decisions must often be understood, validated, and trusted by human operators or engineers. Architectures that provide explicit knowledge representations and interpretable reasoning processes are therefore preferable to purely opaque or black-box models.
4. **Support for learning and adaptation:** A CDT must be able to adapt to changes in the system, in the environment, and in operational conditions. This includes learning new patterns, updating models, and modifying decision strategies over time.

5. **Hybrid symbolic and sub-symbolic processing:** CDTs must combine data-driven learning (typically sub-symbolic) with explicit knowledge, rules, and models (symbolic) to support both adaptation and explainability. Hybrid architectures are therefore particularly suitable for CDTs.
6. **Practical implementation criteria:** Technical accessibility must be ensured, including the availability of source code and documentation, the possibility to modify and extend the architecture, and its current level of maintenance. In addition, the architecture must be easy to integrate into a modular CDT framework and should remain maintainable over the long term.

Table 3.4 summarises the comparison of the main candidate architectures with respect to these criteria.

Criterion	SOAR	ACT-R	LIDA	CLARION
Coverage of core cognitive functions	Medium	High	Very high	High
Online / continuous operation	Medium	Medium	High	High
Transparency and explainability	High	High	Medium	High
Support for learning and adaptation	Medium	High	High	High
Hybrid symbolic / sub-symbolic processing	No	No	Partial	Yes
Technical accessibility (code, docs, maintenance)	High	Medium	Low	High
Ease of integration into CDT framework	Medium	Medium	Low	High
Long-term maintainability	Medium	Medium	Low	High

Table 3.4: Comparison of candidate cognitive architectures with respect to CDT selection criteria

The table shows that LIDA scores very highly on cognitive processes and online operation. From a purely cognitive perspective, it is therefore an excellent candidate. However, it performs poorly with respect to technical accessibility, ease of integration, and long-term maintainability. The available implementations are outdated, poorly documented, and not actively maintained, which makes integration into a CDT framework difficult and risky.

The CLARION architecture was chosen for its modular, integrative framework, which captures essential cognitive processes through the interaction of multiple specialised subsystems. A central principle of CLARION is the distinction between implicit and explicit cognition, represented systematically across its main components. This dual structure enables the architecture to model both data-driven, subsymbolic processes and symbolic, interpretable knowledge within a unified framework. This makes it particularly well-suited to systems that require a combination of learning, reasoning and explanation.

Unlike many other cognitive architectures, CLARION includes mechanisms for motivation and metacognition, as well as perception, learning and action selection. This enables cognitive

systems to not only react to external inputs, but also generate internal goals, evaluate their own performance and regulate their internal processes. These capabilities are important for CDTs, which must operate autonomously, adapt to changing conditions and justify their behaviour in complex, dynamic environments.

Furthermore, CLARION supports both bottom-up learning, where explicit knowledge is extracted from experience, and top-down learning, where rules or instructions are internalised into implicit routines. This bidirectional learning mechanism enables the architecture to adapt to new situations while remaining consistent with domain knowledge and operational constraints.

3.3 CLARION Architecture for CDTs

From cognitive and engineering perspectives, CLARION provides a structured, explainable and adaptable basis for implementing the cognitive function of a CDT. The following subsections (Section 3.3.1) first describe the CLARION architecture and its main components and then explain how these components are integrated into the proposed CDT framework (Section 3.3.2).

3.3.1 Detailed Description of CLARION

CLARION is a cognitive architecture designed to model and implement cognitive processes in artificial agents. Introduced and refined in a series of works by Sun and colleagues (Sun, 2006), it has been applied in cognitive modelling and AI research.

CLARION comprises four interactive subsystems: the [Action-Centred Subsystem \(ACS\)](#), the [Non-Action-Centred Subsystem \(NACS\)](#), the [Motivational Subsystem \(MS\)](#), and the [Meta-Cognitive Subsystem \(MCS\)](#). Each subsystem contributes to a specific set of processes, and their interaction enables the architecture to support key cognitive functions within a single agent. Furthermore, CLARION conforms to the Common Model of Cognition (CMC), which describes cognition through functional components such as perception, working memory, long-term memory, action, learning, and cognitive control. Figure 3.3 shows the correspondence between these CMC functions and the subsystems and representation structures of CLARION.

Each of the four subsystems has a two-level internal structure. The lower level represents implicit knowledge in a distributed, subsymbolic form, while the upper level represents explicit knowledge in a symbolic, localist form. This organisation supports both automatic behaviour based on experience and deliberate behaviour based on explicit knowledge.

The subsystems are not independent. They continuously exchange information: action decisions depend on knowledge representations; learning modifies internal representations over time; and motivational and metacognitive processes influence perception, learning and action selection.

The following sections provide a detailed description of the four subsystems, with the explanations and algorithms summarised from Sun (2006). Firstly, the Action-Centred and Non-Action-Centred subsystems are presented. These are primarily responsible for action control and knowledge representation. Then, the Motivational and Meta-Cognitive subsystems are presented. These provide motivational guidance and regulatory control.

3.3.1.1 Action-Centred Subsystem (ACS)

The ACS has been developed to provide a model of an individual cognitive agent's action selection as it interacts with the environment. This interaction represents the agent's primary mechanism for controlling behaviour. The ACS is the core component of CLARION. The action decision

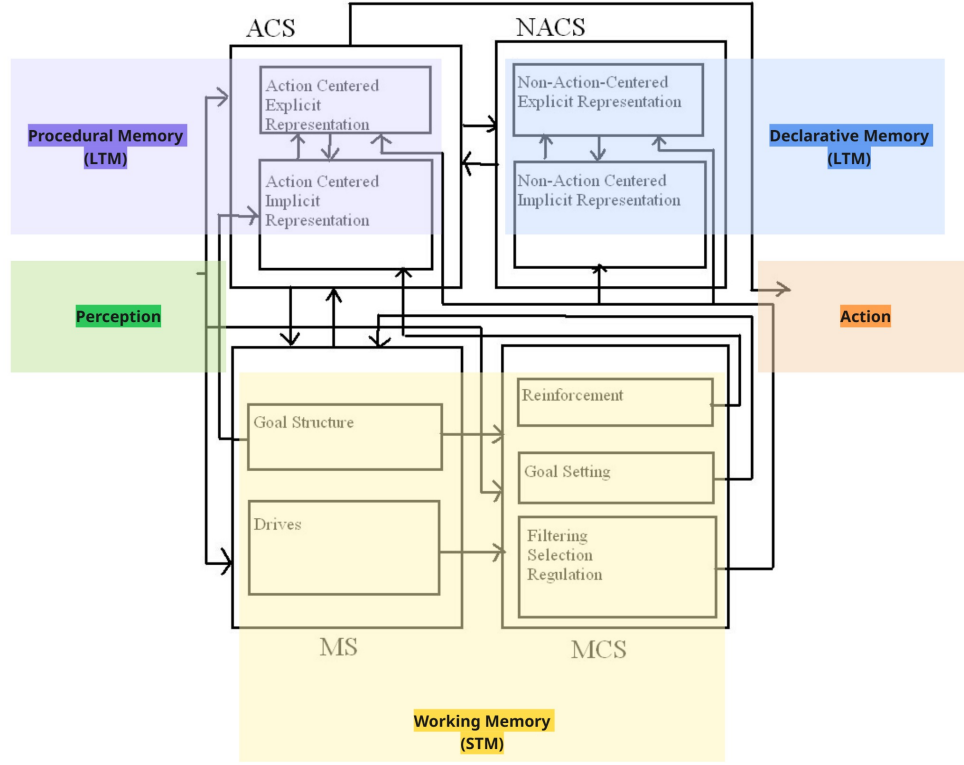


Figure 3.3: Mapping of CLARION to the Common Model of Cognition. Adapted from (Sun, 2007a).

process is initiated as follows: the agent first observes the current world state. Then, two processing levels inside the ACS (implicit and explicit) each independently arrive at an action decision based on their respective knowledge. These two outcomes are then combined in a stochastic manner to produce a final action choice, which is then executed. The action has a significant impact on the world. By comparing the new world state with the previous one, the agent learns Q-learning (Watkins and Dayan, 1992). The cycle then repeats.

Within the ACS, the lower level is the **Implicit Decision Networks (IDNs)**, implemented as neural networks with distributed representations. The upper level is the **Action Rule Store (ARS)**, implemented with symbolic representations.

Overall action decision algorithm

Algorithm 1 describes the recurrent interaction cycle of the ACS. The bottom level (IDNs) evaluates candidate actions through state-action values, whereas the top level (ARS) proposes actions supported by symbolic rules. A stochastic selection mechanism arbitrates between these two proposals to determine the executed action. Learning is performed after each transition: the IDNs are updated via reinforcement feedback, while the ARS is updated through rule extraction and refinement.

Bottom level (IDNs): inputs, outputs, and stochastic choice

The bottom-level input state x comprises three information sources: (1) sensory input, (2)

Algorithm 1 Overall algorithm for action decision making in the ACS

- 1: Observe the current state x .
 - 2: Compute in the bottom level (IDNs) the value of each possible action a_i associated with x :
 $Q(x, a_1), Q(x, a_2), \dots, Q(x, a_n)$.
 Stochastically choose one action according to these values.
 - 3: Find out all the possible actions $b_1, b_2, \dots, b_m$ at the top level (ARS), based on the current state x and the existing rules in place at the top level.
 Stochastically choose one action.
 - 4: Choose an appropriate action by stochastically selecting the outcome of either the top level or the bottom level.
 - 5: Perform the action, and observe the next state y and (possibly) the reinforcement r .
 - 6: Update the bottom level in accordance with an appropriate algorithm, based on the feedback information.
 - 7: Update the top level using an appropriate algorithm (for extracting, refining, and deleting rules).
 - 8: Go back to Step 1.
-

working-memory contents, and (3) the currently selected goal item from the goal structure. Sensory input is organized into multiple input dimensions, each with a set of possible values. Goal information is likewise divided into dimensions, and working memory is also structured dimensionally. Thus, x can be expressed as a set of dimension–value pairs:

$$(d_1, v_1)(d_2, v_2) \dots (d_n, v_n).$$

The bottom-level output is an action selection, grouped into three action classes: working-memory actions, goal actions, and external actions.

Within each implicit network, actions are chosen according to their computed values. A Q value represents the estimated “quality” of taking action a in state x : $Q(x, a)$ reflects how desirable a is in x . At each step, given x , the system computes $Q(x, a)$ for all candidate actions and then selects an action probabilistically using a Boltzmann (softmax) distribution:

$$p(a \mid x) = \frac{e^{Q(x, a)/\alpha}}{\sum_i e^{Q(x, a_i)/\alpha}}, \quad (3.1)$$

where α is the temperature parameter controlling decision randomness.

Top level (ARS): explicit rules and chunks

Explicit knowledge in the ARS is represented as rules and chunks. A rule’s condition mirrors the bottom-level inputs (sensory input, working-memory items, and the current goal), and a rule’s output mirrors the bottom-level outputs (working-memory actions, goal actions, and external actions). The condition part of a rule is treated as a distinct representational unit (a *chunk*), and the conclusion part is also a chunk.

Rules take the general form:

$$state-specification \rightarrow action.$$

The left-hand side (condition) is a conjunction (logical AND) of elements. Each element refers to a state dimension x_i and specifies an allowable value set/range, e.g.,

$$x_i \in (v_{i1}, v_{i2}, \dots, v_{in}).$$

The right-hand side is an action recommendation.

The structure of a set of rules can be translated into a network at the top level. Each value of each state dimension (i.e., each feature) is represented by an individual node at the bottom level (together forming a distributed representation). Those bottom-level feature nodes relevant to the condition of a rule are connected to a single node at the top level representing that condition, known as a *chunk node* (a localist representation). Given a set of rules, a rule network can be wired up at the top level, in which conditions and conclusions of rules are represented by respective chunk nodes, and links representing rules connect the corresponding pairs of chunk nodes.

Bottom-up learning: Rule Extraction and Refinement (RER)

To capture bottom-up learning ((Karmiloff-Smith, 1986)), the *Rule-Extraction-Refinement* (RER) mechanism acquires rules at the top level by leveraging information from the bottom level. The basic idea for bottom-up learning of action-centered knowledge is:

If an action selected by the bottom level is *successful* (i.e., it meets a specified criterion), then an explicit rule is extracted at the top level. During subsequent interactions, the extracted rule is refined by evaluating the outcome of applying it. If the outcome is successful, the rule condition may be *generalized* to increase its scope; if the outcome is not successful, the rule condition should be made *more specific* so that it excludes the current case and becomes more selective.

This refinement process follows a structured decision logic, which is summarised in the flowchart presented in Figure 3.4.

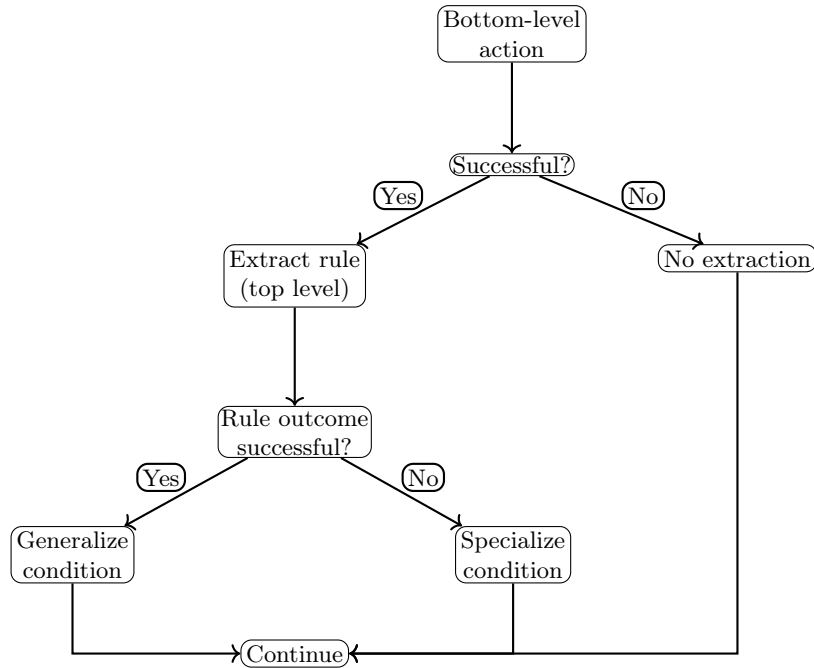


Figure 3.4: Compact logigram of bottom-up rule extraction and refinement.

An agent needs a rational basis for making the above decisions. Accordingly, numerical criteria are defined to determine whether an outcome is successful or not, and these criteria govern whether (and how) the operations above are applied ((Sun, 2001)). In essence, at each step an

information-gain measure is computed to compare rules. The rule-learning operations (extraction, generalization, and specialization) are selected and executed based on this information-gain measure ((Sun, 2006)).

Top-down learning via dual representation

In the opposite direction, the dual representation in the ACS (implicit and explicit) also supports top-down learning. When explicit knowledge (in the form of rules) is available at the top level, the bottom level learns under the guidance of those rules. That is, early in learning the agent relies primarily on top-level rules for action selection. Gradually, as the bottom level acquires more knowledge by “observing” actions directed by the rules (using the same Q-learning mechanism described earlier), the agent becomes increasingly reliant on the bottom level (assuming that the inter-level stochastic selection mechanism is adaptable). In this way, top-down learning emerges.

Stochastic inter-level selection and probability matching

For stochastic selection between the outcomes of the two levels, at each step: with probability P_{BL} , the bottom-level outcome is used. Likewise, with probability P_{RER} , if there exists at least one RER rule indicating an appropriate action in the current state, then the outcome from that rule set (via competition based on rule utility) is used; otherwise, the system defaults to the bottom-level outcome (which is always available). Additional components may be incorporated in an analogous fashion.

The selection probabilities can be adaptive, determined by a process known as *probability matching*: the probability of selecting a component is set based on that component’s relative success rate.

3.3.1.2 Non Action-Centred Subsystem (NACS)

The NACS represents broad, non-action-oriented knowledge about the world in order to support inference. Like the ACS, it uses a dual representational scheme: explicit *associative rules* at the top level and implicit *associative memory* at the bottom level. The ACS governs when and how the NACS is engaged.

At the NACS bottom level, [Associative Memory Networks \(AMNs\)](#) encode implicit, non-action-centered knowledge by learning mappings from inputs to outputs. Such associations can be learned using standard connectionist methods.

At the top level, explicit general knowledge is stored in a [General Knowledge Store \(GKS\)](#). As in the ACS, knowledge is organized into *chunks* defined by dimension-value specifications. Each chunk consists of an identifier plus a set of dimension-value pairs. In the GKS, a *chunk node* represents each chunk and connects to the feature nodes (dimension-value pairs) implemented at the bottom level in distributed form within the AMNs. In addition, the GKS contains links between chunk nodes that encode explicit associations, i.e., *associative rules*. CLARION allows these explicit rules to be learned in multiple ways within the GKS.

Beyond rule-based inference, the [NACS](#) also supports **similarity-based reasoning**, which enables the system to infer knowledge by comparing the current situation with previously stored

knowledge.

In **rule-based reasoning**, inference relies on explicit symbolic rules of the form *condition* $\rightarrow$ *conclusion*. When the current state matches the condition part of one or several rules, the corresponding conclusions are activated. If multiple rules apply, rule selection can be resolved through competition mechanisms based on measures such as rule utility, activation level, or confidence. This reasoning mode is well suited for explicit knowledge, structured domain constraints, and interpretable decision processes.

In **similarity-based reasoning**, inference is performed by comparing a given chunk with other stored chunks in memory. Each chunk can be represented as a vector of feature values, and a similarity measure is computed between the current chunk and candidate chunks. When the similarity exceeds a predefined threshold, the system infers that the corresponding chunk is relevant in the current situation. A commonly used form is a distance-based, normalised similarity function, for example:

$$\text{Sim}(x, y) = \frac{1}{1 + \sum_i w_i |x_i - y_i|},$$

where x and y are feature vectors representing two chunks, and w_i are optional weights reflecting the relative importance of each feature.

Rule-based and similarity-based reasoning are not exclusive. In CLARION, both mechanisms can be combined: similarity-based reasoning provides flexible generalisation when explicit rules are missing or incomplete, while rule-based reasoning provides precision and interpretability when explicit knowledge is available. Their interaction can therefore support more robust and adaptive reasoning in dynamic environments.

Learning in the NACS can proceed in either direction, as in the ACS: *bottom-up learning* can extract explicit top-level knowledge from implicit bottom-level representations, and *top-down learning* can gradually assimilate explicit knowledge into implicit associative memory.

3.3.1.3 Motivational Subsystem (MS)

Having covered CLARION’s primary mechanisms for controlling action (via the ACS and NACS), we now describe how the architecture introduces internal regulation through motivation. Figure 3.5 provides an overview of the MS and its main elements, namely *drives* and a *goal structure*. In CLARION, drives are internal motivational variables that represent the agent’s current needs and priorities. They play a central role because they provide an operational meaning to what counts as a “good” or “bad” outcome for the agent: they determine which situations should be pursued or avoided, and they generate the signals used for evaluating actions. In this sense, the MS explains *what the reward signal represents and where it comes from*, rather than assuming that reward is given externally.

More concretely, the MS influences behaviour by (i) transforming drive states into explicit goals that can be used by the ACS during action selection, and (ii) producing evaluative feedback (reinforcement) that reflects the satisfaction or dissatisfaction of drives. The MS therefore modulates the operation of the ACS (and indirectly the NACS) by setting priorities, shaping goal selection, and defining the criteria according to which actions are reinforced during learning.

A motivational subsystem must capture several key properties of drives (Simon, 1967; Tyrrell, 1993):

- **Proportional activation:** drive intensity increases with the corresponding deficit.

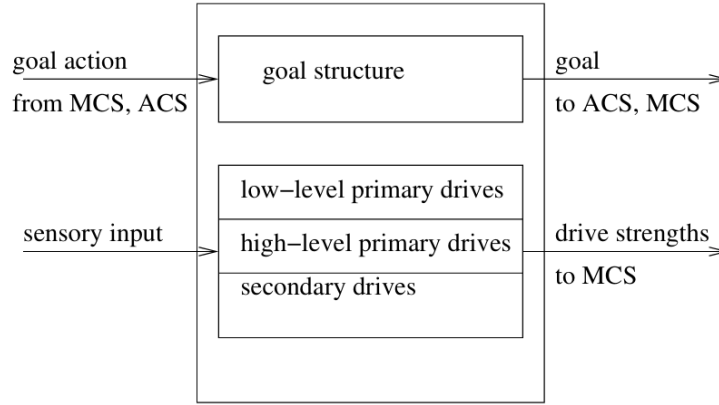


Figure 3.5: The structure of the Motivational Subsystem (MS) ((Sun, 2007b)).

- **Opportunism:** preferences depend not only on deficits but also on available opportunities.
- **Action contiguity:** switching between behaviours is biased by switching costs.
- **Persistence:** drive-reduction behaviour may continue beyond minimal satisfaction.
- **Interruption when needed:** a higher-priority drive can interrupt an ongoing behaviour.
- **Preference combination:** multiple drive-induced preferences can be integrated into a joint preference.

Motivational representation is commonly described as *bipartite* (Simon, 1967; Nerb et al., 1997): internal drive states (e.g., hunger) provide the basis for generating explicit goals (e.g., “find food”), which are then handled by the ACS. In other words, explicit goals are grounded in and depend on implicit internal conditions.

CLARION distinguishes between *primary drives* and *derived drives*. Primary drives are assumed to be fundamental and built-in (hard-wired), and their activation reflects basic needs. Typical examples include (Tyrrell, 1993): **get-food** (driven by food deficit and food-related stimuli), **get-water** (driven by water deficit and water-related stimuli), and **avoid-danger** (driven by danger signals such as proximity, intensity, and certainty). In addition to such built-in drives, CLARION incorporates derived drives, which are acquired and more flexible. They can emerge gradually through conditioning or be established through instruction. For instance, a general tendency to satisfy authority may become a specific motivation to follow instructions, turning compliance into a derived drive.

Finally, based on the current configuration of drives (primary and derived), the MS contributes to setting explicit goals in the goal structure, which are then exploited by the ACS during action selection, as discussed in the next section.

3.3.1.4 Meta-Cognitive Subsystem (MCS)

Metacognition refers to an individual’s knowledge about their own cognitive processes and the outcomes those processes generate. It also involves the active monitoring of these processes

and the resulting regulation and coordination of them, typically in the service of a concrete goal ((Flavell, 1976; Mazzoni and Nelson, 1999)). This conception of metacognition is operationalized within CLARION.

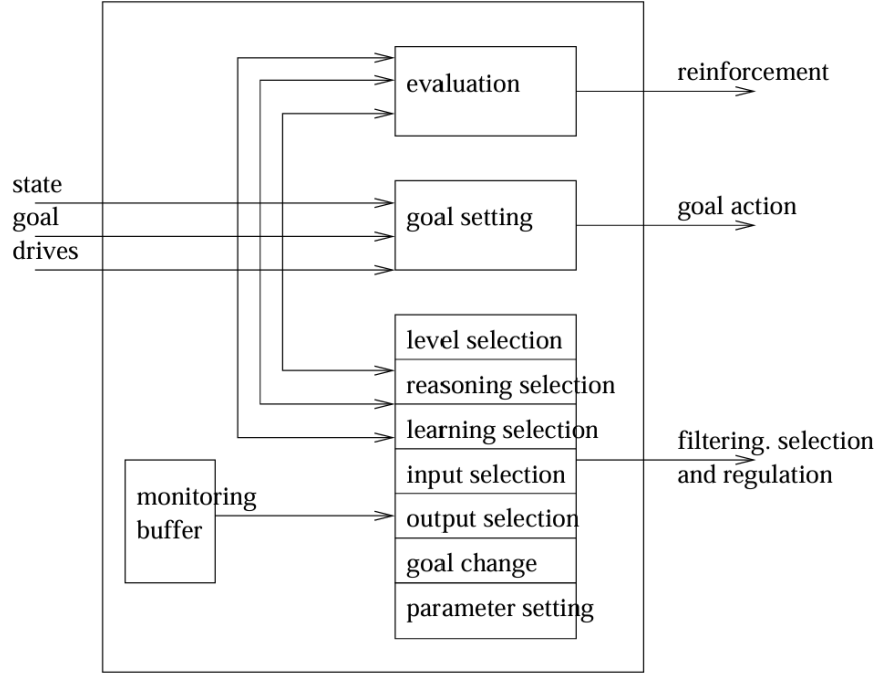


Figure 3.6: The structure of the Meta-Cognitive Subsystem (MCS) ((Sun, 2007b)).

As illustrated in Figure 3.6, the MCS is tightly linked to the MS. The MCS monitors, controls, and regulates cognitive processes with the aim of improving cognitive performance ((Simon, 1967; Sloman, 2000)). Such control and regulation may take the form of setting goals for the ACS, interrupting and redirecting ongoing processes in the ACS and the NACS, and setting key parameters of the ACS and the NACS. Regulation is also implemented by defining reinforcement functions for the ACS on the basis of drive states.

Within this subsystem, multiple types of metacognitive operations are available for different control purposes. These include the following ((Sun, 2006; Mazzoni and Nelson, 1999)):

1. **Behavioural aiming:**

- setting reinforcement functions
- setting goals

2. **Information filtering:**

- focusing/selecting input dimensions in the ACS
- focusing/selecting input dimensions in the NACS

3. **Information acquisition:**

- selecting learning methods in the ACS

- selecting learning methods in the NACS

4. Information utilization:

- selecting reasoning methods in the ACS
- selecting reasoning methods in the NACS

5. Outcome selection:

- selecting output dimensions in the ACS
- selecting output dimensions in the NACS

6. Cognitive mode selection:

- selecting explicit processing, implicit processing, or a combination thereof (with appropriate integration parameters) in the ACS

7. Parameter setting for the ACS and the NACS:

- setting parameters for the IDNs
- setting parameters for the ARS
- setting parameters for the AMNs
- setting parameters for the GKS

Structurally, the MCS can be decomposed into a number of modules. At the bottom level, it includes separate networks such as a goal-setting network, a reinforcement-function network, an input-selection network, an output-selection network, and a parameter-setting network (for learning rates, temperatures, and so on), among others. In a similar fashion, any rules at the top level (when present) may be subdivided correspondingly.

This subsystem may be pre-trained prior to simulating any particular task, in order to capture evolutionarily prepared instincts or knowledge and skills acquired from prior experience.

3.3.2 Integration of Clarion into the CDT environment

This section explains how the CLARION cognitive architecture is integrated into the CDT environment, and how its internal subsystems collectively realize the core cognitive functions required by the CDT.

3.3.2.1 Realization of CDT Cognitive Functions in CLARION

Having motivated the choice of CLARION as the cognitive architecture for human-like cognition and described its internal subsystems, we now explain how they jointly support the cognitive functions within the twin, as shown in table 3.5.

Table 3.5 summarizes how the cognitive functions required by the CDT are instantiated within the CLARION architecture. Each function is realized through one or more specialized subsystems that together provide a complete cognitive loop. Perception and memory are implemented through CLARION's dual representational structure, combining distributed implicit representations with explicit symbolic chunks and rules. Learning occurs at multiple levels

through reinforcement, associative learning, and rule extraction, enabling both adaptation from experience and abstraction of knowledge. Reasoning is supported by a combination of rule-based inference and similarity-based processes, allowing both structured and flexible forms of inference.

Function	CLARION	Subsystem(s)	Explanation
Perception	State encoding	ACS, NACS, MCS	Maps CDT sensory data to state representations; the implicit layer learns recurrent patterns from sensor streams and the explicit layer encodes interpretable concepts as chunks.
Attention	Input filtering and dimensional focusing	MCS	Selects relevant input dimensions from CDT data streams and suppresses irrelevant information before cognitive processing.
Memory	Dual implicit-explicit memory	ACS, NACS	Stores CDT experience as distributed implicit traces and as explicit chunks/rules to support reuse and explanation.
Learning	RL + rule learning	ACS, NACS, MCS	Updates CDT knowledge through Q-learning and rule extraction/refinement, with meta-control of learning parameters.
Reasoning	Rules + similarity inferences	NACS, ACS	Derives CDT inferences using explicit rules and similarity-based retrieval over stored chunks, often in combination.
Action Selection	Stochastic integration	ACS, MS, MCS	Combines implicit and explicit action proposals probabilistically to produce CDT decisions under motivational and metacognitive guidance.
Metacognition	Monitoring/control	MCS	Monitors CDT cognitive performance and regulates processing by tuning parameters, switching modes, and controlling attention and learning.

Table 3.5: Mapping of CDT cognitive functions to CLARION mechanisms

Action selection is handled by the ACS through the stochastic integration of implicit and explicit action evaluations, under the influence of motivational context and metacognitive regulation. Attention and metacognition are provided by the MCS, which filters information, monitors performance, adjusts processing parameters, and dynamically selects cognitive modes. Together, these mechanisms ensure that the CDT is not only capable of perception, learning, reasoning, and action, but also of regulating its own cognitive processes in a human-like and adaptive manner.

3.3.2.2 Functional Coupling within the CDT

It is important to explain how CLARION is operationally coupled with the rest of the CDT. This coupling defines how information, control signals, and learning feedback flow between the DT and the cognitive architecture.

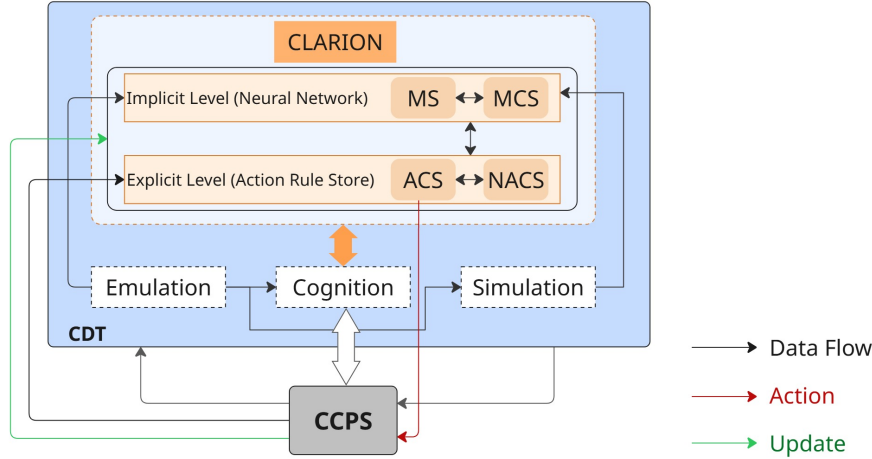


Figure 3.7: Operational coupling between the CDT and CLARION

Figure 3.7 summarises this integration as a closed **perception-action-learning** loop. *Black arrows* represent the *perception/data flow*: observations from the physical or simulated entity are collected by the CDT and transformed into structured state representations that are provided to CLARION, populating the perceptual and working-memory representations exploited by the ACS and NACS. In parallel, CDT context information (e.g., internal drive states and system-level objectives) informs the motivational and metacognitive subsystems.

Based on the current state, goals, and internal control settings, the ACS selects an action through the stochastic integration of implicit and explicit evaluations. The *red arrow* represents the *action/control flow*: the selected action is transmitted back to the CDT control layer, where it is translated into executable commands for the physical entity or the simulation environment. The environment then responds, producing new observations and, when available, performance or reward feedback.

Finally, the *green arrow* represents the *feedback/update flow*: outcomes of executed actions are routed back into CLARION to support learning and adaptation. The ACS updates implicit and explicit action knowledge, the NACS updates associative and conceptual knowledge, the MS may adjust drive and goal priorities, and the MCS may adapt control parameters or processing modes.

3.3.2.3 Role of the CLARION subsystems within the CDT

As described in Section 3.3.1, CLARION has 4 subsystems and each subsystem plays a specific role within the CDT architecture.

Action-Centered Subsystem (ACS)

The ACS is responsible for real-time action selection and learning from interaction. Within the CDT, it receives state descriptions derived from sensor data, working memory, and current goals, and outputs control commands that are translated into actuator-level instructions.

- The implicit level (IDNs) supports fast, reactive behavior and continuous adaptation through reinforcement learning.
- The explicit level (ARS) supports symbolic reasoning, rule-based control, and interpretable decision making.

Together, they allow the CDT to combine human-like intuition and deliberation, enabling both rapid responses and structured reasoning.

Non-Action-Centered Subsystem (NACS)

The NACS provides general knowledge representation and inference capabilities. In the CDT, it supports tasks such as interpretation, prediction, categorization, and completion of partial information. It provides contextual and conceptual knowledge to the ACS.

For example, the NACS may infer high-level situation descriptions from raw sensory patterns, recognize recurring structures, or retrieve relevant associations that inform action choice. This allows the CDT to go beyond reactive control and perform more human-like understanding and reasoning.

Motivational Subsystem (MS)

The MS determines why the CDT acts as it does. It generates and modulates drives, goals, and reinforcement signals based on internal and external conditions. In the CDT, this subsystem links system objectives (e.g., safety, efficiency, exploration, user preferences) to concrete cognitive goals and learning signals.

Thus, rather than optimizing a single fixed reward, the CDT can dynamically shift priorities, balance competing objectives, and exhibit context-sensitive preferences in a psychologically plausible way.

Metacognitive Subsystem (MCS)

The MCS monitors and regulates the operation of the other subsystems. In the CDT, it functions as a supervisory controller that:

- adjusts learning rates, exploration parameters, and reasoning modes,
- switches between implicit and explicit processing when appropriate,
- filters inputs and outputs based on relevance,
- interrupts or redirects ongoing processes when needed.

This allows the CDT to adapt not only what it does, but how it thinks, enabling flexible strategies, resource management, and self-regulation.

Finally, the integration of CLARION enables the CDT to learn on multiple levels:

- The ACS learns from action outcomes via reinforcement.

- The NACS learns associations and conceptual knowledge.
- The MS adapts drives and priorities over longer time scales.
- The MCS adapts control strategies and meta-parameters.

This multi-level learning allows the CDT to evolve over time in a way that reflects both experience-driven adaptation and higher-level cognitive development, closely resembling human learning patterns.

3.4 Chapter Conclusion

We define CDT as an architecture that combines *emulation*, *simulation*, and *cognition*. This supports **RQ2** by explaining what a cognitive layer is and where it is located: cognition is implemented inside the CDT as a functional component that interprets the current situation, reasons on it, learns from feedback, and produces decisions.

A second contribution of this chapter is a selection of a cognitive architecture for the CDT, which contributes to **RQ4**. We reviewed the main families of cognitive architectures (symbolic, subsymbolic, and hybrid) and compared them against practical requirements for industrial CDTs: coverage of core cognitive functions, online operation, adaptability, explainability, hybrid processing, and feasibility of engineering integration. Based on this comparison, **CLARION** is selected as the most suitable architecture because it natively supports a dual representation: an implicit level for learning from experience and an explicit level for symbolic knowledge and reasoning. This makes it possible to combine learning and explainable decision-making within the same system.

We detail then the role of each CLARION subsystem in the CDT context. The ACS supports action selection and RL, the NACS supports knowledge representation and associative/conceptual learning, the MS provides motivation through drive and goal management, and the MCS provides meta-level control (e.g., regulating exploration vs. exploitation and selecting processing modes). Beyond describing these components, the chapter explored how they are operationally coupled to the CDT through a **perception-action-learning loop**: the CDT perception layer produces structured state representations as inputs to CLARION, CLARION selects actions, the CDT control layer executes them in the CDT environment, and execution feedback returns to CLARION to update knowledge, priorities, and control settings. This coupling is essential because it ensures that CLARION is a decision and learning component that remains synchronised with the evolving CDT.

Finally, this chapter prepares the remaining part of **RQ4**: *how can the chosen cognitive architecture be integrated with the chosen knowledge representations?* Chapter 4 addresses this point by introducing the ontology-based semantic layer used to represent concepts, tasks, constraints, and context.

Chapter 4

CDTO Ontology for Cognitive Digital Twins

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Introduction

This chapter addresses the knowledge representation part of the thesis and is mainly linked to **RQ3** and **RQ4**. The work presented in (Al Haj Ali et al., 2026) forms the basis of this chapter. First, it responds to **RQ3** (*Which representation principles and knowledge structures are required to support cognitive interoperability in CDTs?*) by introducing an ontology-based approach to represent the concepts needed for collaboration: agents, tasks, roles, resources, constraints, context, and interaction state. In particular, it proposes **CDTO**⁷ ontology as an ontology that structures these elements within a coherent framework and enables consistent interpretation and exchange of knowledge between heterogeneous actors.

Second, this chapter contributes to **RQ4** (*How can cognitive architectures be integrated with the chosen knowledge representations?*) by explaining how ontology-based knowledge can be operationally exploited inside a CDT. Beyond defining the conceptual model, the chapter details how OWL axioms, inference mechanisms, and SWRL rules support reasoning about system behaviour, task feasibility, and decision-relevant conditions.

The remainder of the chapter is organised as follows. After recalling the foundations required to understand ontology-based modelling (RDF/RDFS, OWL, SPARQL, inference, and SWRL), we present the conceptual design of CDTO, with a particular focus on the Capabilities, Abilities, and Affordances (CAA) model, which provides a structured way to represent what a system can do, how it can do it and the contextual conditions under which actions are possible. We then describe the main classes, properties, constraints, and reasoning rules implemented in the ontology, and explain how they support inference and interoperability in collaborative industrial scenarios. Finally, the chapter describes the inference mechanisms supported by the ontology, and clarifies how CDTO facilitates reasoning about system behaviour. This formalisation establishes a semantic foundation for the cognitive architecture presented in the previous chapter, as well as for the evaluation framework that will be developed later in the thesis.

4.1 Motivation and Role of Ontologies in Cognitive Systems

HCC requires that both agents share a consistent understanding of *entities* (agents, tools, resources), *tasks* (goals, steps, preconditions), and *constraints* (safety, quality, policies). This is particularly evident in HRC, where decision-making must remain coherent under variability induced by the human partner and the evolving shop-floor context. Here, ontologies provide the formal semantic layer that enables this shared understanding.

4.1.1 Ontologies for Collaboration: Semantic Interoperability

An ontology is commonly defined as “an explicit specification of a conceptualization” (Gruber, 1993), meaning that it formally describes the concepts in a domain and the relations among them in a machine-interpretable way. When instantiated with individuals and facts, it becomes a knowledge base (KB) that can be queried and reasoned over. In collaboration contexts, the primary motivation for using ontologies is semantic interoperability, which is the ability of heterogeneous components to exchange information with a shared meaning.

⁷<https://www.ai4c2ps.eu/ontologies/>

For collaboration, multiple modules must coordinate (perception, planning, execution monitoring, safety enforcement, explanation, and external information systems). Without a shared semantic model, each module tends to implement its own representation of tasks and context, leading to duplicated mappings, inconsistent assumptions, and fragile integration. An ontology mitigates this by providing:

- a **shared vocabulary** for agents, roles, actions, and resources;
- a **machine-interpretable structure** that supports consistent grounding of observations into symbolic facts;
- **formal constraints** (axioms) that reduce ambiguity and allow automated consistency checks;
- **explicit relations** enabling traceable inference (e.g., task feasibility, safety violations, role compatibility).

In HRC, this semantic layer is particularly useful because collaboration dynamics require frequent alignment between the robot’s internal state and the human-understandable description of the ongoing task. Ontology-based representations have been proposed to address variability and to support flexible reasoning about collaborative tasks and constraints (Umbrico et al., 2020; Hall et al., 2024).

More generally, (Roxin et al., 2022) demonstrate that semantic web technologies (e.g., OWL-based models and rule-based reasoning) can support interoperable and explainable representations of digital twins, reinforcing the importance of ontology-based knowledge structuring for collaboration and decision support.

Recent work also proposes ontologies explicitly dedicated to collaboration between humans and AI. For example, the principle of *sliding work sharing* has been introduced to dynamically distribute work between human and artificial agents, supported by DT/HDT-based representations, which is directly relevant for modelling bidirectional and explainable collaboration (Karapanagiotis et al., 2025b). Similarly, in the field of robotics, a contextual ontology for human-robot interoperability in dynamic environments formalises notions such as sliding autonomy and task allocation, providing a concrete example of how semantic structures can represent roles, constraints, and decisions during collaboration (Karapanagiotis et al., 2025a).

4.1.2 Why Ontologies Matter in CDTs

A CDT must integrate heterogeneous data streams (IoT, robot controllers, human-related signals) and connect them to task-, process-, and safety-level concepts that are meaningful for collaboration. Ontologies contribute to CDTs in three complementary ways:

- **Semantic integration of data and models:** the ontology provides a conceptual backbone to fuse observations, events, and model outputs into a coherent semantic state.
- **Interoperability across assets and services:** different twins and services can exchange information through shared concepts and mappings, supporting scalable system-of-systems integration.
- **Reasoning and validation:** logical inference and constraint checking enable consistency verification and derivation of implicit knowledge (e.g., detecting unsafe configurations, inferring task progress).

This semantic enrichment is widely identified as a key enabler for advanced digital twin capabilities (including human-centric twins and cognitive twins), especially for manufacturing and HRC (Krupas et al., 2024; Karabulut et al., 2024).

4.1.3 Why an Ontology in Addition to a Cognitive Architecture

Several works have shown that cognitive architectures benefit from explicit semantic layers: (Oltramari et al., 2014) demonstrated how ontologies can be integrated into ACT-R to enrich declarative memory; (Lieto et al., 2018) argued that cognitive architectures require richer conceptual representations to scale to complex domains; and (d’Avila Garcez et al., 2019) emphasized the importance of combining neural and symbolic approaches to achieve robust and explainable intelligence. In our work, using an ontology alongside a cognitive architecture is preferable because it provides:

Externalized and shared semantics (beyond internal memory). Cognitive architectures primarily organize knowledge to support cognitive processing *within* the agent. In contrast, an ontology is designed as a *shared semantic contract* among heterogeneous components and agents. This is essential in collaboration: the same concept (e.g., **handover**, **safety-zone**, **operator-role**) must be interpreted identically by perception, planning, the CDT, and the human-interface.

Formal constraints and verifiable consistency. Ontologies encode axioms and constraints that can be checked systematically (e.g., incompatible states, violated safety conditions, inconsistent task context). This supports robust collaboration by preventing incoherent internal states from propagating into action selection, especially when the semantic state is continuously updated from uncertain observations.

Interoperable interfaces between cognitive and robotic layers. An ontology provides a stable interface for exchanging information between cognitive reasoning and robotic systems (perception/action), reducing architecture-specific coupling. Ontology-driven interfaces have been proposed explicitly to formalize communication between cognitive models and robotic systems (Azevedo et al., 2020).

Traceability and explanation grounded in domain concepts. In collaboration, *why* a robot decided to stop, request confirmation, or change strategy must be explainable in human terms. Ontologies represent decisions in a vocabulary aligned with domain concepts, enabling explanations grounded in explicit relations (task steps, preconditions, constraints) rather than opaque internal structures.

4.1.4 CLARION-Ontology Integration

CLARION is well suited for integrating ontologies because it separates implicit learning from explicit reasoning. The implicit level learns from experience using neural networks, while the explicit level represents knowledge symbolically using chunks and rules (Sun, 2006). Ontologies are integrated into CLARION mainly in the NACS, where they structure the GKS. Ontological concepts correspond to chunks, and relations between concepts are represented as structured links. This gives explicit knowledge a clear semantic organization and prevents it from becoming

inconsistent or fragmented.

In CLARION, [Long-Term Memory \(LTM\)](#) contains both declarative and procedural knowledge. Declarative knowledge includes facts, concepts, and relations defined by the ontology, while procedural knowledge includes rules and strategies stored in the [ARS](#). This separation is consistent with cognitive theories that distinguish between factual knowledge and skills ([Anderson, 2007](#)). Ontologies ensure that both types of knowledge are defined over meaningful concepts instead of arbitrary features, which supports abstraction and generalization.

[Working Memory \(WM\)](#) plays a key role by connecting perception, memory, and action. Outputs from neural perception models are mapped to ontological concepts and stored temporarily in working memory. These representations are then combined with goals and rules to guide reasoning and decision-making. In this way, working memory maintains a conceptual view of the current situation, not just raw sensory data.

Ontologies also support different types of reasoning in CLARION ([Zi et al., 2025](#)):

1. Deductive reasoning is performed by applying symbolic rules and constraints from the ontology.
2. Inductive reasoning arises from learning in the neural components.
3. Abductive reasoning occurs when the system uses structured knowledge to infer the most likely explanation for observed data.

This combination allows CLARION to integrate data-driven learning with structured, explainable reasoning ([d’Avila Garcez et al., 2019](#)).

The following section provides the general semantic foundations required to understand ontology-based knowledge representation. It introduces the core ontology vocabulary (classes, individuals, properties, and axioms), outlines the main semantic representation languages, and explains how knowledge is accessed and enriched through SPARQL queries, inference mechanisms, and rule-based formalisms.

4.2 Ontology Concepts, Languages, and Inference

This section introduces the general foundations required to understand ontology-based knowledge representation. It presents the main ontology vocabulary, the representation languages commonly used in semantic technologies (RDF/RDFS and OWL), and the mechanisms used to query and enrich knowledge (SPARQL, inference, and rule-based formalisms such as SWRL). [Figure 4.1](#) summarises the typical pipeline used in ontology-based systems.

4.2.1 Ontology Vocabulary: Classes, Individuals, Properties, and Axioms

An ontology provides a structured and machine-interpretable description of a domain. Its core elements are:

- **Classes:** concepts that represent categories of entities in the domain.
- **Individuals:** concrete instances of these classes (i.e., entities).

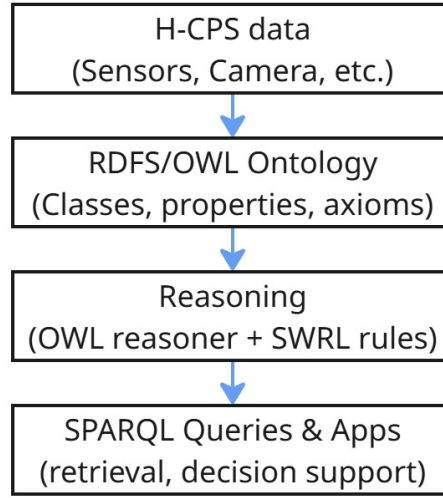


Figure 4.1: Semantic Workflow for Ontology-Based Reasoning.

- **Properties:** relations used to connect entities or to attach data values.
- **Axioms:** formal statements that constrain and define the meaning of classes and properties.

Used together, these elements allow defining not only a vocabulary, but also the intended semantics of a domain. Axioms are particularly important because they support consistency checking and enable automated reasoning.

4.2.2 RDF: Knowledge Representation as Triples

A common way to represent ontology-based knowledge is the [Resource Description Framework \(RDF\)](#)⁸. RDF models knowledge as a graph built from **triples** of the form:

< Subject, Predicate, Object >

The subject and object refer to entities (or literals), and the predicate expresses the relation between them. RDF graphs can therefore represent complex knowledge structures in a uniform way. Figure 4.2 illustrates the triple principle that underlies knowledge graphs used in ontology-based systems.

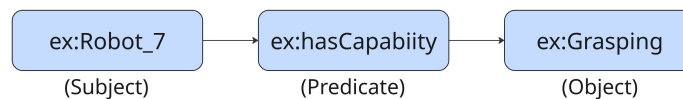


Figure 4.2: RDF representation of knowledge as a triple (Subject-Predicate-Object).

⁸<https://www.w3.org/TR/rdf11-schema/>

4.2.3 RDFS and OWL: Adding Semantics and Constraints

RDF provides a graph model, but additional layers are needed to express richer meaning. [RDF Schema \(RDFS\)](#) extends RDF with basic constructs to define classes and hierarchies (e.g., `rdfs:Class`, `rdfs:subClassOf`) and to constrain properties (e.g., `rdfs:domain`, `rdfs:range`). It also supports typing individuals using `rdf:type`. Figure 4.3 illustrates a simple example of knowledge representation in RDFS.

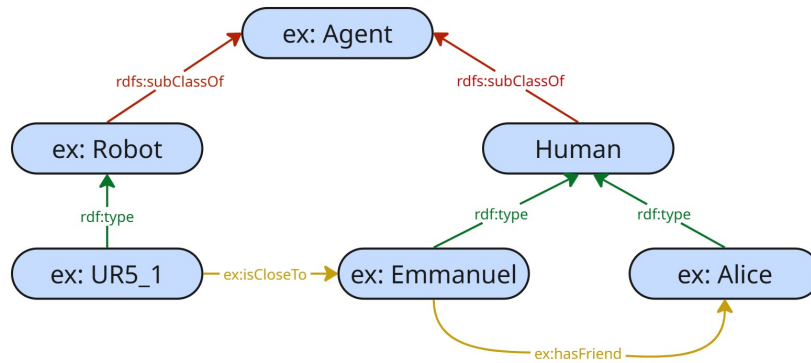


Figure 4.3: Example of knowledge representation in RDFS.

In this example, `ex:Robot` and `ex:Human` are defined as subclasses of the more general class `ex:Agent` through `rdfs:subClassOf`. Individuals are then linked to their classes using `rdf:type`, such as `ex:UR5_1 rdf:type ex:Robot` and `ex:Emmanuel rdf:type ex:Human`. In addition to the taxonomic structure, the graph contains domain relations expressed as object properties, for instance `ex:isCloseTo` between `ex:UR5_1` and `ex:Emmanuel`, and `ex:hasFriend` between `ex:Emmanuel` and `ex:Alice`. Such a representation is useful because it allows basic semantic interpretation and inference. For example, since `ex:Robot` is a subclass of `ex:Agent`, a reasoner can infer that `ex:UR5_1` is also an instance of `ex:Agent`, even if this type assertion is not explicitly stated.

[Web Ontology Language \(OWL\)](#)⁹ further extends RDFS with more expressive constructs to define and constrain knowledge. OWL makes it possible to specify stronger modelling assumptions (e.g., disjoint classes, equivalence, cardinality restrictions) and to support reasoning with a formal semantic basis. The listing 4.1 shows the same knowledge as in the previous RDFS graph, but expressed in the OWL *RDF/XML* syntax, which is a standard serialisation format for OWL ontologies.

```

<owl:Class rdf:ID="Agent" />
<owl:Class rdf:ID="Robot">
  <rdfs:subClassOf rdf:resource="#Agent" />
</owl:Class>
<owl:Class rdf:ID="Human">
  <rdfs:subClassOf rdf:resource="#Agent" />
</owl:Class>

<owl:ObjectProperty rdf:ID="isCloseTo">
  <rdfs:domain rdf:resource="#Agent" />

```

⁹<https://www.w3.org/OWL/>

```
<rdfs:range rdf:resource="#Agent" />
</owl:ObjectProperty>
<owl:ObjectProperty rdf:ID="hasFriend">
  <rdf:type rdf:resource="http://www.w3.org/2002/07/owl#
    SymmetricProperty" />
  <rdfs:domain rdf:resource="#Human" />
  <rdfs:range rdf:resource="#Human" />
</owl:ObjectProperty>

<owl:NamedIndividual rdf:ID="UR5_1">
  <rdf:type rdf:resource="#Robot" />
  <isCloseTo rdf:resource="#Emmanuel" />
</owl:NamedIndividual>
<owl:NamedIndividual rdf:ID="Emmanuel">
  <rdf:type rdf:resource="#Human" />
  <hasFriend rdf:resource="#Alice" />
</owl:NamedIndividual>
<owl:NamedIndividual rdf:ID="Alice">
  <rdf:type rdf:resource="#Human" />
</owl:NamedIndividual>
```

Listing 4.1: Representative extract of an OWL ontology illustrating classes, properties, and individuals.

The first part declares the core classes **Agent**, **Robot**, and **Human**. The subclass relations are encoded using `<rdfs:subClassOf>` so that **Robot** and **Human** inherit from **Agent**. The second part defines two object properties: **isCloseTo** and **hasFriend**. Their intended usage is constrained through `rdfs:domain` and `rdfs:range`, indicating which types of individuals they can connect. In addition, **hasFriend** is declared as an `owl:SymmetricProperty`, meaning that if **Emmanuel** **hasFriend** **Alice** is asserted, a reasoner can infer **Alice** **hasFriend** **Emmanuel**. Finally, the individuals **UR5_1**, **Emmanuel**, and **Alice** are introduced as `owl:NamedIndividual` and typed using `rdf:type`. Their relations are then asserted through property statements (e.g., **UR5_1** **isCloseTo** **Emmanuel** and **Emmanuel** **hasFriend** **Alice**). This OWL representation preserves the same content as the RDFS example while making additional semantics explicit for automated reasoning.

4.2.4 Inference and Reasoning

A key benefit of ontologies is that they support **inference**: deriving implicit knowledge from explicitly stated facts and axioms. A reasoner can, for example:

- infer additional class memberships (classification),
- identify inconsistencies when constraints are violated,
- propagate consequences of hierarchical relations.

In practice, these inference tasks are performed by **OWL reasoners**, i.e., software engines that implement the formal semantics of Description Logics and compute the logical consequences of an ontology. Commonly used reasoners include:

- **HermiT** (([Glimm et al., 2014](#))): an OWL 2 DL reasoner widely used in academic and engineering contexts for ontology classification and consistency checking.

- **Pellet** ((Sirin et al., 2007)): an OWL DL reasoner that supports standard reasoning services and is often used in ontology-driven applications; it also provides support for SWRL rule reasoning in many toolchains.
- **FaCT++** ((Tsarkov and Horrocks, 2006)): an OWL DL reasoner focused on efficient classification and satisfiability checking.
- **ELK** ((Kazakov et al., 2014)): a high-performance reasoner specialised for the OWL 2 EL profile, suitable for large ontologies when lightweight reasoning is sufficient.

The choice of reasoner typically depends on the required expressiveness (e.g., OWL 2 DL vs. OWL 2 EL) and the scale and performance constraints of the application.

4.2.5 SPARQL Queries and SWRL Rules for Knowledge Retrieval

SPARQL is the standard query language for RDF graphs. It allows applications to retrieve knowledge by matching graph patterns, such as:

- retrieving all entities related by a given property,
- finding individuals that satisfy a combination of relations,
- accessing results after reasoning (depending on the system configuration).

OWL axioms provide strong semantics, but some conditional knowledge is more naturally expressed using explicit rules. The [Semantic Web Rule Language \(SWRL\)](#) ¹⁰ complements OWL by allowing **if-then** rules to be stated over ontology concepts. A SWRL rule has the general form:

$$\text{Body (conditions)} \Rightarrow \text{Head (conclusion)}$$

The body specifies conditions that must hold, and the head specifies what can be concluded when these conditions are satisfied. SWRL is useful when modelling needs require a direct rule structure, for example when conclusions depend on the simultaneous satisfaction of multiple relations and context conditions. In practice, SWRL rules are often used together with OWL axioms:

- OWL provides the core semantics, constraints, and class definitions,
- SWRL provides additional conditional derivations to enrich inference results.

4.3 Overview and Conceptual Model of the CDTO Ontology

The CDTO¹¹ ontology is designed as a *modular* ontology that reuses and aligns with existing semantic resources rather than redefining common concepts from scratch. This design choice follows established ontology engineering practice: reusing well-known ontologies improves semantic interoperability, reduces modelling ambiguity, and increases the likelihood that the resulting knowledge base can be shared, queried, and reasoned upon across heterogeneous tools and agents (Gruber, 1993; Noy and McGuinness, 2001; Uschold and Gruninger, 1996). This is particularly important in the context of Human-CPS collaboration because capability assessment, decision

¹⁰<https://www.w3.org/submissions/SWRL/>

¹¹<https://www.ai4c2ps.eu/ontologies/2024/>

support and coordination must be based on consistent definitions of entities such as agents, events, observations, roles and context. Moreover, using imports to split an ontology into modules makes it easier to maintain and extend: each domain module can be updated independently, while external ontologies act as stable, well-supported reference vocabularies for standard tools (reasoners and knowledge-graph systems) (Stuckenschmidt et al., 2009; Gangemi, 2005).

Our ontology adopts a modular design that reuses both established external ontologies and introduces project-specific modules. Figure 4.4 summarises this structure by distinguishing between imported ontologies and modules built by the CDTO and by explaining the dependency links (e.g., “imports”, “based on” and “connected to”) used to ensure a consistent and interoperable semantic model.

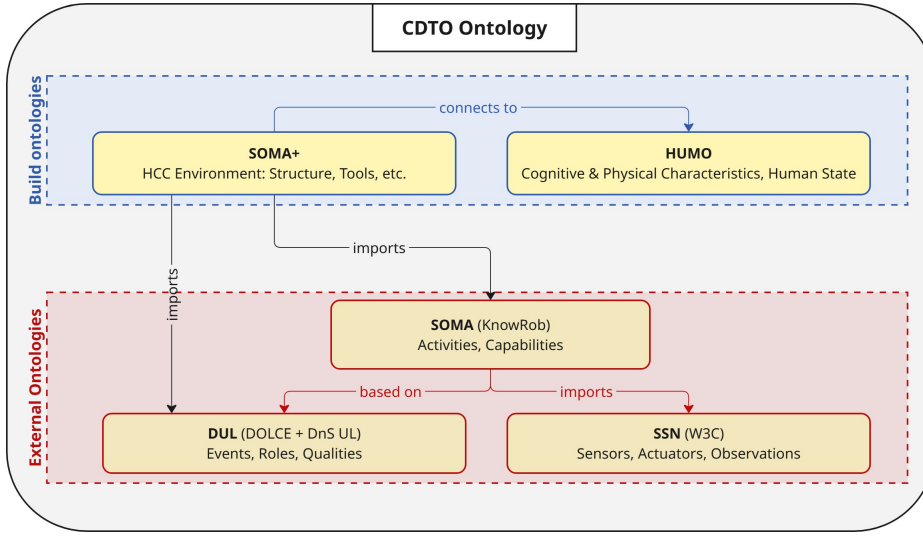


Figure 4.4: CDTO ontology modules and reused external ontologies.

- **SOMA** (Beetz et al., 2018), the foundational ontology of the KnowRob framework, contributes a robotics-oriented conceptualisation of *activities*, *dispositions*, *capabilities*, and *social/contextual notions*. In CDTO, this is useful to ground action-centred knowledge (what an agent can do and how actions relate to situations) in a model that has been validated in knowledge-enabled robotics. SOMA also provides modelling patterns that facilitate the transition from low-level perceptions to higher-level, interpretable action concepts.
- **DOLCE+DnS Ultra Lite (DUL)** (Gangemi et al., 2002) provides upper-level constructs to model general categories such as *objects*, *events*, *qualities*, *roles*, and *descriptions*. DUL is used to ensure conceptual coherence across modules and to provide a principled semantic backbone for representing system entities and contextual descriptions. This is particularly relevant for cognitive interoperability, where different components may model the same phenomenon (e.g., a maintenance event, an operational role, a system state) from different viewpoints that must remain compatible.
- **SSN** (Compton et al., 2012) enables the representation of *sensors*, *actuators*, *observations*, and their relations with the environment. This layer is essential in CPS because cognitive reasoning is ultimately grounded in observations and signals. By relying on SSN, CDTO can represent observation-driven knowledge in a standard way (what is observed, by which

sensor, under which conditions, with which feature of interest), which supports traceability from decisions back to evidence.

This base is enriched with two ontologies on top of it:

- **HUMO** (Naudet et al., 2025) provides a dedicated representation of human-related aspects that are central in . It captures dimensions such as cognitive and physical characteristics, human state or constraints, and human-related context needed for interaction and cooperation. Introducing an explicit human-oriented module supports modelling *human capabilities and limitations* in a way that can be used consistently by CDTs and other agents, which is important for safe task allocation, collaboration, and explainability in industrial settings.
- **SOMA+** captures the domain-specific semantics of our cyber-physical environment. It formalises concepts that are not sufficiently covered by general ontologies, including task structures, tools, resources, and robot- and CPS-specific capabilities. Importantly, SOMA+ provides the semantic foundations required by the cognitive layer of the CDT (e.g., capability modelling, contextual feasibility, and interoperability across heterogeneous agents).

Overall, CDTO follows a layered integration strategy: (i) upper-level semantics (DUL) provide general modelling categories, (ii) domain ontologies (SOMA, SSN) provide established representations for robotics/action knowledge and sensing/observation knowledge, and (iii) project-specific modules (HUMO, SOMA+) introduce the concepts required for cognition, capability modelling, and human-CPS collaboration. This combination supports both reusability and specialisation, while maintaining compatibility with standard semantic web tooling and reasoning engines (Euzenat and Shvaiko, 2013).

Finally, the next section introduces the main concepts and relationships formalised in the CDTO ontology, which provide the semantic basis for representing and reasoning about cognition in Human-CPS collaboration.

4.4 Ontology Classes, Properties, and Constraints

This section describes the concrete structure of the **CDTO global ontology**, which provides a semantic framework to represent cognitive agents, either a human or a CCPS, together with their cognitive functions and collaboration context. CDTO is constructed by integrating and aligning a set of complementary ontologies: **SOMA+**, **HUMO**, and two widely adopted reference ontologies from the literature, **SOMA** and **SSN/SOSA**. Within this global framework, **SOMA+** is the main project-developed module that extends SOMA and captures the concepts specific to the **HCC** environment, in particular the modelling of robots, tools, task-related resources, and their operational context. The SOMA+ module contains **675 classes**, **266 object properties**, and **38 data properties**. It also includes a large set of logical axioms, notably **357 OWL restrictions** and **70 disjointness axioms** (`owl:disjointWith`), which support consistency checking and automated inference. To inspect and validate the ontology structure, we used **Protégé** as the main development and visualisation environment.

The CDTO global ontology is therefore defined by the combination of these integrated modules and their imported semantics:

- <https://www.ai4c2ps.eu/ontologies/2026/SOMA+/> (SOMA+),
- <https://www.ai4c2ps.eu/ontologies/2024/Humo/> (HUMO),

- <http://www.ease-crc.org/ont/SOMA.owl> (SOMA),
- <http://www.w3.org/ns/ssn/> (SSN, including SOSA terms).

4.4.1 Classes of the CDTO Ontology

CDTO is a modular OWL ontology in which the global conceptual space is defined by the combination of:

1. the project-specific classes introduced in **SOMA+** and related project module,
2. the classes reused from established ontologies imported by CDTO (notably SOMA, SSN/-SOSA, and DUL)

In practice, this means that the “classes of CDTO” include both the classes defined in the `core:` namespace and the classes that remain in their original namespaces (e.g., `soma:`, `sosa:`, `dul:`, `sys:`) but are referenced and constrained by CDTO axioms.

The extract in Figure 4.5 illustrates several central branches that are used later in the thesis for capability modelling and task reasoning.

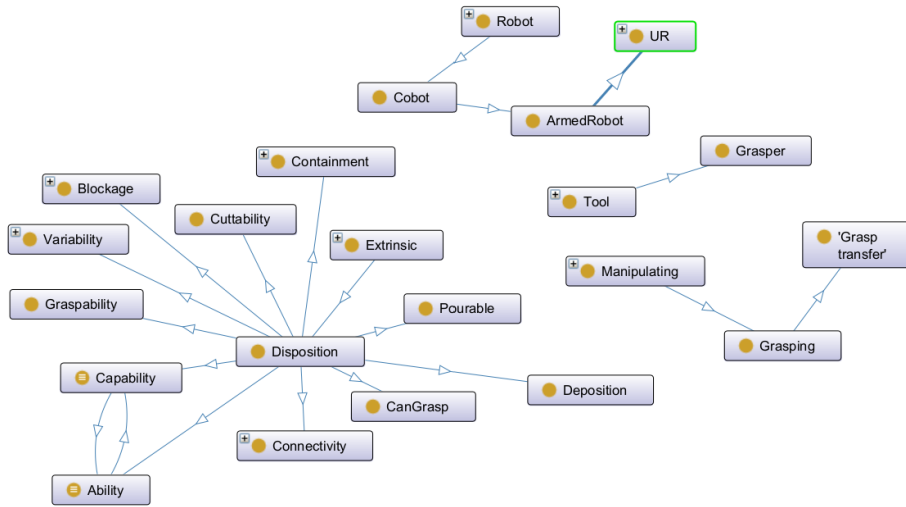


Figure 4.5: Extract from the CDTO ontology.

SOMA+: The `core:` namespace contains the main domain concepts aligned with the application use-case to model industrial HCC elements and the resources required for task execution. Representative examples include:

- **Robot classes:** `core:Robot`, `core:Cobot`, `core:ArmedRobot`, `core:UR`, and other robot specialisations.
- **Tool and fastening-related classes:** `core:Grasper`, `core:Screwdriver`, `core:Screw`, `core:Nut`, `core:Bit`, `core:DriveTipProfile`, and related component classes such as `core:Screw` and `core:Nut`.

- **Sensing outputs and media:** `core:Camera`, `core:Capture`, `core:Record`, `core:Image`, and `core:Video`.

These classes are used to describe the operational world of the CDT at a level that supports reasoning (e.g., linking a recorded output to its sensor, or linking a robot type to the tools it can use).

HUMO classes (human and cognitive layer): HUMO provides the human-centred semantic layer required for Human-CPS collaboration. Its role is to represent, in a structured and interoperable way, the information that a CDT needs about the human actor and the cognitive aspects involved in interaction and decision support (Naudet et al., 2025). Concretely, this layer is built by combining two ontologies: **SysAgentOnto** and **CognitionOntology**. The detailed specification and modelling choices of HUMO are presented in Section 4.5; the following section therefore provides a general description.

The SysAgentOnto defines a compact set of core abstractions for describing an agent/system in terms of *what it is*, *what functions it provides*, and *which abilities it requires*. The main classes defined locally in SysAgentOnto include:

- `sys:GeneralSystem`
- `sys:Function`
- `sys:Interface`
- `sys:Ability`

In addition, SysAgentOnto explicitly reuses well-established upper-level and domain classes (via imported/referenced namespaces), including `dul:Agent`, `dul:Entity`, `dul:Quality`, `dul:Task`, and SOMA notions such as `soma:Disposition` and `soma:Intrinsic`. These abstractions are operationalised through object properties such as `sys:hasFunction`, `sys:hasAbility`, `sys:requiresAbility`, and `sys:hasInterface`, which provide a direct semantic bridge to the capability and cognitive modelling developed later in this chapter.

The CognitionOntology file (<https://www.ai4c2ps.eu/ontologies/2024/CognitionOntology>) provides a richer taxonomy dedicated to cognitive notions. In the OWL file used in this thesis, the ontology defines **71 classes**, including a majority of locally defined classes representing cognitive structures and functions. Representative examples include:

- **Cognitive system and functions:** `cog:CognitiveSystem`, `cog:CognitiveFunction`, `cog:CognitiveAbility`;
- **Perception and sensation:** `cog:PerceptionFunction`, `cog:PerceptionAbility`, `cog:SensationFunction`, `cog:SensationAbility` (and their corresponding components);
- **Attention:** `cog:AttentionFunction`, `cog:SelectiveAttention`, `cog:SustainedAttention` (with related abilities/components);
- **Memory structures:** `cog:WorkingMemory`, `cog:ShortTermMemory`, `cog:SensoryMemory`, `cog:DeclarativeLTM`, `cog:ProceduralLTM`, and specific memory constructs such as `cog:PhonologicalLoop`, `cog:VisualSketchpad`, `cog:EpisodicBuffer`;

- **Decision-making and executive aspects:** `cog:DecisionMaking`, `cog:ExecutiveFunction`, `cog:CentralExecutive`.

CognitionOntology reuses SysAgentOnto relations (e.g., `sys:hasFunction`, `sys:requiresAbility`) and also relies on DUL part-whole modelling (e.g., `dul:hasPart`) to represent how cognitive systems can be decomposed into components and functions.

The next section is therefore dedicated to the *cognitive functions* formalised in this ontology layer and explains which functions are selected and modelled in this thesis.

Reused classes from SOMA: CDTO reuses a large set of robotics-oriented classes from SOMA. These classes provide a stable conceptualisation for action and capability modelling, and CDTO aligns its own classes to them where appropriate. Examples used in CDTO include:

- `soma:Tool` as the general category for tools, together with action concepts such as `soma:Manipulating` and `soma:Grasping`;
- `soma:Disposition` and specialised dispositions (e.g., `soma:Graspability`, `soma:Cuttability`, `soma:Connectivity`), which are used to represent action-enabling properties;
- `soma:Capability`, which provides a domain-level anchor for modelling what an agent/system can do.

Reused classes from DUL (upper-level semantics): CDTO also relies on DUL categories to maintain conceptual clarity across modules, for instance distinguishing entities, processes, qualities, and regions. Representative DUL classes referenced in CDTO include:

- `dul:Entity` and `dul:Object` for general entity modelling,
- `dul:Process` and `dul:Event` for time-dependent phenomena,
- `dul:Quality` and `dul:Region` for attribute modelling,
- `dul:Goal` and `dul:Plan` for decision-relevant descriptions.

Reused classes from SSN/SOSA (observations): To ground semantic reasoning in evidence, CDTO uses the SSN/SOSA conceptualisation of sensing and observation. Commonly used SOSA classes and terms include `sosa:Sensor` and `sosa:Observation`, together with SOSA properties that appear directly inside CDTO class constraints.

4.4.2 Class Hierarchy and Taxonomy

This subsection focuses on the **main taxonomic branches** that structure the global hierarchy and that are directly used later for capability and task reasoning. The CDTO taxonomy is designed to provide:

- a clear entry point through agents and systems,
- action-oriented concepts for task modelling,
- tool/resource specialisations for operational reasoning,
- dispositions as semantic enablers,
- observation-related concepts to ground reasoning in evidence.

Agent-centric branch: A central entry point of the taxonomy is the `dul:Agent` branch, which provides the semantic anchor for actors that can participate in tasks and decisions. The class tree shown in Protégé illustrates a progressive specialisation from generic agents to industrial robot categories. A representative path visible in the hierarchy is:

`dul:Agent` $\rightarrow$ `Physical agent` $\rightarrow$ `Artificial agent` $\rightarrow$ `Robot` $\rightarrow$ `Cobot` $\rightarrow$ `ArmedRobot` $\rightarrow$ `UR`.

This organisation is useful because it supports inheritance-based reasoning: any instance of `UR` is automatically also an instance of `Robot` (and more generally an `Agent`), and therefore inherits the constraints and relations defined at higher levels.

Action and manipulation branch: The global taxonomy also includes action concepts used to represent what agents do. In the ontology extract (Figure 4.5), manipulation-related classes such as `Manipulating` and its subclass `Grasping` appear as part of the action taxonomy. This hierarchy supports abstraction: high-level reasoning can be expressed over `Manipulating` (e.g., a category of tasks), while more detailed reasoning can target a specific action such as `Grasping`. The class `Screwing` in the project namespace is another example of a domain-specific action that is integrated into this action-oriented structure.

Tool and resource branch: Tools and resources are represented through a hierarchy where a general tool superclass is specialised by concrete tool types. In the extract, `Tool` is refined into classes such as `Grasper`, and project-defined tool concepts (e.g., `Screwdriver`) are integrated in the same area. This is important for later reasoning because tools and actions are modelled in a compatible way: the taxonomy makes it possible to query for all available tool types, or to reason at different levels of granularity (e.g., a generic grasper versus a specific gripper model).

Disposition-centric branch: Figure 4.5 also highlights `Disposition` as a key hub in the taxonomy. Several specialised dispositions (e.g., `Graspability`, `Cuttability`, `Containment`, `Pourable`, `Connectivity`) are represented as subclasses of `Disposition`. This taxonomic organisation is not only descriptive: it supports reasoning about feasibility by linking object/environment characteristics to possible actions. For example, if an object is classified under a disposition that implies it can be grasped, this can later support the inference that grasp-related actions are applicable under appropriate conditions.

Sensing and observation branch: The global hierarchy also contains sensing and observation concepts used to connect knowledge to evidence. At the taxonomic level, `sosa:Sensor` and observation-related concepts provide the standard structure, while SOMA+ introduces concrete sensing artefacts such as `Camera`, `Capture`, and `Record`. These classes are integrated in the hierarchy so that sensing outputs (e.g., `Image`, `Video`) remain semantically connected to their production process and sensing device, which is later exploited through OWL restrictions and SPARQL queries.

This structure is used directly in the next subsections to present the ontology constraints, and the inference services supported by CDTO.

4.4.3 Constraints and Consistency Rules

CDTO includes a significant number of OWL axioms that constrain how classes and properties can be used. These axioms serve two practical goals: (i) detecting inconsistent assertions, and

(ii) enabling inference (deriving implicit knowledge).

Example 1: result typing constraints for sensing outputs (SOSA restrictions). The SOMA+ module defines Description Logic (DL) restrictions for sensing-related concepts. In particular, we constrain the range of `sosa:hasResult` and the type of the producing sensor as follows:

$$\text{core:Capture} \sqsubseteq \forall \text{sosa:hasResult. core:Image} \quad (4.1)$$

$$\text{core:Record} \sqsubseteq \forall \text{sosa:hasResult. core:Video} \quad (4.2)$$

$$\text{core:Capture} \sqsubseteq \forall \text{sosa:madeBySensor. core:Camera} \quad (4.3)$$

$$\text{core:Record} \sqsubseteq \forall \text{sosa:madeBySensor. core:Camera} \quad (4.4)$$

where `core:Camera` is aligned with `sosa:Sensor`. These constraints ensure that the ontology remains semantically coherent: a reasoner can flag incorrect modelling (e.g., a `core:Record` returning a `core:Image`) and can also infer expected result types when only partial information is provided.

Example 2: classification patterns based on DUL modelling. The ontology also reuses DUL patterns for classification, such as `dul:isClassifiedBy`. These patterns are used to connect physical objects (e.g., fasteners such as `core:Screw` or `core:Nut`) with classification descriptors (e.g., profile-like classes). In practice, such modelling supports consistent annotation and reasoning over components and their properties.

Example 3: disjointness axioms. The ontology contains multiple `owl:disjointWith` axioms. Disjointness is used to prevent contradictory classifications, which is important for industrial settings where incorrect typing can lead to invalid capability or task reasoning.

These constraints are operational: when CDTO is loaded in Protégé and a reasoner is executed, the reasoner can detect inconsistencies caused by invalid assertions and classify individuals and classes according to the logical definitions. In this thesis, we used the **HermiT** reasoner for consistency checking and classification, and the ontology was found to be **logically consistent** under HermiT.

4.5 HUMO: Human and Cognition Layer in CDTO

As previously mentioned, the CDTO consists of complementary ontological modules dedicated to the application use-case that supports HCC. In this modular design, **SOMA+** captures the cyber-physical environment and the robot/tool side of the collaboration, while **HUMO** provides the missing human-centred and cognition-oriented semantics required to represent the human actor within the same semantic space. Since HUMO introduces new concepts that are not natively provided by the reused robotics and sensing ontologies, we detail here how HUMO structures the human as a system, how it models cognition, and how it connects these notions to SOMA.

4.5.1 Human-System-Cognition Structure and Integration with SOMA

Figure 4.6 summarises the **Human-System-Cognition** core of HUMO. It represents the human as a system composed of three main subsystems (cognitive, physical, and physiological) and

For the physiological system, HUMO attaches it to humans without enforcing the converse implication (i.e., without claiming that any entity having a physiological system must be human):

$$\begin{aligned} \text{soma:CognitiveAgent} &\equiv \exists \text{dul:hasPart.CogSys}; \\ \text{dul:PhysicalAgent} &\equiv \exists \text{dul:hasPart.PhysSys}; \\ \text{Human} &\sqsubseteq \exists \text{dul:hasPart.PhylSys}. \end{aligned} \quad (4.6)$$

Finally, HUMO introduces explicit concepts for **Function** and **Ability** to support task assignment and cognitive modelling. **Function** is modelled as an intrinsic aspect of a system, while **Ability** is modelled as a disposition that is associated with at least one function. The property **hasAbility** is specialised from **dul:hasQuality**, and its range is restricted to **Ability**:

$$\begin{aligned} \text{Function} &\sqsubseteq \text{soma:Intrinsic}; \text{hasFunction} \sqsubseteq \text{dul:associatedWith}; \\ \text{Ability} &\sqsubseteq \text{soma:Disposition}; \text{Ability} \sqsubseteq \exists \text{dul:associatedWith.Function}; \\ \text{hasAbility} &\sqsubseteq \text{dul:hasQuality}; \top \sqsubseteq \forall \text{hasAbility.Ability}. \end{aligned} \quad (4.7)$$

Since **soma:Intrinsic** is a subclass of **soma:PhysicalQuality** in SOMA, functions can be understood as system aspects that depend on the physical manifestation of the system—which remains compatible with both physical and cognitive features. Moreover, **Ability** is explicitly distinguished from **soma:Capability**; this distinction is important for the way HUMO and CDTO later represent task feasibility and assignments (discussed in detail in Section 4.6).

Cognitive abilities are then captured as a specialisation of **Ability** that must be associated with at least one cognitive function. In addition, HUMO allows the inference that any entity possessing a cognitive ability is a cognitive agent:

$$\begin{aligned} \text{CogAbility} &\sqsubseteq \text{Ability}; \text{CogAbility} \sqsubseteq \exists \text{dul:associatedWith.CogFunction}; \\ \exists \text{hasAbility.CogAbility} &\sqsubseteq \text{soma:CognitiveAgent}. \end{aligned} \quad (4.8)$$

4.5.2 Cognition Model and Link to Mental Tasks

Figure 4.7 focuses on the **cognitive interface between HUMO and SOMA**. It highlights the subset of SOMA concepts that are reused in HUMO to represent *mental activities* (in particular **soma:MentalTask**) and illustrates, through examples, how decision-related mental tasks can be semantically mapped to the corresponding cognitive functions introduced by HUMO.

The cognition model discussed in Section 2.1.4 is formalised in HUMO as a structured hierarchy under **CognitiveSystem**. Concretely, HUMO introduces a family of subsystem components **xComponent**, where **x** denotes one of the main cognitive spheres (**Sensation**, **Perception**, **Memory**, **Attention**, **Motor-Action**, **MetaCognition**, **Language**, and **HigherCognitiveFunction**), together with their internal subcomponents as depicted in Figure 2.1. This representation supports a system-oriented view of cognition: each component can be linked to the cognitive structures that are mobilised during task execution, and it enables a principled decomposition of cognition into reusable functional units. Although these components can be associated with brain areas activated during execution, HUMO does not attempt to formalise overlaps between brain regions; this aspect is intentionally left open for refinement by neuroscience specialists.

For each cognitive sphere **x**, HUMO introduces a coherent triplet:

1. a corresponding **xCognitiveFunction**,

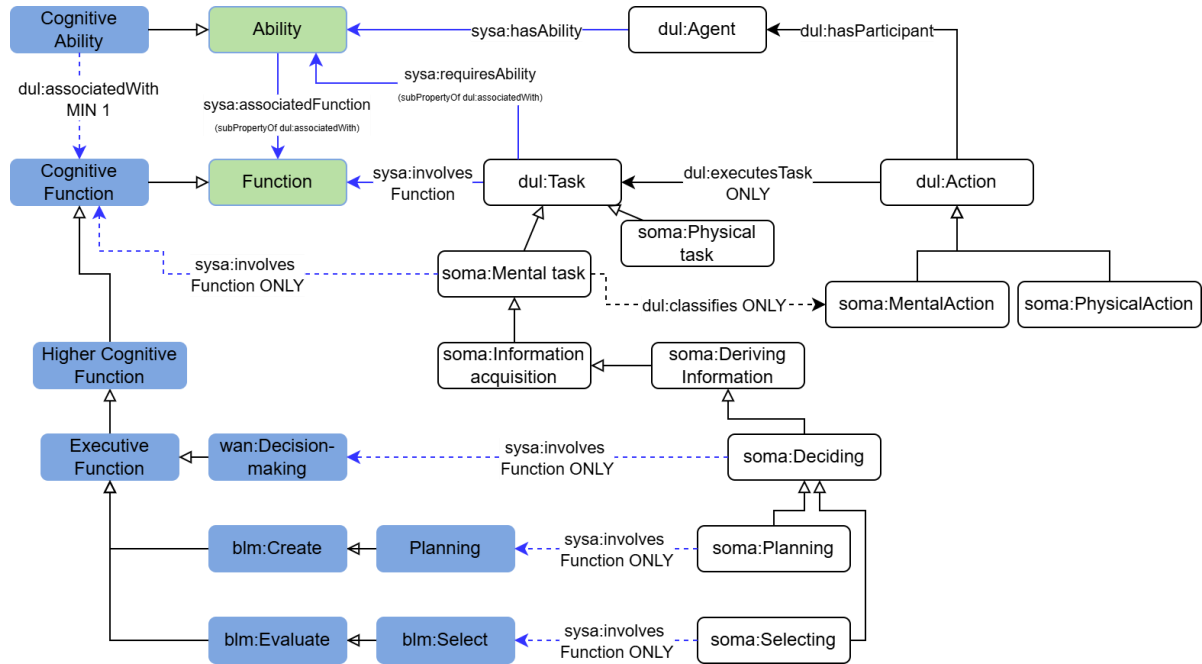


Figure 4.7: Cognitive functions linked to Mental tasks of SOMA in HUMO (Naudet et al., 2025).

2. a corresponding `xCognitiveAbility`, and
3. the `xComponent` that supports them at the system level.

In addition, HUMO distinguishes two major families of components, **GroundCognitiveSystem** and **HigherCognitiveSystem**. This design ensures that cognitive functions and abilities are not isolated labels, but are anchored in a system structure that can be reused for modelling, reasoning, and later capability/decision analyses.

SOMA, on the other hand, does not provide native concepts for cognitive functions, cognitive abilities, or cognitive components. What it does provide is a task-level entry point for mental activities through `soma:MentalTask` and related properties. HUMO exploits this entry point by explicitly linking SOMA mental tasks to the HUMO cognition hierarchy. Since the set of mental tasks provided by SOMA is mainly oriented toward executive and decision-related activities, HUMO maps them primarily to higher cognitive functions. To generalise this mechanism beyond the predefined SOMA task set, HUMO also constrains the task-function relation so that mental tasks can be annotated with the HUMO function vocabulary. This is achieved by specialising the task involvement property as follows:

$$\text{soma:MentalTask} \sqsubseteq \forall \text{involvesFunction.CognitiveFunction},$$

which means that any function involved in a mental task is required to be a **CognitiveFunction**. This alignment is important for the CDT: it allows tasks to be described not only in terms of their observable execution, but also in terms of the cognitive functions they require, which later supports explicit reasoning about cognitive workload, decision support, and the assignment of tasks in human–robot collaboration.

Finally, this task-function alignment motivates the next part of HUMO, where functions and abilities are related to tasks more explicitly. In particular, the following subsection (4.6) details how HUMO models task assignment based on the abilities of an entity, in a way that remains valid for both humans and robots, and for both physical and mental tasks. This formalisation directly prepares the transition to the CAA model introduced later in this chapter.

4.6 Capabilities, Abilities, and Affordances (CAA) Model in HUMO

The previous subsection established a semantic link between **cognitive functions** (introduced in HUMO) and **mental tasks** (available in SOMA). Building on this link, HUMO further clarifies how **functions** and **abilities** are represented and how they relate to the **tasks** that an entity can be assigned. This point is central in collaboration requires a shared, explicit representation of “who can do what”, and it must remain valid for both **humans and CPS**, regardless of whether the task is predominantly physical or mental.

Production engineering perspectives: In production engineering, the move toward re-configurable shopfloors and automated planning has promoted task-oriented programming approaches, in which resources must expose their skills so that they can be matched with production requirements (Brovkina and Riedel, 2019). Within the Product-Process-Resource (PPR/PDR) paradigm, a process can be described as a set of operations whose execution requires specific skills from the resources, while the product defines what is to be assembled (Backhaus and Reinhardt, 2017). In this view, skills provide the bridge between process needs and resource execution capacities, and are often defined as the ability to perform an activity or operation (Cândido and Barata, 2007). More recent work proposes the CCS extension (Capabilities, Skills and Services), explicitly separating capability, skill, and ability which is a separation that was not consistently applied in earlier work (Hammerstingl and Reinhart, 2018; Köcher et al., 2023). In particular, a *capability* is typically defined as an abstract, implementation-independent specification of a function to achieve an effect, while a *skill* is seen as an executable implementation of such a capability (Köcher et al., 2020, 2023). However, the literature still contains divergent definitions, and the term “skill” is sometimes used interchangeably with ability or capability (Gonnermann et al., 2020).

Human sciences perspectives: From work psychology, tasks and activities are defined in relation to goals and execution conditions: a task is a goal to be attained under given constraints, and an activity is the concrete response produced by an individual in order to satisfy the task (Leplat, 1990, 2004). Skills are then understood as the ability to execute a task, and are often treated as specialisations of broader abilities (Fleishman, 1975). These perspectives highlight that ability-related modelling must apply naturally to humans, not only to machines, and must remain meaningful when describing cognitive demands (e.g., decision-making) as well as physical demands (e.g., manipulation).

HUMO modelling choices: HUMO adopts a system-level perspective in which **abilities are properties of an agent/system** and are grounded in **functions**. This allows a unified modelling approach for human and robot agents: an agent can be assigned a task if it has (or can realise) the ability required to execute the function underlying that task. HUMO therefore distinguishes:

- **Function** (what transformation the system can realise),
- **Ability** (a disposition that is associated with at least one function),
- **Task** (a goal to be achieved under given conditions),

and introduces relations such as **hasAbility** (agent $\rightarrow$ ability) and **requiresAbility** (task $\rightarrow$ ability) to enable explicit task assignment reasoning.

4.6.1 Formalisation with SOMA Affordances

SOMA provides a precise formal mechanism to connect **dispositions** to **tasks** through **affordances** and **roles**. In SOMA, **Capability** is defined as a specific kind of disposition associated with agents and tasks, and affordances provide the description layer that links a disposition to the task it enables through the roles played by participants (Beßler et al., 2020). In this mechanism, a **Bearer** carries the disposition, a **Trigger** exploits it, and both participate in the execution of the task described by the affordance. This structure is particularly useful in collaboration settings, because it makes explicit *which participant must have which role* for a task to be feasible.

HUMO reuses this affordance mechanism to keep interoperability with SOMA, while maintaining an explicit distinction between **capability** and **ability**. The underlying rationale is that SOMA’s capability construct captures task-oriented enabling properties at a high level, whereas HUMO’s ability concept is additionally grounded in functions and is used directly for assignment reasoning (e.g., a task requiring an ability, an agent possessing an ability). The resulting modelling is illustrated in Figure 4.8.

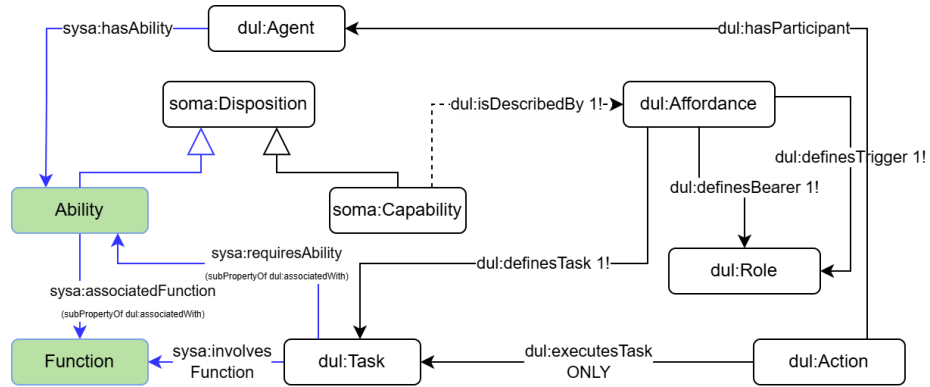


Figure 4.8: Ability-Capability-Task part of the HUMO, integrated with SOMA (Naudet et al., 2025).

In this model, an **Ability** is associated with a **Function** it requires, tasks specify which abilities are required, and agents declare which abilities they have. Furthermore, HUMO expresses that an agent can be considered as having a relevant ability only when the corresponding capability/disposition is present and the affordance structure is satisfied in the situation (i.e.,

the required roles and participants are available). This is captured through the following rule:

$$\begin{aligned}
 \text{hasAbility}(\text{ag}, \text{ab}) \leftarrow & \text{Capability}(\text{ab}) \wedge \text{hasDisposition}(\text{ag}, \text{ab}) \\
 & \wedge (\exists \text{aff}) \text{isDescribedBy}(\text{ab}, \text{aff}) \\
 & \wedge ((\exists \text{bear}) \text{definesBearer}(\text{aff}, \text{bear}) \wedge (\exists \text{obj}) \text{hasRole}(\text{obj}, \text{bear})) \\
 & \wedge ((\exists \text{trig}) \text{definesTrigger}(\text{aff}, \text{trig}) \wedge \text{hasRole}(\text{ag}, \text{trig})).
 \end{aligned} \tag{4.9}$$

This rule operationalises the intuition that ability is not only an internal property of the agent: it becomes *actionable* when the environment provides the right affordances and the agent can occupy the correct role in the task execution context. This perspective directly prepares the next section, where the **Capabilities-Abilities-Affordances (CAA)** model is introduced as a core semantic structure of CDTO for reasoning about feasibility, action selection, and cognitive interoperability in .

4.7 SWRL Rules for Reasoning and Inference Mechanisms

In collaborative assembly, the CDT must determine whether a *human* or a *cobot* (considered as a CPS) is able to perform a given task. This assessment cannot rely only on the static capability descriptions stored in the ontology: it also depends on **contextual factors** such as tool availability, the presence of relevant affordances in the environment, and sequencing constraints between tasks and actions. To support such context-dependent inferences, CDTO is complemented with a set of **SWRL rules** that deduce implicit knowledge from explicitly instantiated individuals. The rules presented below cover (i) *ability acquisition* from capabilities and affordances, (ii) *action feasibility* based on required abilities, (iii) *fallback assignment* to ensure robustness when deviations occur, and (iv) *temporal dependencies* to preserve the correct execution order.

Ability acquisition from capability, affordance, and context: The central rule used in our framework adapts SOMA’s capability-affordance mechanism to collaborative assembly. An agent acquires an ability only when three elements converge: (1) the agent has the underlying capability/disposition; (2) an affordance exists that describes the disposition in terms of roles and task type; and (3) the required roles are instantiated in the current context, including the availability of the appropriate tool. This transition from capability to actionable ability is already formalised by Rule (4.9).

This rule makes explicit the idea that ability is not purely an internal property of the agent: it becomes *operational* only when the environment provides the appropriate affordance structure and the necessary resources (e.g., tools) are available.

Action feasibility from task requirements: Once an ability has been inferred, a second rule determines whether the cobot can perform a specific action associated with a task. The rule checks the task requirements and asserts `canPerform` only when the cobot satisfies the required ability. This provides a direct semantic bridge between task descriptions and executable actions.

$$\begin{aligned}
 \text{Task}(\text{tsk}) \wedge \text{requiresAbility}(\text{tsk}, \text{ab}) \wedge \text{isExecutedIn}(\text{a}, \text{tsk}) \\
 \wedge \text{Cobot}(\text{co}) \wedge \text{hasAbility}(\text{co}, \text{ab}) \rightarrow \text{canPerform}(\text{co}, \text{a}).
 \end{aligned} \tag{4.10}$$

Fallback assignment for robustness: Human-robot collaboration requires robustness to deviations, including incomplete or delayed human actions. When an action is still pending for the operator, the following rule reassigns it to the cobot, provided that the cobot has already been inferred as able to perform it. This implements a simple but effective semantic fallback mechanism.

$$\begin{aligned} \text{Action}(a) \wedge \text{Operator}(op) \wedge \text{performedBy}(a, op) \wedge \text{hasExecutionState}(a, \text{Pending}) \\ \wedge \text{Cobot}(co) \wedge \text{canPerform}(co, a) \rightarrow \text{performedBy}(a, co). \end{aligned} \quad (4.11)$$

Inferring temporal ordering between actions: To preserve the logical sequence of an assembly procedure, temporal relations declared at the task level must be propagated to the action level. Rule (4.12) transfers a **follows** relation between tasks to the corresponding actions that execute them.

$$\begin{aligned} \text{Task}(t_1) \wedge \text{Task}(t_2) \wedge \text{follows}(t_2, t_1) \wedge \text{isExecutedIn}(t_1, a_1) \wedge \text{isExecutedIn}(t_2, a_2) \\ \rightarrow \text{follows}(a_2, a_1). \end{aligned} \quad (4.12)$$

Ordering constraints for hierarchical task decomposition: When tasks are decomposed into subtasks, sequencing constraints must be preserved at the subtask level. Rule (4.13) ensures that the first subtask of a task B is constrained to follow the last subtask of a task A when B is declared to directly follow A .

$$\begin{aligned} \text{Task}(A) \wedge \text{Task}(B) \wedge \text{directlyFollows}(B, A) \wedge \text{hasPart}(A, l_A) \wedge \text{isLastSubtask}(l_A, \text{true}) \\ \wedge \text{hasPart}(B, f_B) \wedge \text{isFirstSubtask}(f_B, \text{true}) \rightarrow \text{directlyFollows}(f_B, l_A). \end{aligned} \quad (4.13)$$

From symbolic inference to behavioural control: Although these SWRL rules allow the ontology to deduce which actions are feasible and under which conditions, they do not by themselves decide *when* an action should be executed. Their operational use therefore requires a cognitive mechanism capable of selecting and learning behaviours. In our framework, this role is provided by the CLARION architecture, which interprets the inferred capabilities/abilities and regulates the activation and learning of action patterns. Chapter 5 will illustrate this reasoning approach through a concrete use-case, where the SWRL rules are executed within the CDTO ontology to support [Human-Robot Collaboration \(HRC\)](#). In particular, it shows how CDTO enables the CDT to infer additional knowledge including newly applicable capabilities for the robot.

4.8 Chapter Conclusion

This chapter provides the semantic foundation of the thesis and directly addresses **RQ3** and **RQ4**. For **RQ3**, we have formalised the knowledge structures needed to represent collaborative situations in HCC, and have defined CDTO as the ontology-based conceptual model that supports a shared and machine-interpretable view of agents, tasks, roles, resources, context, constraints, and interaction state. This semantic layer supports HCC by ensuring that heterogeneous components (perception, planning, execution monitoring, explanation, and external services) rely on consistent meanings and compatible representations.

For **RQ4**, we have explained how ontology-based knowledge is operationally exploited within a CDT and how it interfaces with the cognitive architecture selected in the previous chapter (*CLARION*). In particular, it shows how the ontology structures explicit knowledge (concepts, relations, constraints) and how SPARQL retrieval, OWL axioms, and SWRL rules support decision-relevant inference, thereby connecting learned signals and observations to explicit symbolic reasoning in a form that remains interpretable.

The main contribution of this chapter is the **CDTO ontology**. CDTO extends and integrates SOMA, DUL, and SOSA/SSN with two modules: SOMA+, which captures the HCC environment, and HUMO, which represents human cognitive/physical characteristics and human state.

Nevertheless, **CDTO can be extended** to cover a wider variety of HCC scenarios by enriching the domain vocabulary, adding specialised modules, and refining constraints and rules as required by new contexts. In addition, the integration with CLARION requires use-case validation to demonstrate, in end-to-end experiments, how ontology-structured explicit knowledge impacts collaborative decision-making, explainability, and adaptation.

Chapter 5 builds on these results by introducing a concrete collaboration use-case in which the SWRL rules are executed over CDTO instances to infer new actionable knowledge (e.g., capabilities/abilities that become available to the robot under specific conditions), supporting adaptive task allocation and decision-making in HCC.

Chapter 5

Use Case Implementation and Evaluation of Cognitive Interoperability

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Introduction

This chapter presents the experimental validation of the concepts, architectures, and models introduced in the previous chapters, using an industrial Human-Robot Collaboration (HRC) use case focused on assembling a motor product. The experiments are conducted on the technical platforms and production facilities of the *Atelier Inter-Établissements de Productique Lorrain (AIPL)*, and the studied assembly corresponds to a product manufactured and used in this school-factory. The main objective is to demonstrate how cognition can be implemented in a Cognitive Digital Twin (CDT) to support cognitive interoperability between human operators and robotic systems in a collaborative manufacturing task. In this respect, the chapter contributes to the **whole set of research questions (RQ1–RQ4)**: it provides an initial application of the maturity model of cognitive interoperability to assess the achieved level in the use case (RQ1), operationalises the notion of a CDT and its cognitive functions in a concrete industrial setting (RQ2), exploits explicit knowledge representations through the CDTO ontology to structure context, roles, tasks, and capabilities (RQ3), and integrates a cognitive architecture (CLARION) to enable reasoning, adaptation, and decision-making aligned with these representations (RQ4).

The chapter begins with a detailed description of the HRC use case, covering the industrial assembly process, the respective roles of human and robotic agents, and interaction scenarios. It then explains how the CDTO ontology is instantiated to represent capabilities, abilities, affordances and contextual knowledge, thereby enabling reasoning and inference during collaboration. Subsequently, the integration and configuration of the CLARION cognitive architecture within the CDT is detailed, with a particular focus on Q-learning-based learning mechanisms, including architectural design choices and training procedures.

Finally, the system is assessed using the cognitive interoperability maturity levels defined in previous chapters to determine how far the proposed solution moves beyond semantic interoperability and to position the achieved results within the MMCI framework (*RQ1*).

5.1 Use Case Description: Human-Robot Collaborative Assembly

The experimental validation is conducted through an industrial [HRC](#) use case. This use case provides a realistic framework for evaluating how a [CDT](#) can support the various cognitive functions that enable coordination, decision-making and adaptive behaviour in collaborative manufacturing.

To clearly present the scenario, the section is structured into three subsections: Subsection [5.1.1](#) introduces the AIPL workstation environment and its physical organisation, Subsection [5.1.2](#) describes the motor product, its components, and the assembly sequence, and Subsection [5.1.3](#) explains the roles and responsibilities of the human operator and the robot during the collaborative execution of the task.

5.1.1 Workstation overview and collaborative setting

The use case consists of a collaborative assembly task performed by a human operator and a cobot (UR5e). The cobot is considered as a CPS because it combines physical manipulation

capabilities (robotic arm and end effector), digital control (robot controller and software) and interaction with the surrounding environment. The cobot interacts with the human operator to assemble an electric motor (the product, components and assembly sequence).

Figure 5.1 shows the workstation configuration using two complementary views. The left part of the figure presents a schematic representation of the workspace, highlighting the spatial organisation that supports collaboration. The station is designed around a shared table where the human and the robot can both access the necessary parts and execute their actions. The layout includes separate areas dedicated to (i) the storage of parts, (ii) the assembly process itself, and (iii) supporting tools and elements required during the task. A tablet is also integrated into the workstation, serving as an interaction and supervision support. This interface can be used to guide the operator, display task status, or exchange information with the robotic system during collaboration.

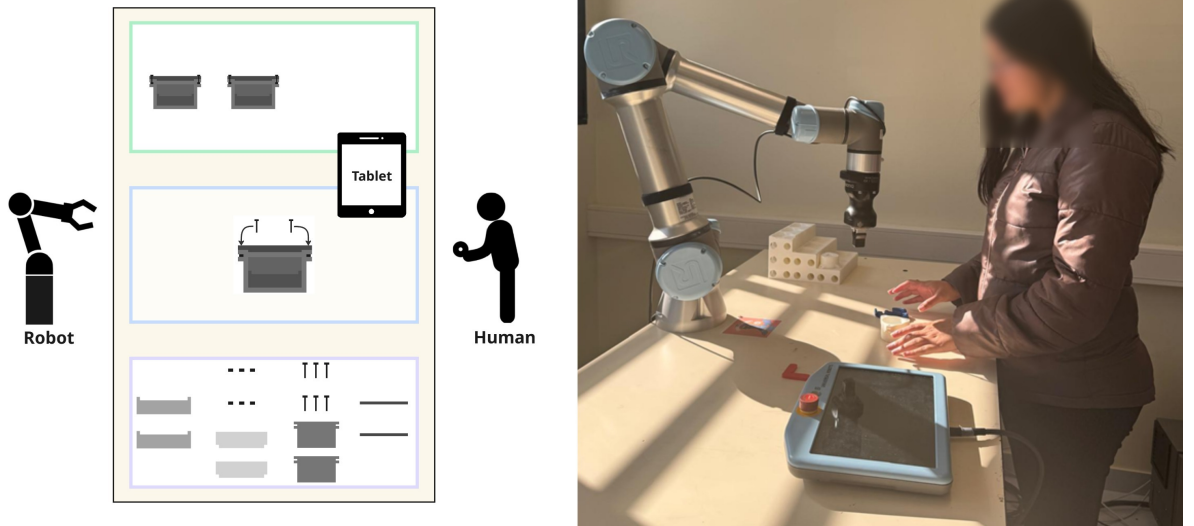


Figure 5.1: Physical setup of the collaborative assembly workstation. Left: schematic layout. Right: UR5 cobot and human operator.

The right part of Figure 5.1 illustrates the real experimental setup at AIPL. The UR5 cobot is mounted next to the working table and is equipped with an end-effector that enables part manipulation. The human operator stands in close proximity to the robot, and the collaboration happens in a shared and accessible space. This physical configuration reflects a typical industrial HRC situation where both agents work on the same product and coordinate their actions sequentially or alternately, depending on the assembly step and the current context.

In addition to the general workstation organisation, Figure 5.2 provides a top view of the working surface and clarifies how the assembly environment is structured into two main functional regions:

- The Stock area (red frame) contains the components before assembly. Parts are arranged in trays or containers to ensure repeatable grasping and reduce operator search time.
- The Assembly zone (green frame) corresponds to the central workspace where the assembly is physically performed. This area is shared during collaboration: the robot may place parts

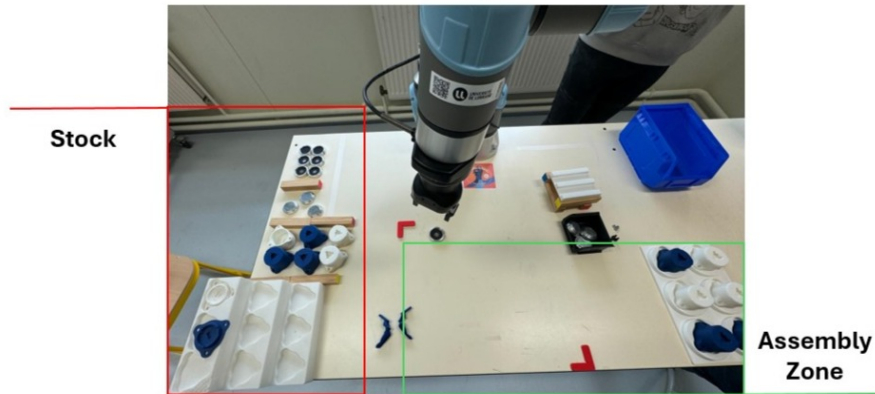


Figure 5.2: Workspace zones for motor assembly: stock (red) and assembly (green).

in this zone and the human may assemble, adjust, and validate the intermediate product states.

This separation between stock and assembly zones is important for collaboration because it supports (i) clearer task allocation, (ii) safer motion planning for the robot, and (iii) easier monitoring of the current assembly state.

5.1.2 Assembly description and sequence of operations

Figure 5.3 summarises the product structure and the assembly logic through a step-by-step representation and the final product. The product is composed of the following elements:

- Rotor
- Stator
- Box (housing)
- Lid
- Nuts (3x)
- Screws (3x)

The assembly is decomposed into four main steps:

1. Step 1: Rotor-stator sub-assembly: the assembly begins by combining the rotor (1) and the stator (2) to create an initial sub-assembly. The process includes inserting one part into the other, followed by a re-orientation (rotation) to reach the correct alignment for the next stage.
2. Step 2: Insertion into the box: the rotor-stator sub-assembly is then inserted into the box (3). This step establishes the main structure of the product and defines the reference orientation for the closure and fastening operations.
3. Step 3: Closing with the lid: the lid (4) is placed on top of the box to close the assembly. Correct placement is important here to ensure proper alignment of the fastening points.

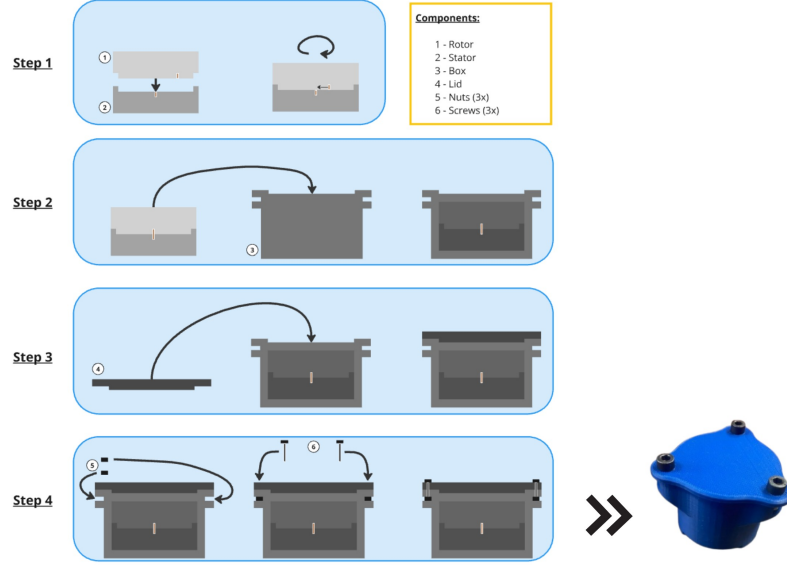


Figure 5.3: Product Description and Assembly Steps.

4. Step 4: Fastening: the assembly is completed by placing the nuts (5) and tightening the screws (6) (three fastening points). This step produces the final assembled motor product and typically requires careful handling and verification.

To provide a more detailed view of the operational structure, Figure 5.4 presents a hierarchical decomposition of the assembly task, including the main phases and the associated elementary actions.

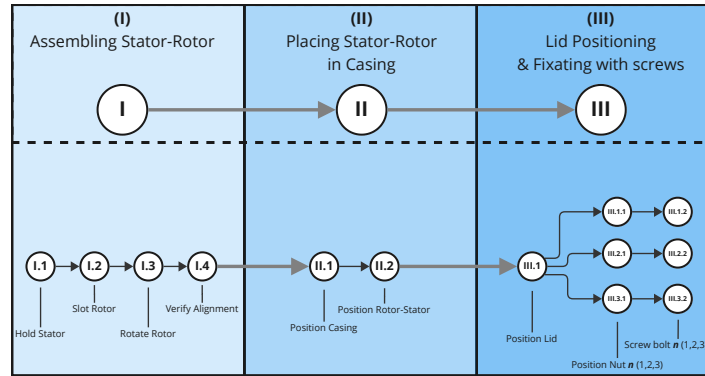


Figure 5.4: Hierarchical decomposition of the collaborative motor assembly process.

The task is structured into three main phases (I–III), each described by elementary operations (e.g., holding, inserting, positioning, and fastening) executed sequentially during collaboration.

5.1.3 Human and Robot Roles in the HRC Scenario

In this use case, the UR5e cobot and the human operator cooperate to assemble the motor product by sharing the same workstation. The assembly structure introduced in Figure 5.4 is

used to define a clear distribution of actions between the two agents.

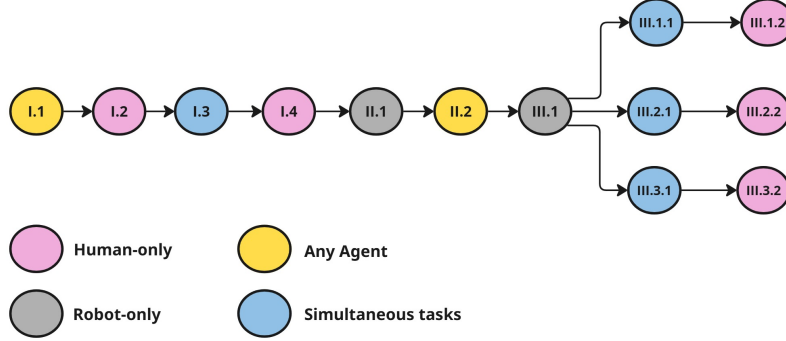


Figure 5.5: Task-agent matching for the collaborative motor assembly process.

Figure 5.5 presents the task-agent matching, where the elementary operations are categorised as human-only, robot-only, executable by any agent, or performed simultaneously. Human-only tasks correspond to operations that require fine manipulation and judgement, such as slotting the rotor (I.2), verifying alignment (I.4), and fastening with screws (III.1.2-III.3.2). In contrast, robot-only tasks are mainly related to stable and repeatable positioning, such as positioning the casing (II.1) and placing the lid (III.1). Some actions, like holding the stator (I.1) and positioning the rotor-stator sub-assembly (II.2), can be executed by either agent depending on the collaboration strategy and the current context. Finally, simultaneous tasks occur during the final phase, where the robot can support the process by positioning nuts (III.1.1-III.3.1) while the human performs the tightening operations, improving efficiency and reducing idle time.

Furthermore, certain tasks are not limited to physical actions alone, because each agent must also perform cognitive processes (2.1.4) such as perception, reasoning, and decision-making to ensure correct execution. This aspect is illustrated in the figure 5.6 through two representative tasks: I.2 “Slot Rotor” and I.3 “Rotate Rotor”.

Each task is structured into two complementary layers: the green area corresponds to cognitive activities (e.g., searching for the component, locating the stator, checking alignment, or deciding the next action), while the yellow area represents the physical activities required to execute the task (e.g., picking the item, moving the arm, applying pressure, or rotating the object). In addition, the red arrows indicate the abilities needed to connect cognitive steps to their corresponding physical actions, highlighting the dependency between perception and manipulation.

5.1.4 Motivation and problem statement

Although the collaborative motor assembly process is defined with a planned task allocation, real industrial collaboration remains dynamic and uncertain. In our use case, the human operator is normally expected to perform the screwing operations during Phase III, especially the actions $III.i.2$ with $i \in \{1, 2, 3\}$, which correspond to the final steps required to complete the assembly. However, during execution, the human may decide not to perform this step for practical reasons (e.g., workload, interruption, preference, or time constraints). For example, starting from motor number 3, the operator may stop handling the screwing subtasks and leave them pending.

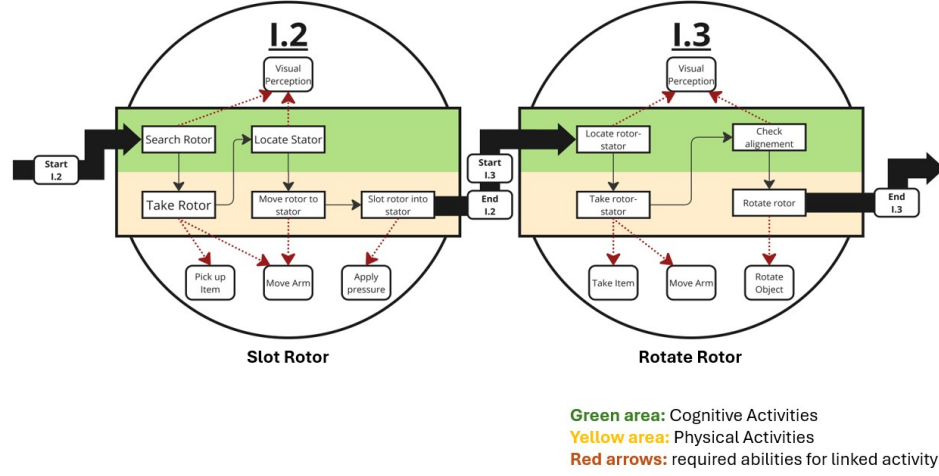


Figure 5.6: Cognitive and physical activity decomposition of an assembly task.

This situation creates a key decision problem for the collaborative system: the robot must determine whether it should take responsibility for the missing task or keep waiting for the human. A simple rule-based robot control is not sufficient in this case, because the decision depends on multiple contextual elements such as task progress, feasibility conditions, tool availability, and the robot's own capabilities. Therefore, the robot requires a mechanism that allows it to interpret the situation, reason about its abilities, and adapt its behaviour.

To address this issue, the proposed solution combines two complementary components. First, the **CDTO ontology** provides a structured knowledge base that represents the tasks, agents, tools, capabilities, and context of the collaboration. This allows the system to check whether the robot has the required capability to execute a missing action and whether the current conditions make the action feasible. Second, when the robot faces a task that is new or not previously learned, the system relies on the **CLARION cognitive architecture** to support decision-making and activate learning mechanisms. In such cases, CLARION enables the robot to learn how to execute the required action and integrate it into the collaboration flow.

5.2 Cognitive Digital Twin for the HRC Use Case

Each agent involved in the collaboration is associated with its own digital representation at the cyber layer. As illustrated in Figure 5.7, the cobot is linked to a CDT that embeds not only data and models, but also integrates cognitive capabilities to enable interpretation, reasoning, and adaptation on the CPS side. In contrast, the human agent is **inherently cognitive**; the associated **HDT** therefore focuses on representing human-related data and models, and on making these available for the collaboration context.

In our work, we focus on the CDT of the UR5e cobot, since it constitutes the core contribution for enabling cognition and adaptation on the CCPS side. The HDT is developed in parallel within the project and is addressed as separate work. The overall objective here is thus to integrate cognitive mechanisms into the CDT in order to support cognitive interoperability between

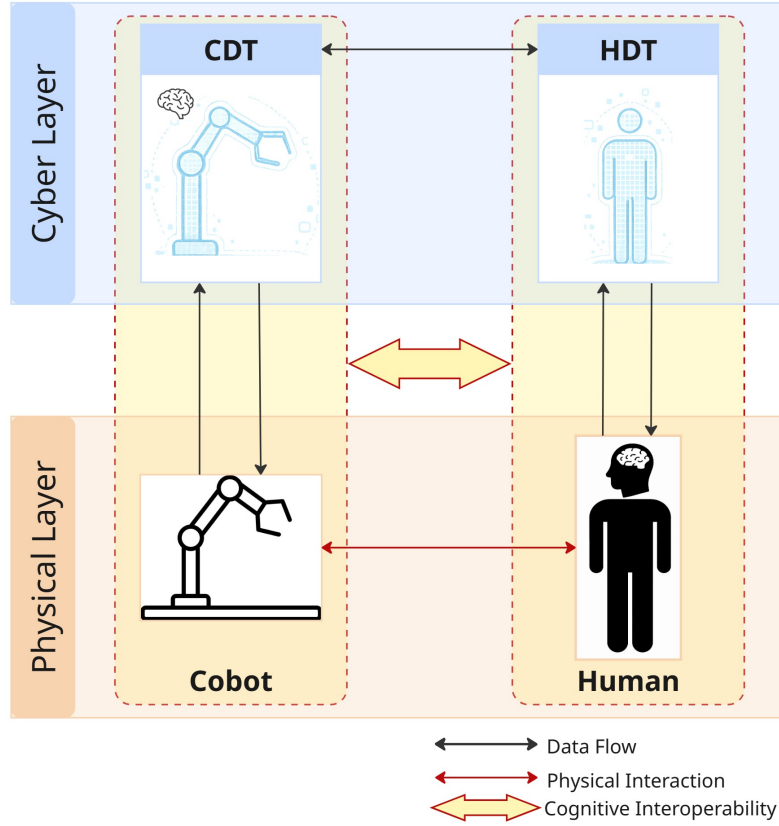


Figure 5.7: Conceptual Architecture for Human-CPS Collaboration.

the human and the cobot during collaborative assembly. Step by step, the following sections will present the implementation of this CDT and will evaluate, at the end of the chapter, whether the proposed solution achieves cognitive interoperability and at which interoperability level it can be positioned. Therefore, this section details the architecture of the CDT developed for the use case.

This thesis is developed within the framework of the AI4C2PS project, which provides a comprehensive end-to-end architecture for connecting the physical collaborative cell to its digital services. We adopt the data formats and interoperability mechanisms used to exchange information (e.g., NGSI-LD for exchanging context information annotated with shared vocabularies) to ensure that our CDT implementation remains consistent with the project.

The CDT receives and exploits two complementary sources of information:

- Data-driven information, collected from the physical space through the IoT platform (e.g., current task state, observed actions, system feedback).
- Knowledge-driven information, provided by the CDTO ontology (introduced in chapter 4, which allows the CDT to interpret the situation using semantic and contextual concepts (e.g., task structure, agent capabilities, object affordances, and expected constraints).

As explained in chapter 3, the CDT is structured around three complementary functions: emulation, cognition, and simulation. These functions operate at the digital space level and interact

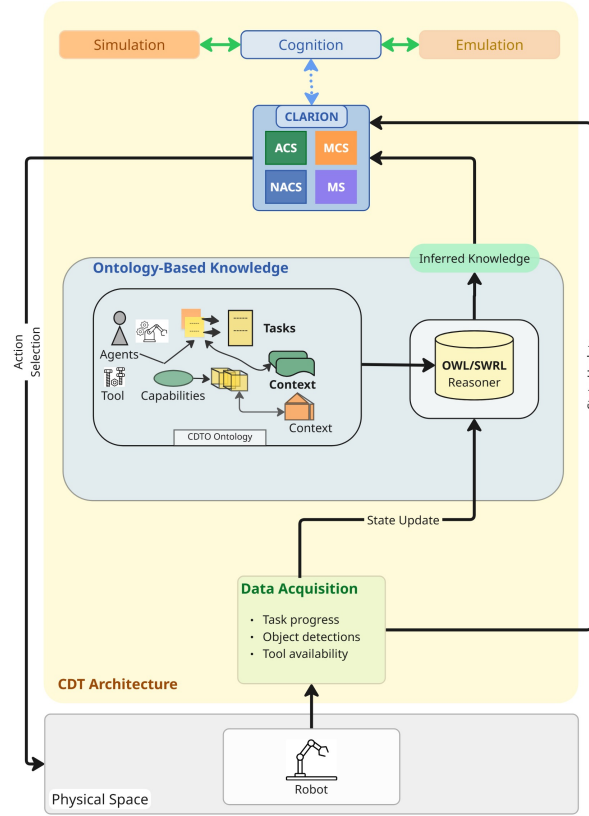


Figure 5.8: Architecture of the CDT where cognition exploits CDTO ontology and is driven by CLARION.

with the physical space through data acquisition and control mechanisms and are represented in the figure 5.8. Our developments cover the **cognition** and **simulation** functions. On the cognition side, we integrated the **CDTO ontology** to structure and exploit semantic knowledge about tasks, agents, and objects, and we connected it to the **CLARION** cognitive architecture cognitive architecture via its subsystems, ACS, NACS, MS, MCS, see section (5.4.1) to support decision-making, task allocation, and action selection based on both perceived state and ontological knowledge. On the simulation side, we implemented a dedicated learning setup to support RL-based acquisition and validation of procedural skills (e.g., screwing) in a safe environment, before transferring the learned policy to execution (i.e., *sim-to-real transfer* / transfer learning). These two functions constitute the core contribution of the CDT implemented in this chapter.

To detail this implementation, the following subsections present, first, the ontology instantiation and the reasoning process used to represent the assembly context and to infer relevant knowledge (Section 5.3). Then, the integration of the CDT cognition function is explained through the configuration of the CLARION cognitive architecture, focusing on how it supports decision-making during collaboration (Section 5.4). Finally, the chapter describes the learning mechanisms and training procedure, highlighting how reinforcement learning is applied to improve the system’s adaptability and performance over time (Section 5.4.3).

This part focuses on the technical implementation that makes the ontology usable in the HRC scenario. The goal is to instantiate the ontology automatically using the data coming from

the environment, which describes the different collaboration steps during the assembly process. In practice, the workflow starts by converting the scenario data into **NGSI-LD** entities for exchange and context management, which are then **mapped into ontology assertions** that can be processed by the knowledge base. Then, the ontology is manipulated using **Owlready2**, a Python library dedicated to OWL ontologies. Owlready2 allows us to load the modular ontology and create instances corresponding to the detected elements (tasks, objects, and context). After instantiation, reasoning is performed using two main mechanisms:

- logical inference with the **HermiT** reasoner to infer implicit knowledge from the available information,
- SWRL rules (presented in Section 4.7) to add additional knowledge required for the cobot.

The choice of **HermiT** is motivated by its suitability for **OWL 2 DL** ontologies and its support for **robust and comprehensive** descriptive logical reasoning. This is necessary for calculating logical implications based on instantiated knowledge, such as ontological classification, instance verification (e.g., class membership) and consistency checking with respect to property restrictions and constraints. In our case, these guarantees are important because inferred knowledge (such as feasibility conditions or deduced abilities) directly impact downstream decision-making. In addition, HermiT is natively supported by Owlready2's reasoning interface, which simplifies integration and enables a fully automated pipeline (instantiation → reasoning → updated knowledge base) without relying on external manual tooling. This approach was chosen because it is flexible, clear to implement, and efficient for our scenario scale, and it supports automatic instantiation and reasoning adapted to the collaborative assembly use case.

5.3 Ontology Instantiation and Reasoning

The CDTO ontology provides the semantic model that enables the system to represent the collaborative environment, interpret perceptions, and deduce the knowledge necessary for decision-making. It structures this context using explicit concepts such as agents, tasks, objects, capabilities, aptitudes, and affordances, which were introduced in Chapter 4.

This part focuses on the technical implementation that makes the ontology usable in the HRC scenario. The goal is to instantiate the ontology automatically using the data coming from the environment, which describes the different collaboration steps during the assembly process. As summarised in Figure 5.9, the workflow starts by converting the scenario information into **NGSI-LD**¹² files that are processed by a Python-based mapping mechanism, which translates context information into ontology-compliant OWL assertions. Then, the ontology is manipulated using **Owlready2**, a Python library dedicated to OWL ontologies. Owlready2 allows us to load the modular ontology and create instances corresponding to the detected elements (tasks, objects, and context). After instantiation, reasoning is performed using two main mechanisms:

- logical inference with the HermiT reasoner (Glimm et al., 2014) to deduce implicit information from the available data,
- SWRL rules (presented in Section 4.7) to add additional knowledge required for the cobot.

¹²<https://ngsild.org>

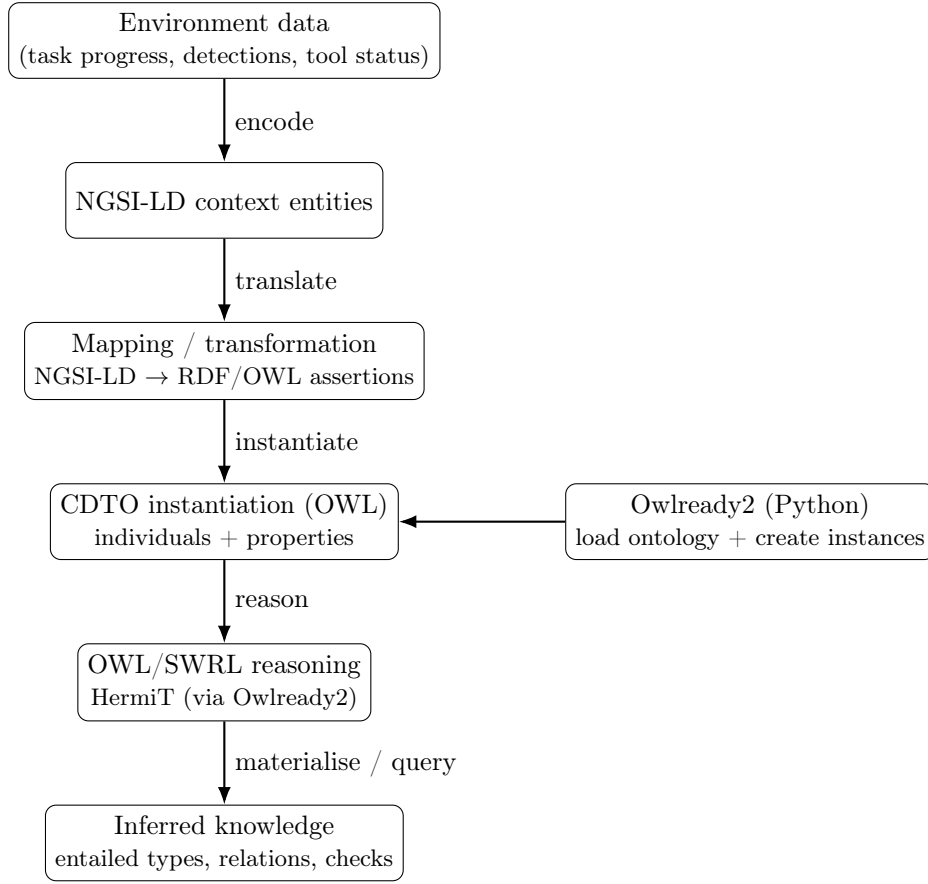


Figure 5.9: Workflow from environment data to CDTO instantiation and inferred knowledge.

5.3.1 Creation of the NGSI-LD input file

The NGSI-LD file aims to encode, in a structured way, the instances and relationships required by the scenario described above. It enables the encoding of the main entities involved in the assembly process, such as the cobot, the human operator, the tools (e.g., the screwdriver), and the different tasks defined in the collaboration workflow. This file is a key element for the automatic instantiation of the ontology used in this work.

5.3.1.1 Structure of the NGSI-LD file

The NGSI-LD file is organised around two complementary parts:

1. The **@context** establishes the mapping between the terms used in the NGSI-LD file and the corresponding IRIs of the concepts defined in the reference ontologies. It ensures that each entity type and each relation declared in the file is aligned with the ontology vocabulary and structure.
2. The **@graph** contains the actual entities describing the scenario. Each entity is identified by a unique [Uniform Resource Name \(URN\)](#), and is associated with attributes that represent data properties (e.g., names or states) as well as relationships linking it to other entities.

A representative extract illustrating this structure is provided in Appendix A (Listing A.1).

This organisation provides a precise representation of the instantiated elements and their interconnections, which is necessary for the correct instantiation of the ontology and the execution of reasoning mechanisms during the collaboration.

5.3.1.2 Modelling the steps of the screwing scenario

The steps of the HRC scenario are modelled using three main ontology concepts: **Workflow**, **Task**, and **Action**. The **Workflow** captures the overall process, each **Task** represents a scenario step, and **Action** elements provide a finer decomposition into executable operations as illustrated in Figure 5.10.

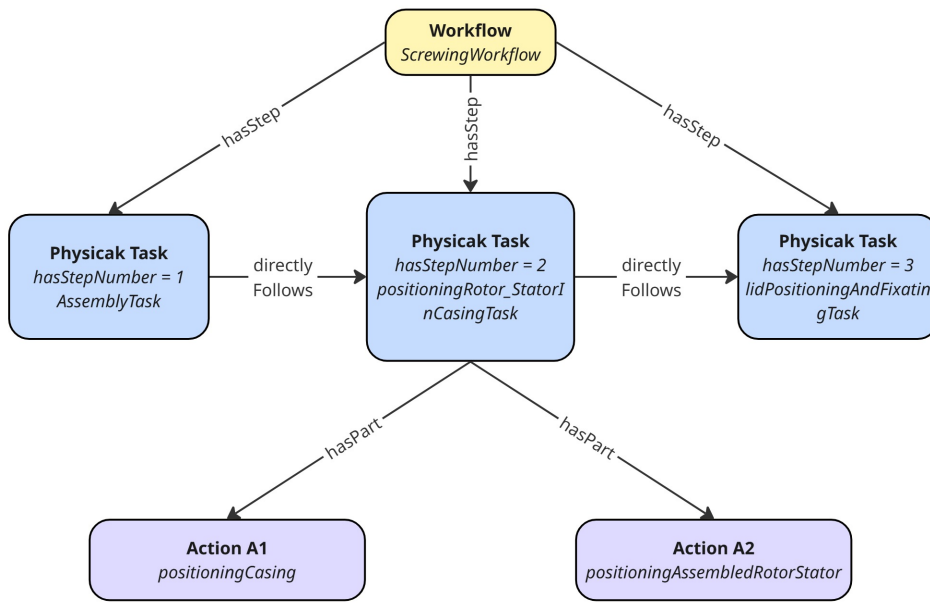


Figure 5.10: Workflow, Task, and Action Structure for the HRC Scenario.

To ensure an explicit and exploitable execution logic, temporal ordering between tasks is represented using **directlyFollows**, while **hasStepNumber** provides an unambiguous step identifier. A representative excerpt illustrating the workflow encoding is given in Appendix A (Listing A.2).

Beyond physical operations, the scenario also involves cognitive activities (e.g., perception and alignment checking), previously discussed in relation to Figure 5.6. To capture these aspects semantically, we represent such activities as **MentalTask** instances, connected to required abilities and cognitive functions through relations such as **requiresAbility** and **involvesFunction**. An example (**SearchRotor**) is provided in Appendix A (Listing A.3). This allows the CDT to maintain a unified representation of both physical and cognitive dimensions of the collaborative process.

5.3.2 Automatic ontology instantiation and inference

Once the NGSI-LD file is produced, the ontology is populated automatically. In practice, the instantiation process follows a robust two-phase strategy:

1. **Instance creation:** each NGS-LD entity in `@graph` is mapped to an ontology class (based on `type`) and instantiated as an individual (named from `id`). At the end of this phase, all entities exist in the knowledge base.
2. **Property and relation assignment:** the instantiated individuals are then enriched with (i) data properties (literal values such as step numbers) and (ii) object relations (links such as `hasPart` or `directlyFollows`).

This separation prevents reference errors and ensures that relations always point to existing individuals. The technical scripts and excerpts corresponding to these steps are provided in Appendix A.

5.3.3 Inferences

The SWRL rules used in this work (e.g., *HasAbilityRule* and *FallbackToCobot*) were introduced earlier (Section 4.7). Here, we focus on the effect of applying them to the instantiated knowledge base in the HRC scenario.

The key objective of the inference process is to infer knowledge that is **not explicitly declared** in the NGS-LD input. In particular, we aim to infer that the cobot possesses the competence to perform screwing, represented as `canScrew`. This competence is not provided as an input attribute; instead, it is derived from semantic conditions encoded in the ontology, such as the cobot's available components/tools, required capabilities, and the affordances associated with the screwing task.

After instantiation, reasoning (OWL semantics + SWRL rules) enriches the knowledge base with inferred facts. To illustrate the added value, we contrast the cobot description *before* and *after* inference. In the declared input, the cobot is described only through explicit information (identity, type, and associated tool), without any competence statement. After inference, the knowledge base contains an additional relation `hasAbility` linking the cobot to `canScrew`, showing that the system can derive operational competences from contextual and structural knowledge. Representative excerpts of both states are provided in Appendix A (Listings A.9 and A.10).

This result demonstrates the interest of ontology-based reasoning for HRC: the system enriches the agent model by adding explicit competences derived from capabilities, tools, and contextual conditions. This inferred knowledge can then be exploited by the CDT and communicated to the CLARION cognitive architecture to support decision-making and task allocation, as will be detailed in the next section.

5.4 CLARION Cognitive Architecture for the use-case

After detailing the CDTO ontology instantiation and inference mechanisms, this section introduces how the **CLARION cognitive** architecture is integrated into the CDT to support decision-making in the collaborative motor assembly scenario. In our use case, CLARION is used as a cognitive layer that exploits both:

- the semantic knowledge provided by the CDTO ontology, including both the explicit knowledge already instantiated (e.g., workflow structure, task constraints, agent capabilities) and the additional knowledge inferred through reasoning,

- the current state of the environment provided by data collection (e.g., object presence, task progress, and human availability).

The goal is to support context-aware decision-making, including action selection and dynamic assignment of upcoming tasks between the human and the robot.

5.4.1 Integration of ontology reasoning with CLARION

Figure 5.11 shows an updated version of the original CLARION architecture. This version has been modified to integrate the CDTO ontology and associated reasoning processes.

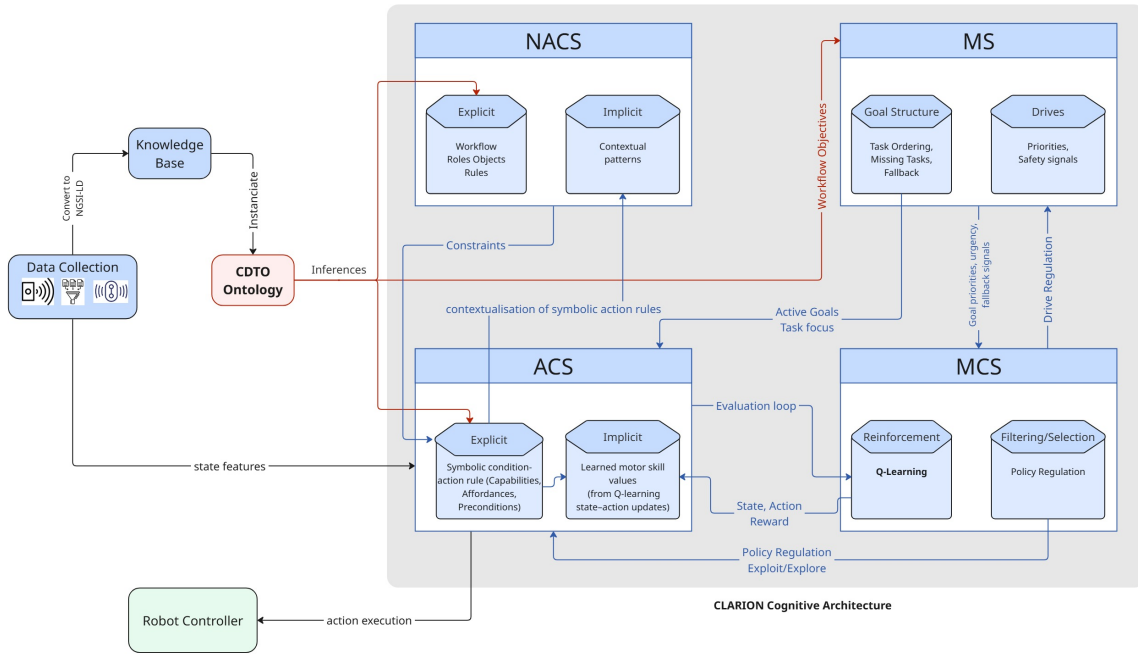


Figure 5.11: Integration of the CDTO ontology with the CLARION cognitive architecture.

The data collection block updates the CDTO ontology using NGS-LD instances that describe objects, tasks, agents, and execution conditions. Ontological reasoning then produces inferences such as task progression, workflow constraints, capability requirements, and feasibility conditions.

These inferred and explicit semantic structures are mapped into CLARION through a feature-based representation and are propagated to its main subsystems (see Section 3.3.1):

- **ACS** is responsible for the procedural aspects of behaviour. Features produced by ontological reasoning, such as available affordances, required capabilities or the presence of appropriate tools, are mapped to the explicit ACS. There, they activate symbolic condition-action rules that filter which behaviours are permissible in the current context. The implicit ACS maintains the numerical action-value functions learned through Q-learning in simulation, enabling the acquisition or improvement of motor skills, such as the helical screwing motion, when new action policies are required.
- **NACS** stores declarative knowledge derived from the ontology, including information on the workflow hierarchy, the temporal relationships between subtasks, the semantic roles of

objects, and their contextual properties. This subsystem provides the CDT with a stable semantic interpretation of the collaborative situation. It also contextualises ACS action rules by applying workflow constraints and contextual patterns, thus promoting consistent behaviour throughout the assembly sequence.

- **MS** organises behaviour around goals derived from ontological relationships, such as **Task** and **directlyFollows**. It prioritises actions, detects missing or pending tasks and manages fallback mechanisms that can allocate human tasks to robots as required. These goals shape the progression of the assembly plan and influence ACS rule activation. Furthermore, the MS communicates urgency, safety, and priority signals to the MCS.
- **MCS** supervises the interaction between symbolic reasoning and learning. It evaluates the performance of the ACS, monitors the success or failure of executed actions, determines when reinforcement learning must be triggered, and regulates the balance between exploration and exploitation. The MCS ensures consistency between declarative knowledge from the NACS and procedural patterns in the ACS, coordinating adaptive behaviour with ontological constraints.

5.4.2 CLARION Configuration

To implement CLARION in software, we used the open-source Python framework **PyClarion** (<https://github.com/cmekik/pyClarion>). PyClarion provides a modular implementation of CLARION, where the main cognitive subsystems are built as interconnected modules and executed in a perception-decision-action loop.

5.4.2.1 CLARION information flow

To operationalise the CLARION architecture, we explored its Python implementation and extracted the underlying information flow across modules. This flow is relatively large, since it includes several interconnected pathways handling perception, memory, goal management, and action execution. For clarity, we therefore present it **part by part**. In this subsection, we focus on the perception pipeline to provide a concrete understanding of how PyClarion modules transform input data into internal symbolic representations. The complete diagrams and explanations of the other cognitive functions (long-term memory, short-term memory, and action selection) are provided in the annex [B](#).

The architecture operates as a recurrent cognitive cycle, where inputs describing the current state of the environment are processed step-by-step to produce an action decision. At each cycle, CLARION receives a set of symbolic inputs representing the situation (e.g., detected objects, task state, or human presence). These inputs are then propagated through a sequence of internal modules that transform raw observations into structured cognitive representations and, finally, into action commands.

This flow can be illustrated with a simple perception example. Assume that perception detects that the rotor is present and reachable in the stock area. This information is received by CLARION as an input feature (e.g., `piece_reached(rotor)=true`). Through the perception pathway, this feature is integrated into the cognitive state and can contribute to the activation

of a higher-level representation such as *rotor_reached*, which may later enable actions such as *grasp rotor* depending on the workflow constraints and the current goal.

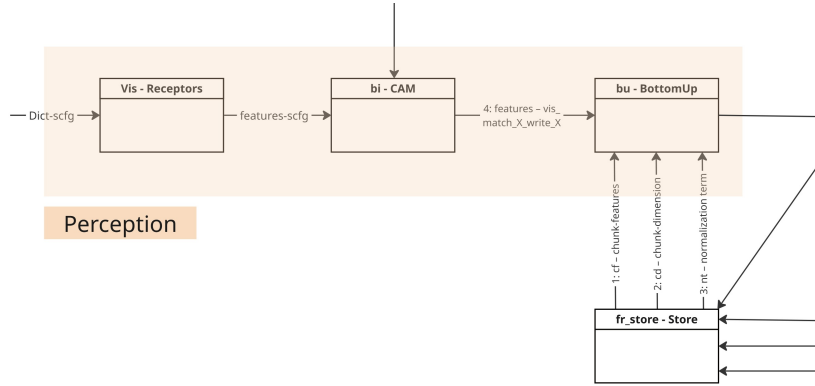


Figure 5.12: PyClarion perception pipeline.

The perception function in CLARION is supported by a set of dedicated modules, as illustrated in Figure 5.12. First, the **Receptors** module converts incoming sensory or semantic data into an internal set of features that can be processed by the architecture. These feature activations are then maintained over time by the **CAM** module, which ensures persistence and allows information to remain available across cognitive cycles. Next, the **bu** (*BottomUp*) module propagates the activation of these features toward higher-level symbolic representations by matching them with stored associations. This mapping relies on a long-term memory structure, the **fr_store**, which contains the predefined correspondences between features and internal representations. Overall, this perception pipeline enables CLARION to transform raw observations into usable symbolic information, supporting the interpretation of the collaborative context before action selection.

To be complete, mapping in pyClarion between the information flow and cognitive functions (long-term memory, short-term memory and action selection) follows the same principle of modular execution. Their detailed diagrams and module-level descriptions are provided in the appendix B, as they mainly support the technical implementation rather than the conceptual contribution of this chapter.

5.4.2.2 Feature and chunk modelling in CLARION

CLARION receives knowledge from the environment and from the ontology in the form of **features**. A feature corresponds to an observable that describes the current state of the collaboration (e.g., *rotor reached*, *human present*, or *step completed*). These features represent the lowest level of information used by the cognitive architecture.

To allow CLARION to reason at a higher level, features are grouped into **chunks**. A chunk is a symbolic unit that represents a meaningful concept for decision-making, such as a task state, a goal, or an action command. In practice, each chunk is associated with one or more features that define when this chunk should be considered active. This mechanism is essential to structure the decision process and to connect perception information to the cognitive rules as seen in Figure 5.13.

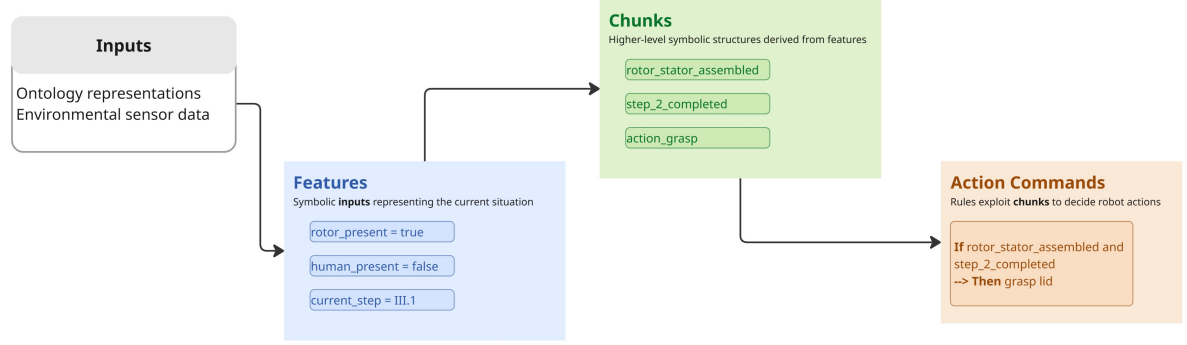


Figure 5.13: Feature and Chunk Modelling in CLARION

In PyClarion, these chunk-feature associations and the decision rules are defined in a dedicated configuration file written in **CCML**. The following example (Listing 5.1) illustrate how chunks are declared and linked to perceptual features, and how they are later used as conditions and consequences to represent scenario progression and action triggering.

```

1 store acs/fr_store:
2
3   chunk rotor_reached:
4     vis#piece_reached rotor_true
5
6   chunk rotor_unreached:
7     vis#piece_reached rotor_false
8
9   chunk set_goal_332:
10    mcs/goals#set-goal 3.3.2
11
12   chunk goal_1:
13    mcs/sl#match-1
14
15   chunk action_grasp:
16    mov#cmd-action grasp

```

Listing 5.1: Examples of chunk-feature associations in the CCML file.

The chunk declarations above follow a consistent logic: perceptual features (**vis#...**) represent the observed state, goal-related features (**mcs/goals#...**) represent goal activation, and command-related features (**mov#...**) represent action requests to be executed by the robot. This structure makes it possible to represent the collaborative workflow as a set of symbolic states and commands that can be manipulated by CLARION.

Based on these chunks, CLARION uses explicit rules to express the progression of the scenario. In this approach, some chunks are used as **activation conditions** (what must be true), while others represent **consequences** (what should be triggered next). The Listing 5.2 below illustrates how a rule can associate a goal state with the activation of an action command.

```

1 store acs/fr_store:
2
3   chunk rotor_reached:
4     vis#piece_reached rotor_true

```

```

5
6 chunk rotor_unreached:
7   vis#piece_reached rotor_false
8
9 chunk set_goal_332:
10  mcs/goals#set-goal 3.3.2
11
12 chunk goal_1:
13  mcs/sl#match-1
14
15 chunk action_grasp:
16  mov#cmd-action grasp

```

Listing 5.2: Example of explicit decision rule encoded in the CCML file.

This chunk-based representation supports a clear decision process: the current situation is first described through features, then abstracted into chunks, and finally used by rules to determine the next action or objective.

Once the CLARION configuration is in place, the CDT can execute *known* tasks through the explicit rule-based pathway, where actions are selected from symbolic condition-action rules grounded in the ontology (capabilities, affordances, tools, and workflow constraints). In our implementation, a “task” is not sent to CLARION as a free-text instruction: it is encoded as a **structured goal and state description**. Concretely, the ontology and the perception stream provide (i) **state features** describing the current situation (e.g., `rotor_present`, `lid_positioned`, `human_absent`) and (ii) **goal features** describing what must be achieved next according to the workflow (e.g., `goal(step_III.1.3)` or `goal(fixate_lid)`). These features are mapped to CLARION *chunks* representing meaningful symbolic states (e.g., `step_II_done`, `lid_fixation_pending`) and used by the motivational mechanisms to keep the current objective active until completion.

An “unknown task” therefore corresponds to a situation where the current goal is well-defined (the workflow indicates what should be done), but the **procedural policy** to achieve it is missing or unreliable for the robot (e.g., no valid action rule applies, or the action has no learned motor routine). In such cases, CLARION can still decide *who should act next* and *what should be attempted* using symbolic knowledge, while the acquisition of *how to do it* must be delegated to the learning pipeline. In our prototype, the triggering and execution of learning is not yet fully automated inside the architecture; therefore, learning is presented as an external training phase that is activated when a missing procedural competence is identified. The next subsection details the learning mechanisms and training procedure used to acquire such procedural skills in simulation.

5.4.3 Learning Mechanisms and Training

Ontology and CLARION determine when learning is necessary and what action should be performed, while the learned policy provides how the motor behaviour is executed. A learning phase is necessary when Clarion is faced with an unknown situation that cannot be resolved based on its internal knowledge and rules, nor through ontological reasoning alone. To this end, we implemented a (D)RL algorithm called Q-Learning that allows the cobot to acquire a missing procedural skill when collaboration requires it. Instead of implementing a complete Q-learning pipeline, we performed simulation learning using a stable policy optimisation method and in-

tegrated the resulting learned policy into the CDT execution loop. Q-learning would update state-action values through repeated interaction with the environment, allowing the cobot to gradually improve its motor behaviour for tasks that cannot be fully specified using symbolic rules.

However, implementing Q-learning in a realistic industrial interaction scenario introduces important technical difficulties. First, the screwing task involves continuous dynamics (force, torque, position), which makes state representation and discretisation non-trivial. Second, the reward function must be carefully designed to ensure safe learning and avoid unstable behaviours. Third, learning directly on the physical robot is not feasible due to safety constraints and hardware wear, which makes simulation-based training necessary. Finally, connecting the learning loop to CLARION in a way that preserves both interpretability and stability requires significant engineering effort, especially when the learning process must be triggered only under specific conditions (e.g., when a task is new or must be executed by the robot for the first time).

The next subsection details the implemented RL setup, including the simulation environment, the state and action representation, the reward structure, and the training procedure used to obtain a reusable screwing skill.

5.4.4 Reinforcement learning for the screwing skill

In the collaborative motor assembly scenario, certain actions require a precise and continuous motor behaviour that cannot be fully defined using only symbolic rules. A typical example is the screwing operation, which combines rotation and vertical translation while respecting physical constraints such as contact force and torque limits.

From a CLARION perspective, this learning mechanism corresponds to the implicit learning pathway supervised by the MCS. When the system detects that an action is required but no reliable motor policy is available (e.g., a new task delegation or a fallback situation), the CDT can trigger a training phase in simulation. Once a satisfactory behaviour is learned, the resulting policy is reused to generate actionable motion commands during execution.

The purpose of learning is to generate a control policy capable of completing the screwing operation efficiently and safely. In our implementation, learning is performed in a simulated environment that models the interaction between a torque-limited screwdriver tool and a screw. The simulation includes both kinematic variables (tool position and rotation) and physical variables (force and torque), so that the learned behaviour remains meaningful for real robotic execution.

Technically, the learning module is implemented as a simulated environment (Gymnasium interface), while the learning mechanism itself is embedded in CLARION. More precisely, Q-learning is implemented in the implicit layer of the ACS, where state-action values $Q(s, a)$ are updated through repeated interaction with the environment. Action selection is then guided by these learned values (e.g., $\arg \max_a Q(s, a)$ with an exploration strategy), enabling the cobot to acquire and improve procedural motor skills under realistic constraints.

5.4.4.1 Environment modelling

The screwing process is represented through two main physical components: (i) a **screw model**, and (ii) a **screwing tool model**. The screw is described by its geometric parameters (length and pitch) and by a torque profile that depends on its state: before reaching the end of travel, the required torque is constant, while after the maximum insertion depth it increases linearly to represent tightening. The tool is modelled as a torque-limited screwdriver equipped with a spring behaviour, which produces a downward force when the tool compresses against the screw head.

This model captures an important constraint of screwing tasks: rotation is only effective when the tool is in contact with the screw, and the applied torque must remain below a maximum value to avoid damage or unsafe behaviour.

The policy controls the screwing behaviour through a discrete action set composed of four actions:

- **Screw (clockwise rotation)**
- **Unscrew (counterclockwise rotation)**
- **Move down** (translation toward the screw)
- **Move up** (translation away from the screw)

Even though the underlying motion is continuous, this discrete formulation provides a clear and interpretable action set, which matches the basic decisions required for screwing: approach contact, rotate to insert, release pressure if saturated, and adjust the tool position when necessary.

At each decision step, the agent receives a numerical observation vector describing the current tool and screw interaction state. The observation is defined as:

$$\mathbf{s} = [z_{tool}, \theta_{tool}, \tau, F]$$

where:

- z_{tool} is the vertical tool position,
- θ_{tool} is the tool rotation angle,
- τ is the torque applied during screwing,
- F is the contact force produced by the tool compression.

This representation provides both geometric and physical feedback, allowing the policy to learn a behaviour that remains consistent with the expected physical interaction.

Figure 5.14 illustrates a typical RL trajectory and highlights the evolution of the main variables used in the observation vector. The first plot shows the vertical position of the screwdriver and the screw along time, reflecting the progressive insertion. The second plot presents the rotation behaviour, where the screw angle increases as insertion progresses, while the screwdriver performs repeated rotational motions. Finally, the third plot reports the contact force and the applied torque, showing that interaction constraints emerge naturally from the simulated contact

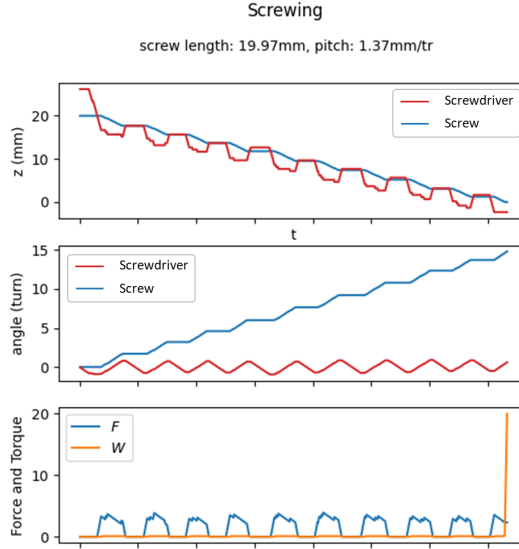


Figure 5.14: RL traces of the screwing task showing tool-screw interaction: vertical position, rotation progress, and resulting force/torque signals over time.

and tightening phases.

The reward function is designed to encourage efficient and stable completion of the screwing operation. It combines several components:

- a small negative time penalty to promote fast completion,
- a positive reward proportional to the progress in screw insertion (increase of screw angle),
- a penalty if the tool reaches saturation limits (angle or vertical bounds),
- a penalty when unscrewing occurs under contact force (unsafe behaviour),
- a strong positive terminal reward when screwing is successfully completed,
- a strong negative reward in case of timeout without completion.

This shaping strategy guides the policy toward a safe sequence: approach the screw, maintain contact force, rotate progressively, avoid saturations, and stop once the tightening condition is reached.

5.4.4.2 Training procedure

Training is performed through multiple episodes, where each episode runs for a fixed number of timesteps. The screwing skill is learned using Q-learning as a standalone training module: the agent iteratively updates the state-action value function $Q(s, a)$ from interaction tuples (s, a, r, s') , and selects actions using an exploration-exploitation strategy (e.g., ϵ -greedy) until a satisfactory policy is obtained. In the current prototype, this Q-learning training is executed **offline** and **is not automatically triggered nor supervised by CLARION**. Instead, CLARION and the ontology provide the decision context (i.e., identifying that the screwing skill

is required), while the learned policy is trained separately in simulation and then reused during execution.

To improve robustness, an optional randomisation mode is applied at reset: the screw length, screw pitch, and the initial tool position can be slightly varied. This prevents overfitting to a single configuration and improves generalisation across similar screwing conditions. At the end of training, the policy is saved as a reusable model, which can later be loaded for execution or evaluation.

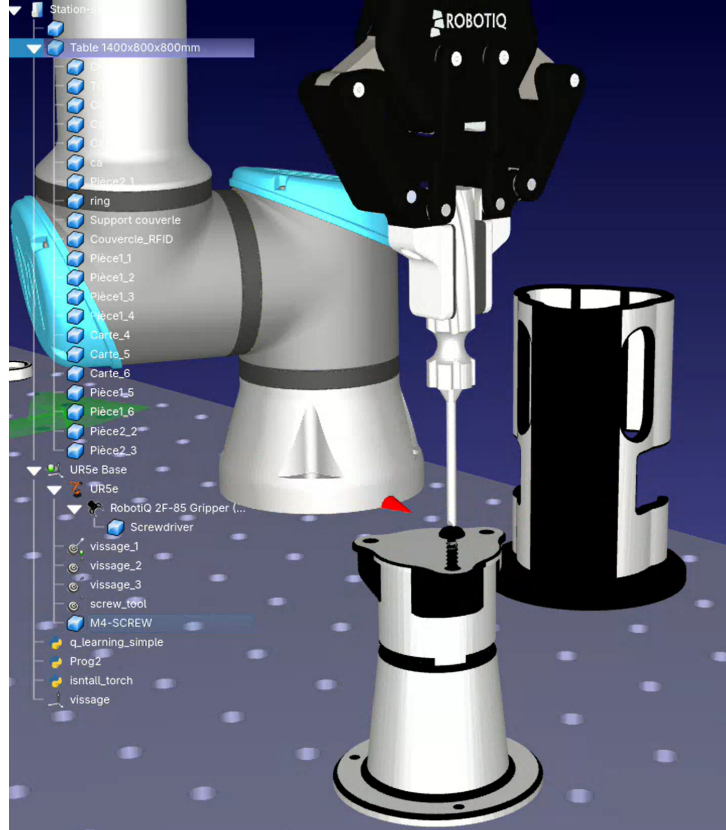


Figure 5.15: RoboDK simulation setup used to validate the screwing policy on the UR5e cobot equipped with a screwdriver tool.

To validate the learned behaviour in a realistic robotic setup, the screwing task is integrated into a RoboDK simulation environment as illustrated in 5.15, where the UR5e cobot is equipped with a screwdriver tool and positioned above the motor casing. This virtual setup reproduces the main execution constraints (tool alignment, approach motion, and workspace configuration), and allows testing the learned policy before transferring it to the CDT workflow.

5.4.4.3 Integration with CLARION decision-making

Symbolic decision-making remains handled by the CLARION subsystems (workflow interpretation, goal management, and rule-based selection). When CLARION selects an action that requires a learned motor routine (e.g., screwing), the system can invoke the learned policy to generate a sequence of low-level control steps until the action is completed. In this way, reinforcement learning complements the cognitive architecture by providing adaptive procedural

knowledge, while maintaining interpretability at the symbolic level.

For this use case, once learning has converged, the CDT updates the robot’s internal representation, integrating the new `hasAbility_Screw` capability into the cobot’s operational knowledge. The cobot can then autonomously perform the missing screwing action, restoring continuity to the collaborative workflow.

5.5 Evaluation framework and metrics

This section evaluates the cognitive interoperability reached in our HRC use case using the Maturity Model of Cognitive Interoperability (MMCI) defined in Chapter 2/Section 2.4. In this evaluation, cognitive interoperability is assessed **between two coupled agents**:

- **<CPS, CDT>**: the cobot (CPS) together with its CDT.
- **<Human, HDT>**: the human operator together with the HDT as the cyber representation of the human for collaboration purposes.

Although the HDT is not developed in this thesis (it is addressed by parallel work within the same project [Gaffinet et al. \(2025a\)](#)), it is part of the conceptual use-case architecture. In the MMCI perspective, the HDT is the element that should allow the human side to be represented in a machine-processable way, so that alignment can be evaluated between **<Human, HDT>** and **<CPS, CDT>** using shared representations, interaction traces, and measurable indicators. In practice, the evaluation relies on the available observable evidence in the use case: system logs, state updates, workflow execution traces, interaction events).

Table 5.1 summarises the evaluation grid used to assess the maturity of cognitive interoperability in the HRC use-case. For each MMCI level, the table lists the corresponding indicators, states whether they are supported by the current implementation (*Yes/Partial/No*), and provides the concrete evidence available in the use-case (e.g., semantic representations, workflow execution traces, decision logs). The goal of this grid is to make the assessment transparent and reproducible: the maturity level is determined as the highest MMCI level for which the required indicators are evidenced in the collaboration between **<CPS,CDT>** and **<Human,HDT>**, while explicitly noting when the available evidence mainly reflects CDT-side capabilities.

Table 5.1: MMCI indicators and applicability in the use-case. The evaluation targets cognitive interoperability between **<CPS,CDT>** and **<Human,HDT>**; when the evidence is mainly internal to the CDT, this is stated explicitly.

MMCI indicator (Chapter 2)	Applied?	Evidence / explanation in the use-case
Level 0 — Semantic interoperability (baseline)		
Shared ontology / mappings / constraints (consistent meaning of exchanged knowledge)	Yes	CDTO ontology instantiated as NGSI-LD entities provides a shared representation of agents, tasks, objects, and relations; CDT modules use the same identifiers/relations to process information consistently.

Continued on next page.

Table 5.1 continued.

MMCI indicator (Chapter 2)	Applied?	Evidence / explanation in the use-case
Conformance / rule-based semantic consistency checks	Yes	Ontology constraints and inference rules support machine-processable semantics (e.g., consistency of task/object relations and workflow constraints within the CDT knowledge base).
Level 1 — Shared situation awareness		
State agreement (shared, time-stamped operational state)	Partial	The CDT maintains a synchronised operational state (task progress, object state, human availability) through perception updates and inference. Alignment with <Human,HDT> depends on what the HDT exposes/receives; in the implemented prototype, evidence mainly comes from CDT-side traces.
Event detection latency (deviation/alert detection speed)	Yes	Deviations from the expected workflow are detected when perception updates and inferred states indicate pending/failed steps; timestamps in logs allow measuring detection latency in the cyber layer.
Short-horizon projection agreement (near-future expectation consistency)	No / Not evidenced	The traces demonstrate updated state and deviation detection, but do not provide an explicit shared prediction model (or measurable divergence) exchanged between <CPS,CDT> and <Human,HDT>.
Level 2 — Shared / mapped mental models		
Shared task structure (workflow / precedence constraints)	Yes	The assembly procedure is modelled as an ordered workflow with precedence constraints (e.g., directlyFollows) and explicit step identifiers, supporting a structured representation of task organisation.
Role and responsibility model (who does what, when, and why)	Yes	Ontology-based modelling of agent capabilities/roles supports assignment of steps to human vs cobot; the system can represent which steps are expected from which agent and under which feasibility conditions.
Coordination quality indicators (hand-off failures, coordination errors)	Partial	Execution traces can be analysed for pending tasks, skipped steps, and takeover events; however, a systematic taxonomy/metric computation for coordination errors is not yet established in the current evaluation protocol.
Level 3 — Intent and reasoning alignment		

Continued on next page.

Table 5.1 continued.

MMCI indicator (Chapter 2)	Applied?	Evidence / explanation in the use-case
Goal alignment / goal-driven adaptation (plan repair, re-allocation)	Partial	CLARION supports goal-driven task allocation and fallback decisions (e.g., takeover when a human step is pending and feasible for the robot), but mutual goal alignment with <Human,HDT> is not fully evidenced.
Rationale traceability (evidence/constraints used for decisions)	Partial	Constraint-based feasibility checks exist (capability/affordance compatibility), but explanations are not systematically exposed as human-interpretable rationales enabling mutual reasoning alignment.
Plan-repair latency (time to revise plan under deviation)	Yes (CDT-side)	Decision logs and workflow traces allow measuring how quickly the CDT revises allocation after detecting a deviation; this remains mainly a cyber-side metric in the current prototype.
Level 4 — Joint decision-making with metacognitive regulation		
Explicit uncertainty/confidence outputs available to the partner	No	No calibrated confidence or uncertainty estimates are produced and shared as part of the operational loop.
Dynamic autonomy regulation based on risk/uncertainty (ask/confirm/act policies)	No	No implemented policy for requesting confirmation when uncertain, or for adapting autonomy based on risk/confidence.
Error monitoring and proactive interception linked to confidence	No	The prototype does not yet implement metacognitive monitoring that triggers verification/recovery strategies based on confidence/error likelihood.

MMCI level reached in the use-case. Based on Table 5.1, the use-case demonstrates **Level 2 (Shared/Mapped Mental Models)** as the highest level for which the core indicators are evidenced. Elements of **Level 3** are **partially** present through goal-driven adaptation and feasibility-based takeover decisions in the CDT, but mutual intent and reasoning alignment with <Human,HDT> is not fully demonstrated. **Level 4** is not reached because explicit uncertainty/confidence modelling and metacognitive regulation policies are not implemented.

5.6 Chapter Conclusion

This chapter reports on the implementation and evaluation of an industrial use case of HRC, based on a collaborative engine assembly process. Its contribution consists of making the concepts of the thesis operational in a concrete setting by (i) structuring the knowledge of the scenario with the CDTO ontology (**RQ4**) and (ii) prototyping a partial CDT focused on the *cognition* and *simulation* functions necessary for decision support and skills acquisition in the

use case (**RQ2** + **RQ3**).

On the semantic side, the chapter shows how the scenario is encoded in NGSI-LD and is used to automatically instantiate the ontology, so that agents, tasks, tools, workflow constraints, and execution context are represented in a unified knowledge base. Reasoning mechanisms (descriptive logical inference and SWRL rules) are then used to deduce additional facts that are not explicitly provided by the input stream, and thus in our use case deducing that the cobot can perform a given operation such as **canScrew**. This in turn provides the CDT with an explicit and explainable representation of the states.

On the cognitive side, the chapter does not claim a full CLARION implementation. Instead, it presents an integration-oriented configuration that illustrates how ontological content (including inferred statements, but also declared semantic structures such as workflow and constraints) can be mapped into CLARION-inspired representations (features/chunks/goals) to support task interpretation and decision logic in the use case. In parallel, the simulation function is demonstrated through a Q-learning setup in simulation for the screwing skill, showing how a procedural policy can be obtained and reused during execution; however, this learning is not fully embedded and automatically orchestrated inside CLARION.

Finally, cognitive interoperability is addressed in the evaluation stage: the chapter applies the maturity model measures defined in Chapter 2 in order to position what has been achieved (and what has not) in relation to the expected levels and criteria. The assessment also highlights that the use case **does not reach level 3 and above**, mainly because learning and policy updates are not managed automatically in CLARION (there is no closed loop where the cognitive architecture triggers training, updates procedural knowledge and reuses it online). As a result, the interoperability achieved remains below full cognitive interoperability as defined by the model, even though the essential prerequisites (semantic anchoring, explainable inference, and cognitive decision logic) are in place.

Several aspects remain open and are identified as limitations of the current implementation. First, the pipeline is not yet fully automated end-to-end between ontology updates, CLARION structures, and execution control. Second, the learning loop is not integrated inside CLARION (i.e., CLARION does not automatically trigger training, manage policies, and update internal procedural knowledge), which directly limits the maturity level that can be claimed. Third, the experimental validation is centred on the screwing-related part of the assembly, and the generality of the approach should still be assessed on an additional collaborative use case beyond screwing.

Chapter 6

General Conclusion

6.1 Summary of the thesis

This thesis is conducted in the context of the **AI4C2PS** project, which aims to overcome the limitations of *semantic interoperability* in HCC by extending it toward **cognitive interoperability**. The main objective of this work is therefore to *establish the conceptual, methodological, and computational principles enabling cognitive interoperability between humans and CPSs within a CPE*.

To achieve this objective, the thesis follows a progressive approach, starting from conceptual foundations and then moving toward formalisation and implementation. First, a comprehensive state of the art clarifies how cognition is understood in psychology and cognitive science, how it is transferred into industrial systems, and how the notion of cognitive interoperability is introduced across different domains. This review highlights a major gap: despite increasing references to “cognitive” collaboration in Industry 5.0, there is no rigorous operational basis explaining what cognitive interoperability concretely means for human-machine or machine-machine cooperation. Our work contributes to answer a series of research questions, which we detail in the following:

RQ1. How can cognitive interoperability within CPE be formally defined and characterised?

Our first contribution is a formal conceptualisation of cognitive interoperability, defined as a higher interoperability level that goes beyond semantic alignment. Building on cognitive psychology and team cognition theories, cognitive interoperability is characterised as the capability of human and CPS agents to mutually align cognitive processes, not only exchanging meaning, but also converging toward compatible *representations of the situation, mental models, intentions, and decision rationales*. To make this concept operational, we propose a structured framework defining the core dimensions of cognitive interoperability (shared awareness, shared understanding, reasoning alignment, and decision co-regulation). This led to the development of a Maturity Model of Cognitive Interoperability (MMCI), organised into successive levels that describe a measurable transition from semantic interoperability to deeper cognitive alignment. The MMCI provides both a scientific foundation and a practical tool for assessing the progression of collaboration capabilities between humans and CPSs in industrial settings.

RQ2. How can the concept of a cognitive layer be defined for CPSs, and how can concepts such as cognitive systems, CCPS and CDTs be theoretically defined?

The second research question focuses on defining what it means for a system to be *cognitive* in the industrial domain and how such cognition can be integrated into CPSs and DTs. To address this, we introduced a clear conceptual distinction between:

- **Cognitive systems**, defined as systems capable of perception, learning, memory, reasoning, and goal-oriented decision-making;
- **Cognitive Cyber-Physical Systems (CCPSs)**, defined as CPSs augmented with cognition enabling autonomous adaptation and interpretation; and
- **Cognitive Digital Twins (CDTs)**, defined as DTs explicitly embedding cognitive mechanisms that support reasoning, explanation, and anticipation.

This theoretical foundation is built upon an explicit modelling of cognitive functions relevant to industrial collaboration (e.g., perception, attention, memory, learning, reasoning and decision-making). A key outcome is that cognition is not treated as an abstract “AI capability”, but rather as an organised functional layer that enables an agent to build internal models of the environment and of its partners in interoperation, and to use these models to adapt its behaviour. We position the DTs as the most suitable “host” for cognition, allowing CPS entities to become cognitively interoperable through their CDTs even when the physical system remains non-cognitive.

RQ3. Which principles of representation and architecture are needed to make cognition functional in CDTs, and how can ontologies and cognitive architectures contribute to this process?

The third research question addresses the implementation challenge: once cognition is defined, what computational structures are required to make it functional in CDTs, and how can symbolic and sub-symbolic approaches be combined? In a HCC context, we have focused on:

1. **explicit knowledge representation** (to structure goals, tasks, constraints, agents, and context), and
2. **cognitive architectures** (to organise reasoning, decision-making, and learning into coherent mechanisms).

To fulfil the representation requirement, we propose the CDTO ontology, designed as a shared semantic and cognitive foundation for collaborative industrial tasks. This ontology integrates key concepts related to humans, systems, cognition, and abilities, and supports reasoning through formal constraints and inference rules. It therefore enables a CDT to store knowledge and to derive new knowledge about the collaboration state, the actors’ capabilities, and the context of action.

To address the architectural requirement, we configure and prototype the integration of the CLARION cognitive architecture as the decision layer within the CDT for the HRC use case. This choice is aligned with the hybrid orientation of the AI4C2PS project, since it allows us to combine (i) **symbolic structures** provided by the ontology (workflow knowledge, roles, constraints, and inferred feasibility/abilities) with (ii) **CLARION-inspired decision representations** (features, chunks, and goals) used to support action selection.

In our implementation, CLARION exploits ontological knowledge together with perception-derived state features to support task interpretation and decision support, for instance by selecting the next action and reallocating responsibility when the collaboration deviates from the nominal workflow. Concerning learning, we do not implement a fully integrated CLARION learning loop. Instead, we develop and validate a Q-learning training setup separately in simulation to acquire a procedural screwing skill, and we reuse the resulting policy during execution.

Finally, the thesis validated the proposed concepts and mechanisms through an implementation in a human-robot collaborative assembly scenario, where the CDT supports decision-making and adaptation during a dynamic task (screwing operation). The evaluation compared a baseline semantic workflow execution with the proposed cognitive architecture-enabled approach, and assessed the achieved collaboration level using the MMCI metrics. The results confirmed that adding a cognitive layer through ontology-driven reasoning and CLARION-based decision mechanisms supports higher levels of cognitive interoperability, particularly in situations requiring adaptation, anticipation, and coherent joint decision-making.

In the end, we have three contributions: (1) a definition and preliminary maturity model of cognitive interoperability, (2) a theoretical framework for cognition in CPS, CCPS and CDTs, and (3) a computational approach combining ontologies and cognitive architectures to make cognition operational in industrial DTs; thereby providing a foundational step toward robust human-centric collaboration in Industry 5.0.

6.2 Perspectives and Future Directions

Although this thesis provides conceptual, methodological, and computational foundations toward cognitive interoperability in CPE, several perspectives remain open and constitute promising directions for future work.

A first perspective concerns the proposed MMCI. While the model provides a structured characterisation of progressive cognitive interoperability levels, it still requires further consolidation through the definition of explicit and verifiable metrics associated with each level. Future work should therefore focus on designing quantitative and qualitative indicators that allow the assessment of cognitive interoperability in a measurable and reproducible manner. Such metrics would make it possible to validate the maturity levels through experiments, industrial benchmarks, and comparative evaluations across different collaborative contexts.

A second perspective relates to the implementation of the CDT. In this thesis, the work mainly focused on the cognitive layer, its representation mechanisms, and the integration of a cognitive architecture to support reasoning and decision-making. In addition, a first part of the simulation function was already addressed within the proposed CDT approach. Nevertheless, the overall CDT vision still requires further development, particularly regarding the emulation function, which was not considered in this work. Future developments should therefore aim at integrating emulation capabilities, enabling the CDT to reproduce more accurately the dynamic behaviour of the physical system in real-time and to support enhanced predictive reasoning based on realistic system evolution. Such an extension would strengthen the CDT as a comprehensive tool for anticipation, monitoring, and decision support in cyber-physical enterprise environments.

Finally, concerning the validation use case, future work should enhance the integration of the learning mechanisms within the proposed cognitive architecture. While the CLARION-based approach demonstrated the feasibility of introducing cognitive reasoning and decision capabilities, the learning part of the architecture was not fully exploited in the experimental setup. A relevant continuation would therefore be to implement and evaluate the learning component of CLARION within the CDT, allowing the system to improve its behaviour over time through experience and interaction. This would make it possible to study adaptive cognitive interoperability, where collaboration progressively evolves based on accumulated knowledge, learned strategies, and contextual adaptation.

Appendix A

Implementation Listings (NGSI-LD and Ontology Processing)

A.1 NGSI-LD input format

```
1 {
2   "@context": [
3     "https://uri.etsi.org/ngsi-ld/v1/ngsi-ld-core-context-v1.jsonld",
4     "https://schema.lab.fiware.org/ld/context",
5     {
6       "Cobot": "https://www.ai4c2ps.eu/ontologies/2024/core#Cobot",
7       "Operator": "https://www.ai4c2ps.eu/ontologies/2024/core#Operator",
8       "Screw": "https://www.ai4c2ps.eu/ontologies/2024/core#Screw",
9       "Screwdriver": "https://www.ai4c2ps.eu/ontologies/2024/core#Screwdriver",
10      "Screwdriver": "https://www.ai4c2ps.eu/ontologies/2024/core#Screwdriver",
11      "Screwdriver": "https://www.ai4c2ps.eu/ontologies/2024/core#Screwdriver",
12    },
13   "@graph": [
14     {
15       "id": "urn:ngsi-ld:Cobot:ur5",
16       "type": "Cobot",
17       "name": "ur5",
18       "hasPart": {
19         "type": "Relationship",
20         "object": "urn:ngsi-ld:Hand-E:handE_Gripper"
21       }
22     }
23   ]
24 }
```

Listing A.1: Representative extract from the NGSI-LD file illustrating the structure of the context and the graph.

```
1 {
2   "id": "urn:ngsi-ld:Workflow:screwingWorkflow",
3   "type": "Workflow",
```

```

4  "name": "screwingWorkflow",
5  "hasStep": [
6    {
7      "type": "Relationship",
8      "object": "urn:ngsi-ld:PhysicalTask:assemblyTask"
9    },
10   {
11     "type": "Relationship",
12     "object": "urn:ngsi-ld:PhysicalTask:
13       positioningRotor_StatorInCasingTask"
14   },
15   {
16     "type": "Relationship",
17     "object": "urn:ngsi-ld:PhysicalTask:lidPositioningAndFixatingTask"
18   }
19 ],
20 {
21   "id": "urn:ngsi-ld:PhysicalTask:positioningRotor_StatorInCasingTask",
22   "type": "PhysicalTask",
23   "name": "positioningRotor_StatorInCasingTask",
24   "hasPart": [
25     {
26       "type": "Relationship",
27       "object": "urn:ngsi-ld:Positioning:positioningCasingTask"
28     },
29     {
30       "type": "Relationship",
31       "object": "urn:ngsi-ld:Positioning:
32         positioningAssembledRotorStatorTask"
33     }
34   ],
35   "directlyFollows": {
36     "type": "Relationship",
37     "object": "urn:ngsi-ld:PhysicalTask:assemblyTask"
38   },
39   "hasStepNumber": {
40     "type": "Property",
41     "value": "2"
42   }
43 }

```

Listing A.2: Representative excerpt of the NGSI-LD file illustrating the screwing workflow steps.

```

1  {
2    "id": "urn:ngsi-ld:MentalTask:searchRotorTask",
3    "type": "MentalTask",
4    "name": "searchRotorTask",
5    "hasStepNumber": {
6      "type": "Property",
7      "value": "1.2.1"
8    },
9    "isExecutedIn": {

```

```

10     "type": "Relationship",
11     "object": "urn:ngsi-ld:Action:searchRotorAction"
12 },
13 "requiresAbility": {
14     "type": "Relationship",
15     "object": "urn:ngsi-ld:PerceptionAbility:visualPerception"
16 },
17 "involvesFunction": {
18     "type": "Relationship",
19     "object": "urn:ngsi-ld:Perceiving:visualPerceptionFunction"
20 }
21 }

```

Listing A.3: Modelling of the cognitive task *SearchRotor* in NGSI-LD.

A.2 Ontology loading and merging (Owlready2)

```

1 from owlready2 import *
2
3 world = World()
4
5 # Load each ontology separately
6 onto_ssn = world.get_ontology("onto/ssn.rdf").load()
7 onto_sosa = world.get_ontology("onto/sosa.rdf").load()
8 onto_soma = world.get_ontology("http://www.ease-crc.org/ont/SOMA.owl#").
9     load()
10 onto_core = world.get_ontology("onto/CDTO-core.owl").load()
11 onto_agent = world.get_ontology("onto/SysAgent.owl").load()
12 onto_cognition = world.get_ontology("onto/CognitionOnto.owl").load()
13 onto_dul = world.get_ontology("http://www.ontologydesignpatterns.org/
14     onto/dul/DUL.owl").load()
15
16 # Main ontology (imports the others conceptually)
17 onto_main = world.get_ontology("onto/AI4C2PS.owl").load()

```

Listing A.4: Loading ontologies with Owlready2.

```

1 merge_classes(onto_core, onto_main)
2 merge_classes(onto_ssn, onto_main)
3 merge_classes(onto_sosa, onto_main)
4
5 merge_properties(onto_core, onto_main)
6 merge_properties(onto_ssn, onto_main)
7 merge_properties(onto_sosa, onto_main)

```

Listing A.5: Copying classes and properties from imported ontologies into the main ontology.

A.3 Automatic ontology instantiation from NGSI-LD

```

1 def create_instances_ngsi_ld(graph_data, onto_main):
2     instances = {}
3     for obj in graph_data: # main loop
4         obj_id = obj.get("id")
5         obj_type = obj.get("type")
6
7         # use the last part of the ID as the name
8         obj_name = obj_id.split(":")[-1]
9         if not obj_id or not obj_type:
10             continue
11
12         if isinstance(obj_type, list):
13             primary_type = obj_type[0]
14             other_types = obj_type[1:]
15         else:
16             primary_type = obj_type
17             other_types = []
18
19         # Find the class in the ontology
20         class_obj = find_class(primary_type)
21         if class_obj:
22             inst = class_obj(obj_name) # create instance with the
23                                     # obj_name
24             instances[obj_name] = inst
25
26         # ... Rest of the function
27     return instances

```

Listing A.6: Excerpt of the instantiation script (instance creation).

```

1 # Function to add properties to instances
2 def add_properties_ngsi_ld(graph_data, instances, onto_main):
3     # Loop through each entity in the NGSI-LD graph
4     for obj in graph_data:
5         # Get the corresponding instance from the first instantiation
6         # step
7         obj_id = obj.get("id").split(":")[-1]
8         inst = instances.get(obj_id)
9         if not inst:
10             continue
11
12         # Loop through each property of the object
13         for prop, val in obj.items():
14             # Find the corresponding property in the ontology
15             prop_obj = world.search_one(iri="*" + prop)
16             if prop_obj is None:
17                 continue
18
19             # Check if the property is an object property (a
20             # relationship)
21             if isinstance(prop_obj, ObjectPropertyClass):
22                 # Get the ID of the target instance
23                 target_id = val.get("object").split(":")[-1]
24                 target_inst = instances.get(target_id)

```

```

23         # Add the relationship to the instance
24         if target_inst:
25             getattr(inst, prop).append(target_inst)
26
27         # Check if the property is a data property (a literal value)
28         elif isinstance(prop_obj, DataPropertyClass):
29             # Get the literal value from the data
30             literal_value = val.get("value")
31
32         # Add the literal value to the instance
33         prop_obj[inst].append(literal_value)
34

```

Listing A.7: Excerpt of the script for adding relations and properties.

```

1 import json
2
3 with open("scenario.json", "r") as f:
4     data = json.load(f)
5     graph_data = data.get("@graph", data)
6
7 with onto_main:
8     instances = create_instances_ngsi_ld(graph_data, onto_main)
9     add_properties_ngsi_ld(graph_data, instances, onto_main)

```

Listing A.8: Ontology instantiation process from the NGSI-LD file.

A.4 Inference illustration (before vs. after)

```

1 {
2   "id": "urn:ngsi-ld:Cobot:ur5",
3   "type": "Cobot",
4   "name": "ur5",
5   "hasPart": {
6     "type": "Relationship",
7     "object": "urn:ngsi-ld:Hand-E:handE_Gripper"
8   }
9 }

```

Listing A.9: Excerpt of the NGSI-LD input file before inference.

```

1 {
2   "id": "urn:ngsi-ld:Cobot:ur5",
3   "type": [
4     "Entity",
5     "Cobot",
6     "Agent"
7   ],
8   "name": "ur5",
9   "hasAbility": {
10     "type": "Relationship",
11     "object": [

```

```
12     "urn:ngsi-ld:Ability:canScrew"  
13   ],  
14 },  
15 "hasPart": {  
16   "type": "Relationship",  
17   "object": "urn:ngsi-ld:Hand-E:handE_Gripper"  
18 }  
19 }
```

Listing A.10: Excerpt of the NGSI-LD output file after inference.

Appendix B

Additional CLARION functional pathways

This annex complements Section 5.4.2.1 in Chapter 5 that focuses on the perception pipeline by providing additional technical diagrams of the CLARION information flow implemented with PyClarion. The following figures present the internal module organisation for the other functions of the architecture, namely long-term memory, short-term (working) memory, and action selection. These diagrams support the technical understanding of how CLARION processes symbolic inputs and generates executable decisions within the CDT.

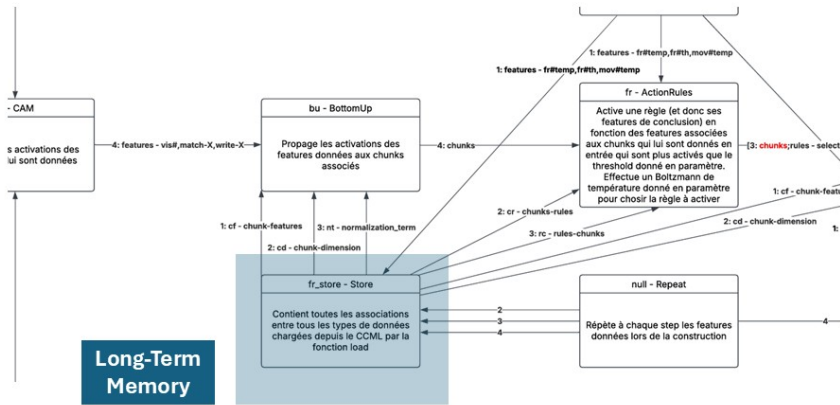


Figure B.1: PyClarion Long-Term Memory pipeline.

The **Long-Term Memory (LTM)** function in PyClarion in Figure B.1 is mainly supported by the **fr_store** module, which stores persistent knowledge required for decision-making. Concretely, this module contains the associations between **features**, **chunks**, and **explicit rules** defined in the CCML configuration. These stored associations allow CLARION to retrieve mean-

ingful symbolic knowledge from incoming perceptions and to maintain stable behavioural structures across time.

From a functional perspective, LTM plays two key roles in the architecture. First, it enables the *BottomUp* pathway (**bu**) to activate relevant chunks based on the current set of feature activations, by matching perception with stored chunk-feature patterns. Second, it supports the selection of decision rules through the **fr** (*ActionRules*) module, since rule activation depends on the chunks that are currently active. In this sense, LTM provides the memory backbone that connects perceptual information to symbolic representations and action selection.

In CLARION, this long-term storage includes both declarative and procedural memory:

- Declarative memory is represented through ontologies that structure knowledge into chunks and concepts.
- Procedural memory is encoded in the rules specified in the .ccml file, guiding behavior through learned action mappings.

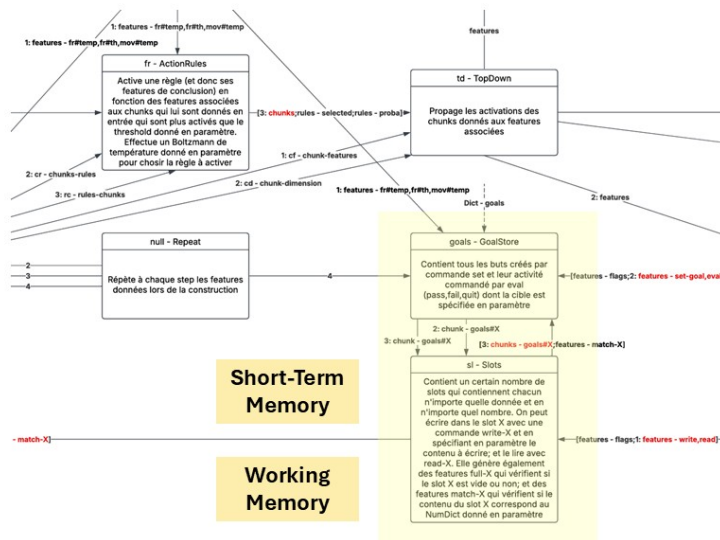


Figure B.2: PyClarion Short-Term Memory pipeline.

The **Short-Term Memory (STM)** function in Figure B.2, also referred to as working memory, is implemented in PyClarion through two complementary modules: **GoalStore** and **Slots**. The **GoalStore** module maintains temporary goals that are active during execution. These goals are typically set by internal commands (e.g., **set-goal**) and can evolve over time depending on execution outcomes (e.g., success, failure, or completion). In parallel, the **Slots** module acts as a temporary *scratchpad memory* that can store symbolic elements for a limited duration. Slots can be written to or read from, allowing the architecture to keep important intermediate information available across multiple cognitive cycles.

In the HRC use case, STM is important because it allows CLARION to preserve the active objective of the assembly process and to maintain short-term contextual information that supports consistent decision-making. For example, it can store the current step goal, the pending task to be achieved, or a temporary state needed to coordinate successive actions. Overall, this mechanism supports real-time reasoning by ensuring that the decision process is continuously guided by the relevant goal and context.

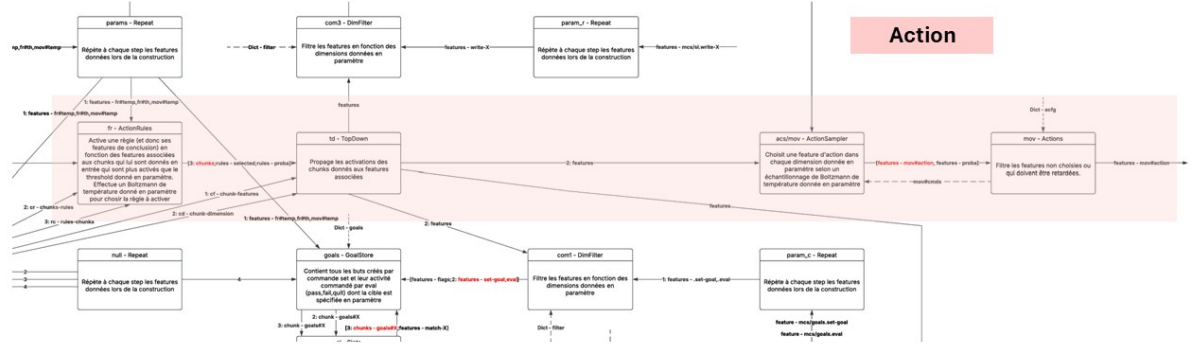


Figure B.3: PyClarion Action Selection pipeline.

The **Action** function in Figure B.3 is executed through a sequence of modules that translate internal decisions into executable actions. This process begins with the **fr** (*ActionRules*) module, which activates explicit rules based on the currently active chunks. These rules define which actions are relevant or permitted in the current context. The resulting feature activations are then propagated through the **td** (*TopDown*) module, which reinforces and spreads activation toward the most suitable action representations.

Next, the **acs/mov** (*ActionSampler*) module selects an action feature in each available dimension by using a probabilistic sampling mechanism (Boltzmann sampling). Finally, the **mov** (*Actions*) module filters or delays actions that are not selected, ensuring that only the chosen command is transmitted to the execution layer. This chain enables CLARION to generate context-appropriate actions that remain consistent with the current goal, workflow constraints, and available knowledge.

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